PM@Siemens Guide

Version 6.4

Version 6.4, updated Oct 2025

This is the original English version.
It always takes precedence over any translation.

All mentioned functions are describing roles (e.g. PM – Project Manager) and respective responsibilities – not individuals. Descriptions are displayed as gender neutral.

All changes of PM@Siemens Guide 6.4 compared to 6.3 are applicable/binding since Oct 15, 2025 for each new project.

If you notice any errors or discrepancies, please inform us about these through our functional mailbox (project.business.excellence@siemens.com) and initial the subject line with ''PM@Siemens Guide''.

The content of the PM@Siemens Guide is the Annex to Siemens Circular No. 200.
For the current version of the PM@Siemens Guide, questions, and more information, please see the PM@Siemens homepage.

PM@Siemens Guide

General notes

The term “Organizational Units” in this PM@Siemens Guide refers to P&L Units and their Business Units.

The term “Project Manager” refers to both Project Manager Bid (PM Bid) and Project Manager Execution (PM Execution). The same applies for Commercial Project Manager and other key roles.

The PM@Siemens Guide is a framework of processes and regulations that can be specified and supplemented by more rigorous requirements from the Organizational Units released through implementation guidance.

Each sub-chapter of chapter 2 and 3 is introduced by a question to interest the readers in the topic, regardless of their role, and to allow them to consider their contribution to it.

Throughout the PM@Siemens Guide, links to other chapters, collaboration platforms, Siemens’ circulars, and intranet pages are used. These are marked in blue and are underlined.

1 Siemens – A Company with Project Excellence

1.1 Preface PM@Siemens Guide

Dear Colleagues,

We are a focused technology company that can combine the real and digital worlds like no other company. We leverage this unique capability to support our customers in mastering their digital transformation and revolutionizing entire industries – industries that form the backbone of our economies and shape our everyday lives.

Our goal is to create sustainable value for our customers with our products and solutions. Our project business is, and will remain, a central component in this effort. Our projects break new ground again and again and are often highly complex. We weigh the risks and, we always learn – since given we have a Growth Mindset. PM@Siemens has supported us in successfully implementing our projects for many years.

Change in the world is accelerating, and that in turn is changing the needs of our markets. New business models like Software-as-a-Service and the cloud are becoming increasingly important for us and our customers. We’re taking these factors into account in our ongoing own transformation and are adapting our project business to the changing requirements of digitalization. The revised PM Guide 6.4 is designed to support you in continuing to maintain our company’s project Management at an outstanding level. This guideline will empower you and your team to make the right decisions and achieve the best possible results.

Let’s continue to collaborate to shape the future, offer our customers the right technologies at the right time with the right processes, and so ensure our ongoing success together.

Roland Busch Chief Executive Officer	Ralf P. Thomas Chief Financial Officer	Judith Wiese Chief People and Sustainability Officer

1.2 PM@Siemens Guide Architecture

The PM@Siemens Guide is designed to ensure clear guidance for project sales and execution. It addresses all relevant professional areas of project Management[1]. It will support the Chief Executive Officers (CEOs) and their key functions as well as Project Managers (PMs) along with their teams to strive for excellence in project business.

The Milestone Model represents the core of the PM@Siemens framework and is supplemented by the other key elements (Figure 1), including the Definitions and Abbreviations (Annex 1: Definitions and abbreviations), the PM Career Model and Development for Project Manager and Commercial Project Manager (Annex 2: Career Model and Development for Project Manager and Commercial Project Manager) and the Siemens Project Management Plan (SPMP).

Figure 1: Key elements of the PM@Siemens Guide[2]

The PM@Siemens Guide describes the expectations for excellent project business and the mandatory minimum requirements that need to be achieved in two dimensions.

In the first dimension (Figure 2), key activities can be found. These are allocated to professional areas of the project business and project Management that are essential to be established and continuously utilized throughout each project's lifecycle.

Therefore, the term project business comprises all activities that are essential to executing customer projects from an organizational perspective. In this guide, project business topics represent the organizational framework. They describe how the Organizational Unit enables the acquisition and execution of customer projects (Chapter 2). To ensure an efficient value contribution to each individual project, the organizational role of a Responsible Business Manager (RBM) is also defined.

[1] Siemens Organization Handbook and Siemens Business Conduct Guideline

[2] Under consideration of ISO21500

Project Management is the application of methods, tools, techniques, and competencies to an individual project[3]. The detailed project Management deliverables are described under Project Management Topics (Chapter 3). The Siemens Project Management Plan is the instrument that the PM uses to plan and monitor their projects (Chapter 4).

In the second dimension (Figure 2), detailed requirements are described, allocated to specific Milestones and adapted to our core project types (Solution, Service and Small Projects). They are established in a logical order based on clear inputs, outputs, and recommended activities (PM@Siemens Milestone Models (Chapter 5)).

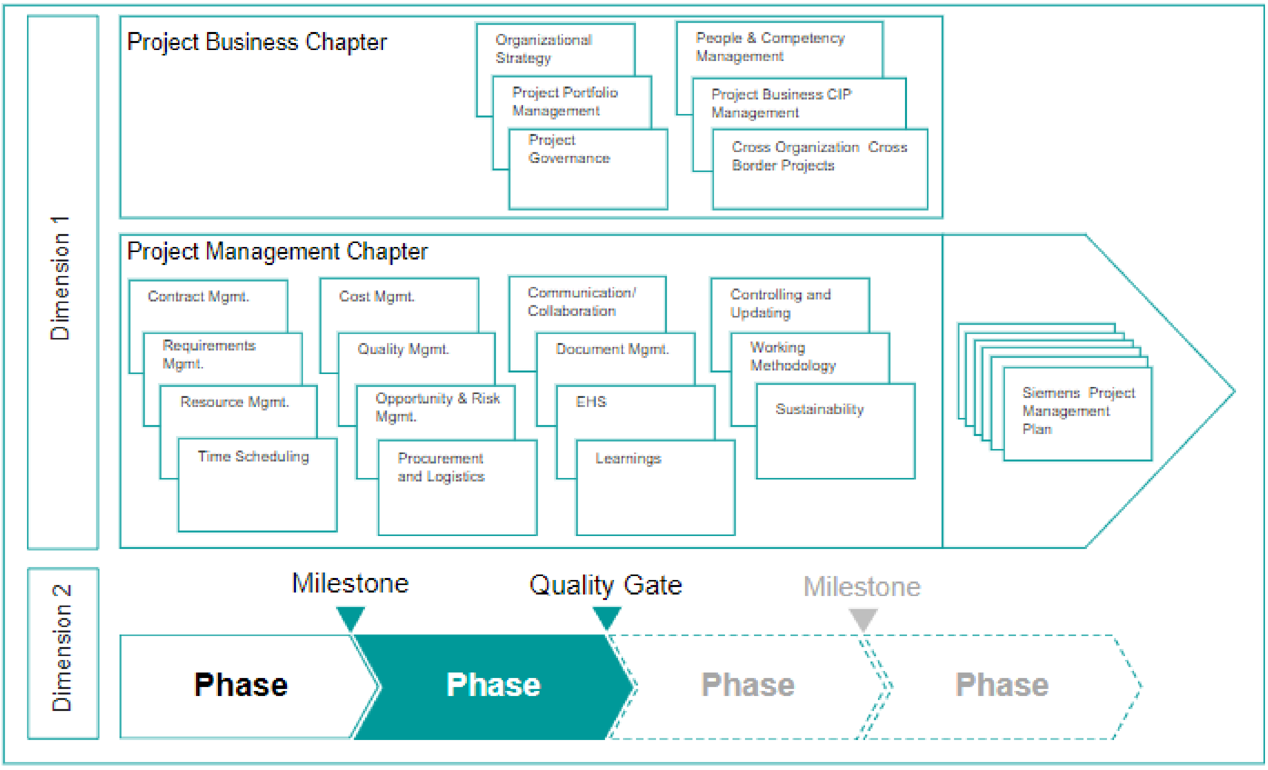

Figure 2: Two-dimensional approach to project excellence

The lifecycle of a project is characterized by a series of phases through which the project passes until the completion of all obligations. Each phase ends with a Milestone that is set as a checkpoint to evaluate the achievement of the defined outputs and to guide the project in a structured and efficient way through its lifecycle. Selected key Milestones are marked as Quality Gates, which are used to verify the achieved maturity of the outputs. The underlying objective of these gates is to proceed only once the requirements have been fulfilled, or clear actions have been defined and assigned to close the gaps.

The Organizational Units, which establish a collaborative working between the key support functions and the RBM and the individual project, fulfill essential preconditions to achieve the defined project benefits.

The Annex 1: Definitions and abbreviations explains and details the requirements described in the PM@Siemens Milestone models.

[3] Source: ISO21500

1.3 Applicability

The PM@Siemens Guide governs the project business at Siemens from an organizational perspective and its implementation within projects. An essential part of the described input, relevant activities, and output is the result of experience-based proven practices within Siemens, but also in adherence and conformity with international standards for project Management.

1.3.1 Definition of a Customer Project

A customer project consists of planned, coordinated, and controlled tasks, with start and end dates, undertaken to achieve unique and defined objectives that result in deliverables conforming to specific contractual customer requirements.

Such a project includes significant manpower effort (internal and subcontracted) directly attributable to the contract but does not include manpower directly included in the cost of a standard product[4]. The Organizational Units can, according to their needs, define a precise threshold for “significant manpower”.

1.3.2 Mandatory Use Cases

This PM@Siemens Guide is mandatory for all Organizational Units and Affiliated Companies of Siemens AG (according to Circular No. 200), irrespective of their specific role in a project as main contractor, subcontractor, partner, member or lead of a consortium or joint venture.

This PM@Siemens Guide is binding for all projects according to the above-mentioned Siemens customer project definition (incl. sub-projects and projects where the customer is another Siemens business with a clear solution character) which are categorized as projects of the categories[5] A, B, C, and S with a volume ≥ 1.0 million EUR.

For category S projects < 1.0 million EUR use of the PM@Siemens guide is recommended as good practice.

For all other Siemens internal projects e.g. Research & Development, IT implementation, CF A Audit, SFS and consultancy the PM@Siemens Guide is recommended to be utilized for guidance.

Please note that as much as the rules of an Organizational Unit deviate from the detailed requirements as defined in the PM@Siemens Milestone models for the project types, such deviations can only be more, not less, rigorous.

The utilization of a Siemens Project Management Plan (SPMP), referenced in this guide, is not mandatory for category S projects.

Project Excellence requires specific certification levels ensuring only suitably qualified Project Managers (PMs) and Commercial Project Managers (CPMs)[6] are assigned to the relevant category of projects.

---

[4] E.g. engineering manpower to design a catalogued product to be sold under standard conditions

[5] PM@Siemens project categorization tool

[6] See Annex 2: Career Model and Development for Project Manager and Commercial Project Manager

1.4 Target Group

The roles and functions listed below are the target group for the PM@Siemens Guide. All of them have an essential stake in contributing to the success of the project business.

- RBM: the PM@Siemens Guide provides the RBM with a general understanding of the principles of project Management and supports them in their leadership role.
- PM: the PM@Siemens Guide provides the PM with a general understanding of the principles and proven practices of project Management and supports their dedicated role and responsibility.
- Project team members (e.g. CPM), Sales Experts, Line Managers, and involved key support functions: the PM@Siemens Guide provides them with a set of common, proven, and optimized project standards and practices as the minimum mandatory requirements to apply to projects.
- Project Management Excellence (PME) function, in some cases called Project Management Office (PMO): the PM@Siemens Guide provides them with a set of corporate core project Management standards and practices that should be used to detail business specifics and processes as a basis for guiding their Organizational Unit's project business.
- Chief Executive Officer (CEO): the PM@Siemens Guide provides them with an understanding of the elements of the project Management framework to ensure its implementation through key roles/functions such as PME and RBM.
- Chief Financial Officer (CFO): the PM@Siemens Guide provides them with an understanding of the elements of the project Management framework to ensure its implementation through commercial key roles/functions.
- PM@Siemens Coordinator (Businesses & Services/Lead Countries): the PM@Siemens Guide supports them to implement the PM@Siemens methodology in the Organizational or Regional Units and to provide guidance with respect to its application.

2 Global Project Business Excellence

The CEO at each of the headquarters of a P&L Unit or Business Unit, which is involved in project business, has the responsibility[7] to ensure an adequate project environment and to ensure key roles are filled. This supports them in implementing the organizational framework and establishing the preconditions for achieving the defined organizational benefits.

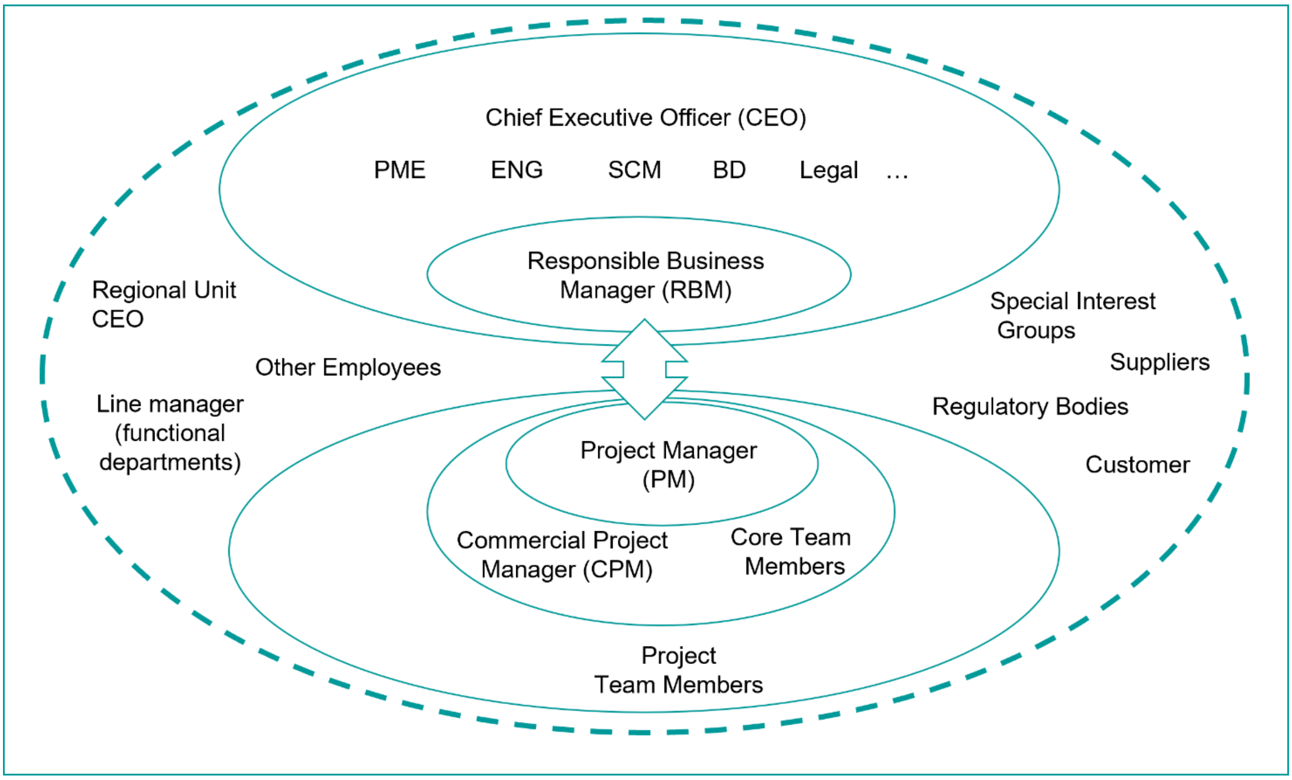

Figure 3: Interaction between organizational and project environment

The CEO needs to ensure that the framework as described in the following chapters is applied through PME, Engineering (ENG), Supply Chain Management (SCM), and other key support functions to enable the RBM to perform their role. (Figure 3)

The role of the RBM ensures the immediate interaction of the Organizational Unit with the project. They:

- secure the fulfillment of the project business's strategic targets as defined by the CEO
- support the CEO in making strategic and operational project portfolio decisions
- establish a framework through which organizational standards for project Management can be implemented and continuously evolved
- drive the assurance of the in-time allocation of necessary competencies in individual projects
- foster continuous improvements in projects and enable a learning culture
- develop PMs
- assure an effective and efficient collaboration and adherence to the principles of cross-organization and cross-border projects

[7] Other responsibilities of the CEO are described in the business mandate, which is part of the Siemens Organization Handbook

2.1 Business Strategy for Project Business

"How can I ensure the contribution of my project teams to the implementation of my business strategy for project business?"

A clear understanding of the value proposition leads to a successful implementation of the business strategy for project business[8] and an early identification of the essential capabilities needed (Figure 4).

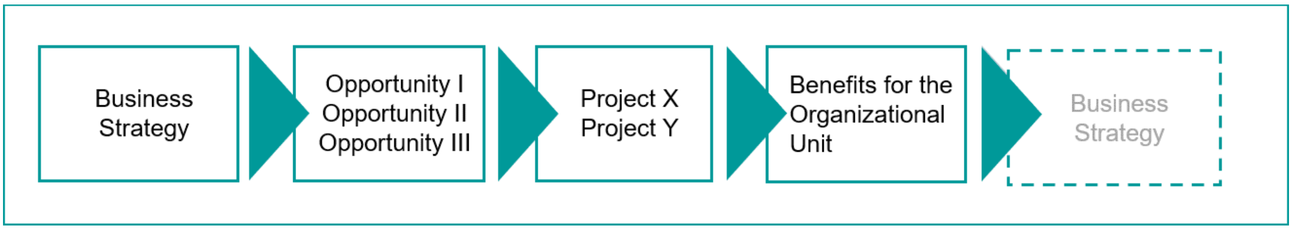

Figure 4: Business strategy implementation[9]

Based on this strategy, the CEO can address several areas of the project portfolio and project types as they:

- manage the project portfolio (Chapter 2.2)
- provide guidance (Chapter 2.3) in strategic action fields (for example, market penetration strategy, claim and negotiation strategies)
- provide input for competency Management (Chapter 2.4)
- evaluate the implementation of the business strategy
- promote the PM@Siemens approach and tailoring it to business needs.

The RBM contributes to excellence in project business and project Management as they:

- communicate the tailored business strategy for project business to project teams
- use the business strategy for project business as the basis for identifying opportunities and executing projects
- utilize information from won and lost bids and execution projects to improve the business strategy for future bids and projects.

A concise, well-understood, and implemented business strategy for the project business is a prerequisite for the organization to outperform the competition.

[8] See Siemens Organization Handbook

[9] Under consideration of ISO21500:2012(E)

2.2 Project Portfolio Management

"How can I implement project portfolio Management in order to promote my business?"

For the project portfolio to contribute to the strategy of the organization, the CEO:

- define the framework for governing project portfolios, including criteria for selection and prioritization
- direct project portfolios in line with the business strategy of the Organizational Unit
- define objectives and targets for the individual portfolios and continuously optimize their elements
- challenge bid proposals and decides which bids to pursue and their strategic fit
- respond to significant portfolio risks
- define a review system based on standardized key indicators.

The RBM contributes to the implementation of the portfolio as they:

- implement the framework for governing project portfolios
- manage the stakeholders in a collaborative way
- monitor and influences the efficient utilization of all assigned resources
- exploit the synergies of projects managed within a portfolio
- identify and respond to opportunities and risks of a portfolio
- contribute to the development of the project portfolio e.g. by using new technical solutions.

Professional project portfolio Management fosters the exploitation of global benefits for the organization.

2.3 Project Business Governance

"How can I establish an organizational environment that supports project excellence and a culture of ownership?"

To establish a basis for excellent project business, the CEO:

- implement a framework for managing and improving organizational standards in alignment with business strategies and market and environmental needs
- establish a uniform structured project data repository that emphasizes the business processes critical to the value chain (for example, SCM)
- ensure that the targets from key support functions such as procurement & logistics and/or engineering support the targets of individual projects
- create awareness of Project Security, Cybersecurity (incl. Product and Solution Security (PSS), Information Security, Cybersecurity risks in the supply chain), Data Privacy (DP), Sustainability- and 'Environment, Social, Governance' (ESG) requirements and their impact on projects
- hold key support functions accountable to providing their deliverables on time, at high quality, and within budget
- drive a culture of honesty, openness, trust, and ownership,
- demand and empower the PMs to act as entrepreneurs.

The RBM equips the PM with a sound basis for his projects as they:

- implement a decision-making framework for managing projects
- implement a monitoring approach to gain transparency of the projects' status
- regularly verify process compliance and driving Quality Gate decisions
- ensure that issues can be escalated to support the PM
- provide the projects with timely access to the relevant expertise and mandatory input
- contribute to make-or-buy decisions for the projects
- support supplier evaluation and Management in the organization
- drive adherence to legal policies and avoid unacceptable terms and conditions
- communicate defined change orders and claim strategies from the organization
- provide defined security practices and access to dedicated experts
- foster awareness and support the PM on Compliance related aspects including Collective Action in project business.

A framework that supports a culture of ownership and trust and ensures minimum standards lays the necessary foundation for proactively minimizing and managing risks and nonconformance costs, creating opportunities, maximizing agility in projects and improving project profitability.

2.4 People and Competency Management in Project Business

"How can I have the right people in the right place at the right time?"

To sustain high competitiveness in the market, the CEO:

- establish a culture of trust and openness
- understand personnel development to be a regular business activity
- regularly derive future project Management competency needs from the business strategy and project portfolio
- drive and implement career paths.

The RBM equips projects with the right competencies as they:

- provide role descriptions and job descriptions for at least the project core team members[10]
- implement Strategic Workforce Planning (SWP), including gap analysis of required and available skills (including soft skills)
- inspire people to continuously develop project Management skills
- offer appropriate training programs and verify their effectiveness
- develop employees' hard skills and human skills
- assure retention of relevant competencies
- ensure the timely availability of the resources and competencies required by the projects
- align the personal targets in the Growth Talks with project targets, at least for the core team
- utilize a performance-related reward system.

Strategic people development and the timely deployment of appropriate staffing prepares the organization and project teams for the challenges of today and tomorrow.

[10] See also Annex 1: Core Team

2.5 Project Business Continuous Improvement Process (CIP)

“How can I strive for excellence and treat deviations as an opportunity to improve?”

To continuously improve the maturity of the Organizational Unit, the CEO:

- explain the need for change using respectful and binding communication
- align improvements with the needs of potential customers and other relevant stakeholders
- stimulate willingness to improve at all levels in a persistent way
- create an environment for managing improvement activities and projects.

The RBM drives continuous improvement as they:

- demand feedback, identify the ultimate root causes of deviations, and assess their impact, involving key functions and all relevant stakeholders
- utilize an established process to optimize workflows, standards, tools, and templates
- implement value adding and risk mitigating improvements
- maintain a push-and-pull exchange of experiences (Lessons Learned) between the organization and individual projects
- regularly evaluate the adequacy of project Management standards (prerequisites to the Siemens Project Management Plan), and update these accordingly
- utilize relevant project business assessments (Maturity in Project Management Assessments) or audits to identify potential gaps and implement derived improvements.

A mindset of continuous improvement fosters a learning organization, increasing performance and customer satisfaction.

2.6 Cross-Organization and Cross-Border Projects

"How can I set up and manage projects across organizations or Regions and also comply with applicable business regulations and foster effective collaboration?"

Respecting the boundary conditions of the involved entities, CEO:

- provide guidance on the collaboration model regarding roles and clear allocation of worldwide responsibilities
- ensure early partnerships are established with relevant organizations and regional units to reach agreement on the work split and the system of collaboration (business model) in a timely manner
- nominate, assign, and empower an overall PM (PM Bid/PM Execution) and World Project Controller (WPC) in the case of cross-organization and cross-border activities (split contract, silent and open consortium).

The RBM provides the organization with the basis for effective cross-organization and/or cross-border collaboration to ensure:

- the implementation of the New Collaboration Model (NCM)
- the adherence to tax regulations, in particular the arm's length principle and Third Party Intermediaries collaboration
- the adherence to Compliance regulations[11]
- the involvement of the delegation center and following the delegation guidelines for international projects (Delegation Overview and Policies Circular)
- the application of the Process Adjustment Tax Advice and Compliance (PATAC) process.

Adherence to cross-organization and/or cross-border boundary conditions is key to sustainable business in the international environment.

[11] See Compliance Handbook

2.7 Summary of Contributions to Projects from the Organizational Environment

The CEO and RBM, supported by the key roles[12], must ensure the timely availability of the following essential contributions to projects. These contributions are derived from the deliverables described in the previous Global Project Business chapters.

The contributions below are clearly allocated to specific Milestones in the PM@Siemens Milestone models in Chapter 5:

- Organizational strategy
- Nomination of a Responsible Business Manager
- Process framework
- Project overview
- Standard work breakdown structure (WBS)
- Role descriptions
- Historical values
- (List of) core competencies
- Reference solutions
- (If applicable) Cross-Organization cooperation agreement
- Official appointment of Sales Expert/Bid Manager
- Expectations on fulfillment of Milestones and Quality Gates and on documentation of related decisions
- Lessons Learned from previous projects
- Project Organization Chart (generic)
- Standard cost breakdown structure (CBS)
- Limits of Authority (LoA) document requirements
- Claim strategy from the organization
- Existing (frame/standard) contracts with suppliers
- Involvement of certified PM from execution phase
- Standard contract terms and conditions and non-disclosure agreements (NDAs)
- For Cross-business projects: Appointed and empowered Lead PM and World Project Controller (WPC)
- Empowerment for the PM and CPM as well as official appointment of other project team members
- Official release of Sales Expert/Bid Manager
- Siemens Project Management Plan Procedures
- Nonconformance Management procedure from organization
- Official release of the PM, CPM, and rest of project team
- Official nomination of the responsible person for warranty/service
- Official release of the person responsible for warranty/service.

[12] See Figure 3

3 Project Management Topics

The PM is the entrepreneur of their project. In the sales phase, they transform customer needs into a feasible solution or concept. Once the project is awarded, they implement the contracted scope, satisfying the customer, and striving for a lasting relationship. A key condition to achieve this within the agreed targets is the empowerment of the PM by the RBM.

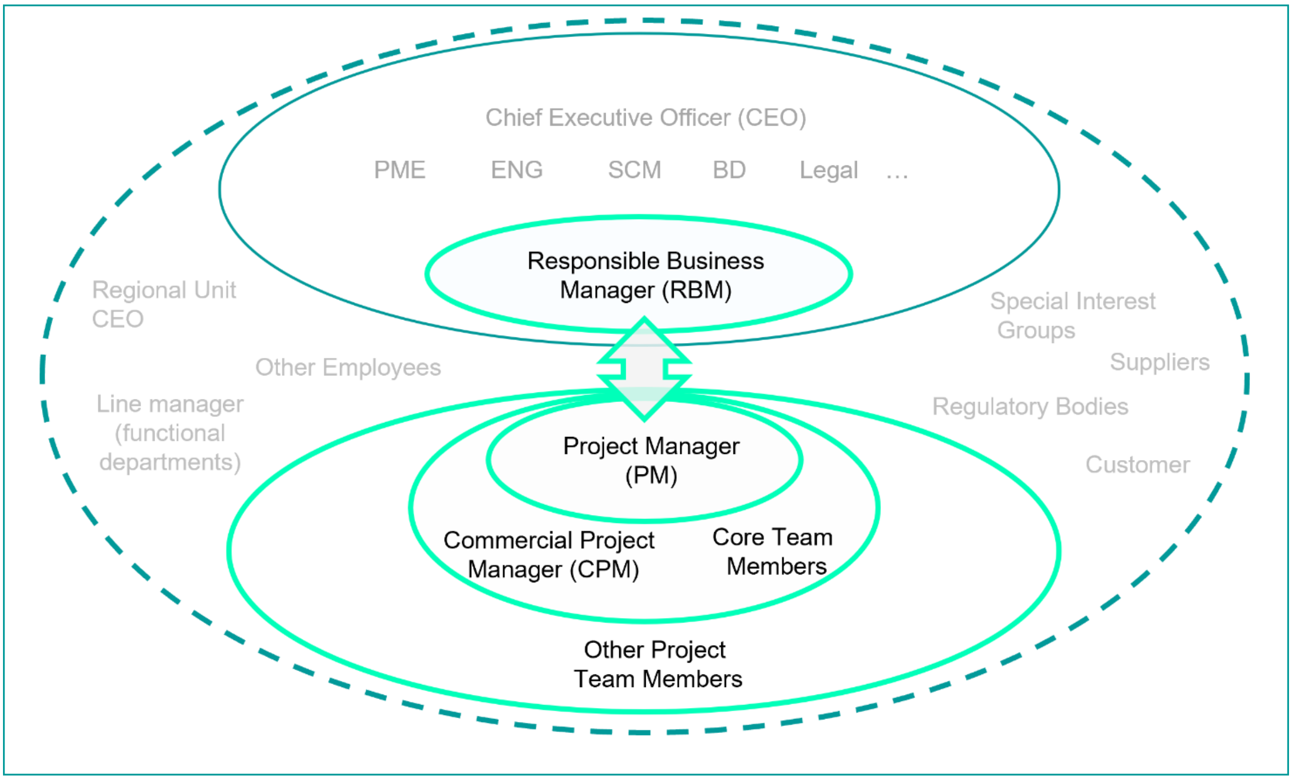

Figure 5: Interaction between RBM and PM

A relationship of trust and close collaboration between the RBM and the PM is essential for the successful fulfillment of the project. (Figure 5)

Utilizing the contributions from the organization (Chapter 2.7), the PM together with the CPM and other members of the team is responsible for ensuring that…

- a well-drafted contract communicates a clear scope for execution (Contract Management)
- stakeholder requirements are captured, communicated and progressed throughout the project lifecycle (Requirements Management and Deliverables)
- the core team and other resources are available in time and of the required quality (Resource Management)
- the status of the project is known at all times and what steps/actions are required for its on-time completion (Time Scheduling)
- a reliable cost calculation exists, facilitating the optimization of the financial results (Cost Management)
- project deliverables are of the highest quality (Quality Management)
- exposure to risks is minimized and opportunities are actively exploited (Opportunity Management and Risk Management)
- procurement and logistics maximize the value contribution to the project (Procurement and Logistics Management)

- stakeholder Management and associated collaboration activities are demonstrated throughout the project (Communication and Collaboration)
- project work result documentation is created, stored, and made accessible on time (Document Management)
- a zero-harm culture is established (Environmental Protection, Health Management and Safety)
- continuous improvement is demonstrated throughout the project (Learning in Projects)
- project results are monitored and controlled (Controlling and Updating)
- that projects are delivered in a sustainable way (Sustainability Management in Projects)
- adequate capabilities, activities, and skills, to properly address cybersecurity as part of the delivered product and the required infrastructure are established (Cybersecurity)
- the minimum standards with respect to the project type are being met (

- Working Methodology).

The above-mentioned project Management topics need to be considered throughout all phases of the project lifecycle and are complemented by detailed requirements from the PM@Siemens Milestone models (Chapter 5).

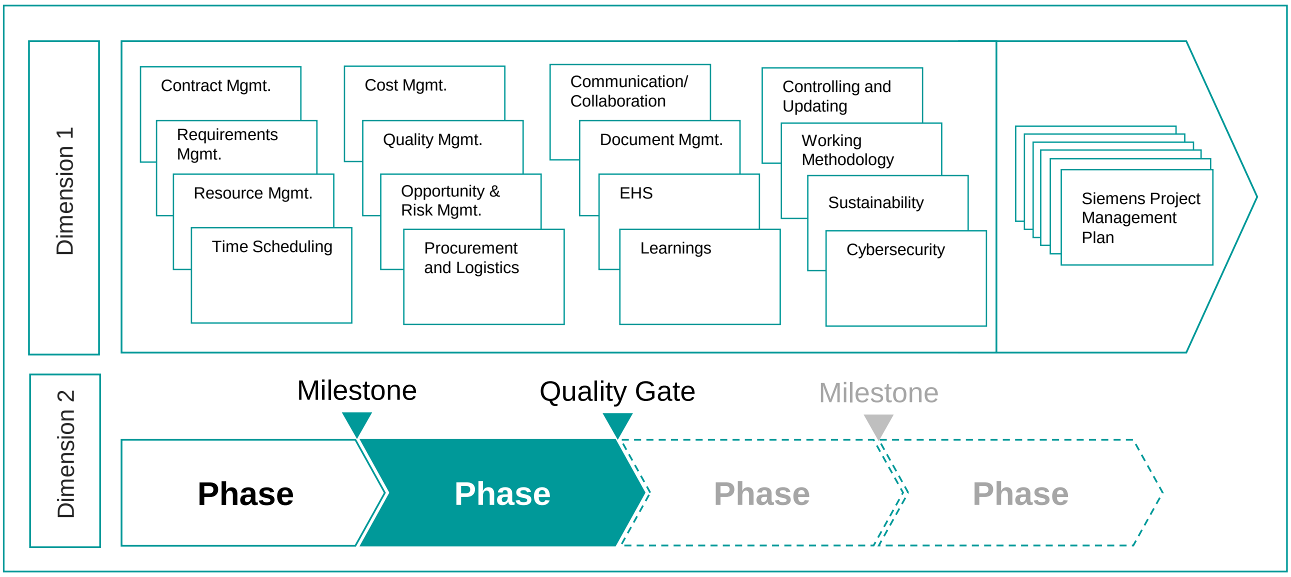

Figure 6: Project Management topics and their relation to PM@Siemens Milestone requirements

3.1 Contract Management

"How can I ensure that the contract includes a clear scope and legal basis upon which the project can be delivered with an acceptable risk tolerance?"

The PM deals with contracts daily to define Siemens' rights and obligations toward customers, partners, and subcontractors as they:

- establish consistent and unambiguous contracts
- involve Contract Management and Legal expertise as well as Compliance, Siemens Financial Services (SFS), insurance, and tax at the early stages of the project
- create awareness of all project team members and enforcing the Siemens position on contractual obligations and rights.

To ensure the contract provides a solid foundation for project execution – that is in the best interests of all parties – the PM:

- form a multidisciplinary bid team, including legal and execution expertise
- comment on all contract documents and considers internal and external regulations
- define negotiation strategies to reach agreements in line with the LoA mandate
- validate the offered scope with all contractual partners prior to contract signature
- conduct a contract analysis workshop (or contract reading) with all core team members to achieve a common understanding of the contractual rights and obligations at project start
- develop and document project change orders and claim strategies
- safeguard adherence to all project contracts
- implement continuous change Management to ensure that all changes to the project baselines are assessed, approved, documented, applied, and tracked
- ensure the identification, analysis, assessment, assertion, and defense of all incoming and outgoing claims to and from the customers, partners, and subcontractors.

A well-negotiated contract utilizing Siemens' expertise – as well as a committed project team actively working with the contract – will lead to customer satisfaction and is a major defense against Non-conformance Costs (NCCs).

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.2 Requirements Management and Deliverables

"How can I manage all stakeholder requirements successfully?"

The PM are responsible for establishing a shared understanding of stakeholder requirements and obtaining their commitment as they:

- identify requirements as a means of addressing contractual deliverables, needs, and expectations
- identify non-functional requirements and achieve a shared understanding
- consider legal and other internal and external boundary conditions
- integrate requirements Management into progress monitoring and controlling
- focus the project efforts on agreed deliverables.

To fulfill the requirements of all project deliverables, the PM:

- identify, analyze, and prioritize requirements
- evaluate, document, and communicate the feasibility of the proposed scope (for example, technical, legal) and examine the interface to product development and suppliers
- negotiate and agree the project scope with the customer, suppliers, and other stakeholders
- use the scope to establish an adequately detailed WBS, time schedule, resource plan, and cost calculation
- address the work split, contributions, obligations, and Milestones and adequately documents these in the requirements Management plan
- break down requirements and documents them in the requirements traceability matrix
- perform an impact and requirements traceability analysis
- verify, validate, and document the fulfillment of requirements end-to-end.

Clearly defined and transparent requirements developed with the stakeholders which take all needs and expectations into account are fundamental to attaining the project goals and achieving acceptance.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.3 Resource Management

"How can I assure the correct placement of the core team roles and availability of other resources at the right time?"

The PM fosters an ownership culture as they:

- assign work packages[13] to project roles or responsible departments
- optimize the utilization of project team members
- lead the project team.

To utilize project resources responsibly and sustainably, the PM:

- obtain commitments and reach agreement on the selection of necessary resources
- produce the resource Management plan (human and other resources)
- establish a project organization with the required roles
- control and manage the necessary resources, adjust if required, and optimize their utilization
- ensure that adequate competencies are always available during the project, in coordination with line Management
- ensure that resources such as special or long-lead items are available on time and provide regular feedback to the project team members as well as their managers
- appraise the project team and continuously support their growth.

A professional project team that shares a common goal will enhance collaboration and improve performance by efficiently utilizing all resources.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

[13] including targets for time, quality, and costs

3.4 Time Scheduling

"How can I ensure an up-to-date status of my project along with the time and task details to complete?"

The PM continuously deal with time scheduling from the early sales phase to the end of the project lifecycle as they:

- support the timely allocation of appropriate resources
- provide transparency on progress and time to complete the project to all relevant stakeholders
- foster proactive and early risk detection.

To successfully complete the entire project scope on time, regardless of its complexity, the PM:

- create and maintain the time schedule, and additionally document it in the Siemens Project Management Plan
- develop and set up a structured, logical, and feasible time schedule that reflects the WBS and consider interdependencies of the different activities and Milestones[14]
- prepare the first project time schedule prior to bid approval, involving implementation experts
- produce an indicative a rough long-term time schedule with detailed controlling of the short-term part
- derive work packages that are clear and achievable within the given project control cycle
- highlight and monitor the critical path for the project team so that they can pay special attention to the most critical activities
- create and archive a customer sign-off baseline after the contract is signed
- align the time schedule with the allocation and utilization of the project resources, including contributing parties
- keep the current time schedule continually up to date and synchronized with ongoing calculations and the project scope.

Conscientious time scheduling, combined with cost and quality Management, is a key driver of project success and of promoting a spirit of schedule adherence.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

[14] PM@Siemens (see 5. Milestone Models) and contractual milestones as minimum requirement

3.5 Cost Management

"How can I establish and maintain a reliable cost calculation and strive for the optimization of the forecasted result?"

The PM are responsible for optimizing the costs, profit, and cash flow, taking the project Management environment into consideration, throughout the entire project lifecycle as they:

- create cost, profit, and cash-flow transparency for the offer, order entry, and forecast calculations
- continuously monitor and control the current cost calculation in line with the time schedule and the project scope
- provide a basis for the fulfillment of the project's cost and profit targets
- foster early detection of cost and cash-flow deviations, followed by the definition and implementation of appropriate measures.

To ensure that the entire project scope, regardless of its complexity, is reflected in the calculation and completed within budget, the CPM supports the PM in ensuring that:

- the offer calculation is reliable and is aligned with the WBS, and that it utilizes the experience of experts from the execution phase
- the order entry calculation is a stringent continuation of the offer calculation
- the order entry calculation is agreed with all involved disciplines (as the baseline for the project)
- a standardized CBS is used throughout the entire project lifecycle
- the risk contingencies adequately reflect potential changes to the project
- adequate methods are applied to guarantee that the costs in the current cost calculation reflect the actual work performed and a reliable cost estimate to completion
- all commercial aspects are considered (e.g. cash flow, financing, bank guarantees, insurance, and customs and taxes)
- the Financial Reporting Guidelines (FRGs) and Implementation Guidelines (IGs) are consistently applied.

Disciplined cost Management, in conjunction with the Management of time and quality, is an essential prerequisite to proactively securing the project's financial targets.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.6 Quality Management

"How can I ensure that the project deliverables are appropriate for their contracted purpose?"

The PM proactively strives to secure a flawless acceptance of the project deliverables by the customer. This may include involving professionals who challenge the project and should be realized within a culture of ownership, where quality is based on the contribution of all project team members.

Key elements of quality Management in projects are:

- planning and defining customer- and business-oriented quality targets as well as appropriate measures to fulfill them, and documenting them in the quality plan
- efficient implementation of the relevant processes tailored to business needs
- a Quality Manager in Projects (QMiP) who continuously challenges conformity with the quality goals and to contribute to Milestone achievement.

To achieve the defined quality goals, the PM:

- establish a quality plan and describe the targets and means necessary to meet the plan in line with the quality requirements
- create transparency through planning and performing PM@Siemens Milestones and Quality Gates
- plan and conduct appropriate reviews for key contributions (for example, design reviews, tests, customer documentation)
- produce a quality report on and monitor defined measures
- apply quality methods with the customer, authorities, suppliers, and internal contributors to prevent Non-conformities.

Project quality Management is an essential prerequisite to achieving overall customer satisfaction and minimizing the cost of non-conformance.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.7 Opportunity Management and Risk Management

"How can I best exploit project opportunities and minimize exposure to risks?"

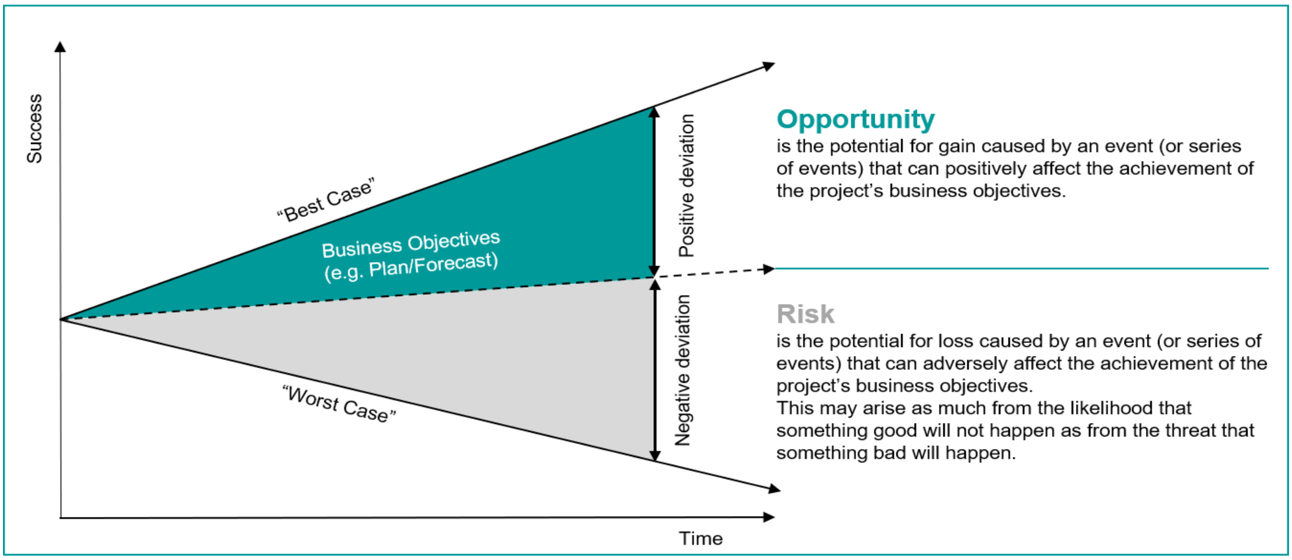

Figure 7: Opportunity and Risk[15]

3.7.1 Opportunity Management and Risk Management – General

The PMs must start Opportunity and Risk (O&R) Management early in the sales phase and have to transparently continue with such throughout the entire project lifecycle as they:

- assume the role of an opportunity manager and a risk manager or delegate the roles to dedicated team member(s)
- identify and define potential opportunities and risks
- apply response strategies and ensure ownership of opportunities and risks
- provide transparency and foster psychological safety [16] to drive exchange of information and experience within the project team, involving all relevant stakeholders (within and outside of the project team).

Figure 8: Siemens O&R process[17]

---

[15] Source: (Link) (page 215)

[16] TOC - Psychological Safety for Leaders | My Learning World (siemens.com)

[17] In accordance with ERM process/guide (page 213)

3.7.2 Opportunity Management

The PMs apply the following Opportunity Management process, supported by the opportunity register:

1. **Identify**: detecting opportunities as early as possible (starting in sales phase and continuing throughout the project life cycle) from all potential sources, utilizing past experiences, expert knowledge, and by conducting regular opportunity workshops with respective stakeholders. This includes a proper opportunity description and assignment of an opportunity owner.
2. **Assess**: performing a detailed analysis (qualitative and quantitative) and evaluating the financial impact, reflected in the current cost calculation in accordance with the Financial Reporting Guidelines (FRGs), Implementation Guidelines (IGs), and the Compliance Handbook (chapter 5.3 Compliance in Project Business). In addition to the evaluation of the opportunities, they must be prioritized.
3. **Respond**: defining a response strategy for each opportunity (e.g., pursue, exploit, share, enhance, ignore), and breaking them down into concise, tangible actions, which are then assigned to individuals, considering fallback strategies and managing opportunities.
4. **Monitor**: regularly reviewing, in line with the reporting frequency, the status of opportunities as well as response actions, including the gross-profit impact.
5. **Report (& escalate)**: reporting the status as part of project performance reviews, if necessary, escalating where Management involvement is needed.
6. **Sustain and continuously improve**: following up on the implementation of opportunities and identification of improvements for the Opportunity Management process within the project team. Providing feedback to the business and other (also future) projects (“lessons learned”). Considering and utilizing interfaces to Risk Management, and respective Sales function, Contract Management (Chapter 3.1), Enterprise Risk Management (ERM), and continuous improvement (Chapter 2.5 and Chapter 3.12).

Opportunities shall not only be derived from cost decreases (incl. non-utilized risk contingencies), but also from areas such as Contract Management during sales and execution phase (incl. Change Orders and Claims), value engineering, productivity increases, supplier competition, automation, additional business (e.g., service contract, upselling), occasionally NCC avoidance and many more.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.7.3 Risk Management[18]

Risks shall not only be derived from technical topics, but also from areas such as financial, legal, Compliance, Cybersecurity incl. PSS and Information Security and Cybersecurity risks in the supply chain, Generative AI Risks[19], Data Privacy, ESG, EHS, Project Security and many more.

The PM apply the following risk Management process, supported by the risk register:

1. **Identify**: detecting risks as early as possible (starting already in sales phase and continuing throughout the project life cycle) and from all potential sources, utilizing past experiences, expert knowledge, and by conducting regular risk workshops with respective stakeholders. This includes a proper risk description with a cause-effect-chain and assignment of a risk owner.
2. **Assess**: performing a detailed analysis (qualitative and quantitative) and evaluating the financial impact, reflected in risk contingencies in accordance with the Financial Reporting Guidelines (FRGs), Implementation Guidelines (IGs), and the Compliance Handbook, chapter 5.3 - Compliance in Project Business.

[18] In accordance with ISO 31000

[19] In accordance with the Generative AI Risk Management (GARM) process

3. **Respond**: defining a response strategy for each risk (avoid, reduce, transfer, retain & “watch”), and breaking them down into concise, tangible response actions, which are then assigned to individuals, considering fallback strategies and managing residual risks.
4. **Monitor**: regularly reviewing, in line with the reporting frequency, the status of risks as well as response actions, including the cost impact.
5. **Report (& escalate):** reporting the status as part of project performance reviews, if necessary, escalating. Additional risks that cannot be managed on project level should be addressed in Enterprise Risk Management (ERM) according the ERM methodology.
6. **Sustain and continuously improve**: following up on the mitigation of the risk and identification of improvements for the risk Management process within the project team. Furthermore, consideration for and feedback into the business and other, also future, projects (“lessons learned”), considering and utilizing interfaces to Opportunity Management, and: Contract Management (Chapter 3.1), Enterprise Risk Management (ERM), Risk and Internal Control (RIC), New Collaboration Model (NCM), Non-conformance Cost (NCC) Management with root cause analyses, and continuous improvement (Chapter 2.5 and Chapter 3.12).

Professional Risk Management that addresses the above-mentioned topics is essential to the fulfillment and improvement of the project’s goals and consequently of the business targets.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.8 Procurement and Logistics Management

"How can I decide on the right sourcing and maximize the value contribution to the project?"

The PM utilize procurement expertise to determine the best option to fulfill the specific project requirements. They initiate procurement support starting in the early sales phase and lasting through to the end of the project.

The following activities provide the basis for positive value contribution:

- analyzing customer requirements proactively considering make-or-buy strategies and total cost of ownership (TOCO)
- identifying and selecting the most appropriate suppliers, developing a reliable logistics approach.

To achieve the targeted value contribution, the PM:

- derive from the requirements baseline and, define appropriate work packages, products, materials, or services to be (out-) sourced
- identify critical procurement-related requirements and focus on managing them
- develop a supply chain/logistics concept for obtaining and transporting the necessary materials
- push to obtain binding offers for key items from suppliers prior to bid approval, considering appropriate periods of commitment in line with the project schedule
- avoid single suppliers, if possible; otherwise, mitigate inherent risks must be mitigated
- strive for an adequate risk sharing with third parties
- obtain regular detailed progress reports from suppliers.

Excellent project procurement and logistics has a fundamental impact on project success. It is therefore imperative to carefully judge the capabilities of the suppliers and establish a target-oriented collaboration[20].

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

[20] Supply Chain Management Procurement Compendium

3.9 Communication and Collaboration

"How can I establish relationships and maximize the positive contribution of my stakeholders for the benefit of the project?"

The PM work with their team to deal with internal and external stakeholders daily as they:

- establish a culture of trust, living an open communication and collaboration
- proactively identify the project stakeholders and their needs and expectations
- constantly interact with the stakeholders in line with the defined strategies and measures
- prepare and maintain the communication and collaboration Management plan, focusing on communication and stakeholder Management.

To allow collaboration within the project to flourish, the PM:

- collect and document information about the main project stakeholders
- analyze the interests and attitudes of the individual stakeholders and potential impacts of their behavior
- define and implement adequate communication content, methods, and frequency
- manage the stakeholders by applying defined measures
- fulfill the information needs of the stakeholders involved.

Collaboration Management with stakeholders fosters their consent and strengthens their contributions to fulfill project targets.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.10 Document Management and Configuration Management

3.10.1 Document Management

"How can I manage the project documents in a manner that supports the success of the project?"

The PM ensure professional document Management throughout the entire project lifecycle as they:

- make available up-to-date, standardized, and consistent documentation of project results and supporting documents
- achieve appropriate quality of the documents, considering the needs of the stakeholders
- provide central and straightforward accessibility for the relevant stakeholders.

To support collaboration and foster transparency through project documentation, the PM:

- use a uniform structured project data repository for hard copies and electronic files, aligned with the WBS
- define and implement document creation, content, filing, and transfer requirements, aligned with the stakeholders
- plan and tracks the progress of internal and external project documentation, including status, versioning, and release
- control, correlate, and maintain documentation, specifications and physical attributes of the customer deliverables
- review the documentation prior to its release
- ensure that the exchange of relevant documents with stakeholders is recorded and traceable
- retrieve binding documents in a timely manner
- document requirements for physical security, Cybersecurity incl. PSS and Information Security and Cybersecurity risks in the supply chain as well as Data Privacy, legal, export control, customs, by involving respective experts and archiving.

Fit-for-purpose project documents and documentation is essential to the successful fulfillment of project requirements and targets.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.10.2 Configuration Management

“How can I ensure identification and traceability of my project deliverables (configuration items: Hardware, Software, Documents)?”

The PM are responsible that all project specific configuration items can be identified and traced at any time.

To ensure that changes of configuration items can be identified, assessed and monitored at any time, the PM:

- defines project specific related responsibilities
- makes sure a project Configuration Management Documentation is available, incl. e.g. rules for identification of configuration items and their structure, definition of outputs (in alignment with their organization and customer requirements).

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.11 Environmental Protection, Health Management and Safety

"How can I ensure zero harm to my team and partners?"

The PM are responsible for complying with environmental protection, health Management, and safety (EHS) procedures, and they strive for:

- zero harm for our customers, employees, and contractors
- zero occupational illnesses
- zero incidents impacting the environment
- compliance with the applicable EHS requirements, for example, internal standards and local laws.

To adequately ensure the fulfillment of EHS requirements, the PM:

- act as a role model regarding EHS and actively create adequate EHS awareness throughout the entire project lifecycle
- involve an Environment, Health and Safety Manager in Projects to support adherence to EHS requirements
- prepare an EHS Management plan during the sales phase, and describes how EHS will be delivered during the execution phase (EHS plan)
- involve appropriate professionals to support the setup and implementation of the EHS plan.

Proactive, well-coordinated, and consistent Management of EHS is a foundation of our corporate social responsibility.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.12 Learning in Projects

"How can I foster learning throughout the project and support the organization in continuously improving?"

The PM support the exchange of experiences within the project team and organization, fostering a culture of continuous improvement as they:

- utilize Lessons Learned consciously
- regularly extract Lessons Learned from the project and provide these to the organization
- support rapid learning by promoting trustful communication of failures, inside and outside of the project team.

To facilitate a continuous exchange of experience throughout the project lifecycle between the project stakeholders and their interfaces, the PM:

- collect and apply Lessons Learned through collaborative means
- implement routines to improve collaboration within the project team, e.g. Retrospective
- provide Lessons Learned to the organization as part of, for example, kick-off meetings, Project Acceleration by Coaching and Teamwork (PACT(light)) workshops, win/loss analyses
- conduct project closure – involving customers and suppliers as well as other stakeholders – and concentrate on the collection and aggregation of Lessons Learned.

The systematic utilization of experiences, as well as providing input from the project, contributes to continuous improvement and fosters a learning organization.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.13 Controlling and Updating

"How can I monitor and control the project on a continuous basis?"

The PM are responsible for monitoring the project performance against the project plans and taking corrective actions to proactively strive towards delivering the scope while optimizing the project results within a dynamic environment. This requires balancing time, cost, and quality, and appropriately managing risks and opportunities, starting from the early sales phase (Figure 8). The PM will be supported by the CPM and his core team in performing these responsibilities.

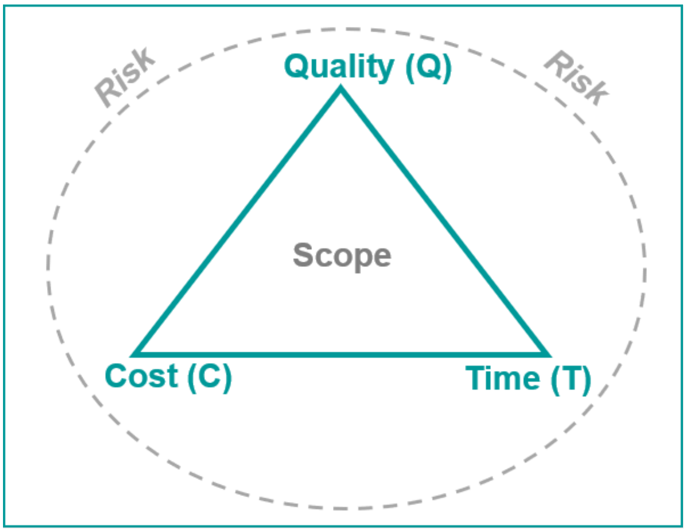

Figure 8: Realizing the scope under the constraints of time (T), cost (C), and quality (Q)

Key elements of controlling and monitoring are:

- set of parameters (KPIs) to ensure transparency about project's status/progress, incl.:
  - order entry, as well as plan and concurrent expectation for revenue, costs, and gross profit
  - deliverables as per scope of work (SoW)
  - change Management (incl. change orders and claims)
  - Non-conformance Costs (NCC) significant risks and opportunities
  - cash-, asset-, guarantee-, currency Management
  - tax concept (incl. NCM and transfer pricing)
  - project financing (e.g., Letter of Credit (L/C), Export Credit Agency (ECA))
  - accounting method (Percentage of Completion vs Completed Performance Method)
  - current cost calculation according to FRG incl. EAC (Estimate at completion) and ETC (Estimate to completion)
  - Division of Responsibilities (DoR)
  - quality indicators, EHS indicators
  - Milestone Trend Analysis (MTA), if applicable
- controlling, monitoring, and updating information in individual plans, documented in the Siemens Project Management Plan
- continuously managing changes to the baselines, including documentation and approval if necessary
- applying the defined controlling requirements including content, frequency, parameters, and KPIs
- provide at least a quarterly report to fulfill also external reporting requirements according to FRG.

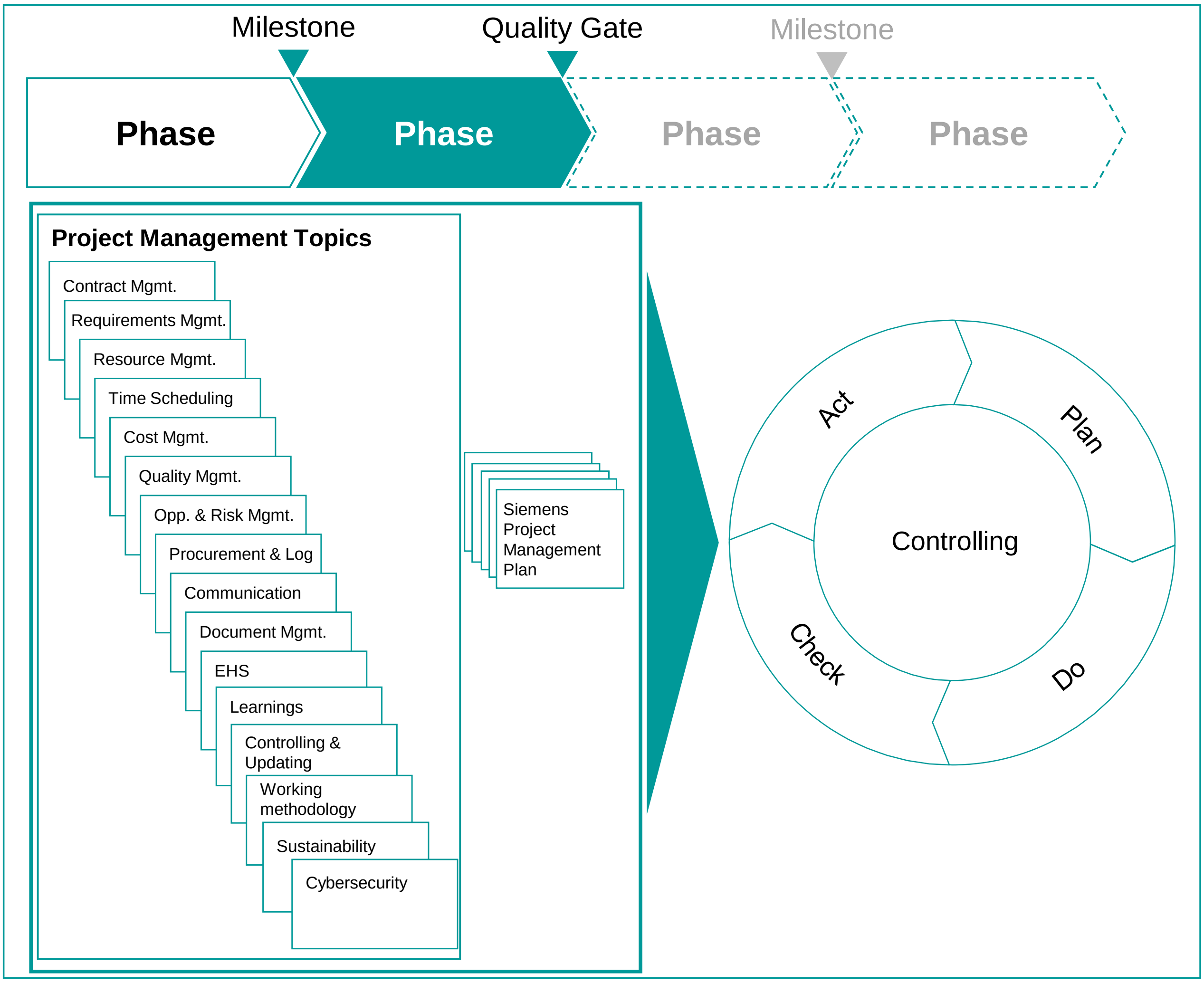

Figure 9: Project Management update cycle

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.14 Sustainability Management in Projects

"How can I ensure that projects are delivered in a sustainable way?"

Sustainable Project Management is the planning, monitoring and controlling of project delivery and support processes, with consideration of the social, environmental and economic aspects throughout the entire project life cycle.

Key elements of sustainable Project Management in projects are:

- optimizing the project's resources
- realizing benefits for stakeholders
- reducing negative impacts for affected people (e.g. communities) and environment
- delivering results in a transparent, fair and ethical way that includes proactive stakeholder participation and project planning.

To achieve sustainability in projects, the PMs:

- include environmental, social, human, and economic aspects in the project targets
- consider contractual and legal obligations like stricter environment and social regulations for the project execution (e.g. relevant regulations and laws or customer requirements) in the project planning activities
- ensure that sustainability areas of concern are part of the Risk and Opportunity Management processes throughout a project's life cycle
- execute the project with a view on environmental, social and economic aspects, i.e. consider the impact and foster digital technologies e.g. to reduce waste and prevent pollution (e.g. by automation and digitalization incl. augmented reality). This may also include corresponding collaboration with suppliers, partners and implementation at customer side.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

3.15 Cybersecurity in Project Management

"How can I identify and minimize risk by ensuring an adequate level of cybersecurity in my project?"

The PM are responsible to properly address cybersecurity and software-related risk as part of the project (including a technical solution and its individual components), related infrastructure, data, and information within the remit of their project.

This includes:

- understanding and managing of cybersecurity risks for both the customer (e.g., delivered product, data) and Siemens (e.g., operated infrastructure, data)
- planning of required project specific security activities, adaption to project scope, and planning of budget and time implications
- planning of required skills in the project team, including required project-specific security related training of project members
- vulnerability monitoring within the project scope
- incident handling in the project
- planning a secure infrastructure (if required in the project).

To meet the requirements of cybersecurity within projects the PM ensures that the following steps are taken care of, as a minimum:

- assess the cybersecurity risks (IT, OT and PSS) and customer security and contractual requirements
- clarify (technically/ legally) the responsibility for cybersecurity and data ownership (which elements or activities are within Siemens area of responsibility, which are within the customer's or partners' area of responsibility)
- assess opportunities to increase the overall system security by additional cybersecurity packages and services to the customer.

Based on the identified risk:

- involve cybersecurity expertise – as needed - to support the assessment and planning of required PSS activities and project deliverables. Create a cybersecurity concept, which comprises the following:
    - cybersecurity customer requirements, internal requirements (e.g., SFeRA), and applicable legal/regulatory requirements and how they are addressed in the project
    - the project-specific PSS concept detailing all necessary security activities for the project. See PSS for further guidance and PSSKB for shared best practices
    - the protection concept for project-specific Siemens-operated infrastructure and data protection and whether this meets the relevant security requirements in alignment with the Siemens Asset Classification & Protection (ACP) process and SFeRA
- update and implement the cybersecurity concept throughout the project. This includes to check whether cybersecurity requirements are met by 3rd party vendors (e.g. used components, applications, platforms, services) or subcontractors, which are added to the project during implementation.

3.16 Working Methodology

"How can I identify the best methodology to execute my project?"

The PM must have a clear understanding of which project Management methodology is ideally suited for delivering the project.

Traditional Project Management (e.g., sequential execution) and agile methodologies are mainly discussed as alternative options. For customer projects in many cases a mixed (hybrid) approach with overarching sequential planning in combination with some integrated agile elements are more beneficial, as it combines the planning reliability of a sequential approach with the flexibility of an agile approach to adapt the scope.

PM@Siemens continues to rely on an overarching sequential process for delivering customer projects.

To decide on the correct approach, the PM must have a clear picture of the

- the complexity, and clarity of the project scope
- size, duration and type of project
- customer expectations and contract requirements
- available team resources and competencies.

Depending on the project analysis, the Stacy matrix can provide a first estimation about which project Management methodology is most suitable.

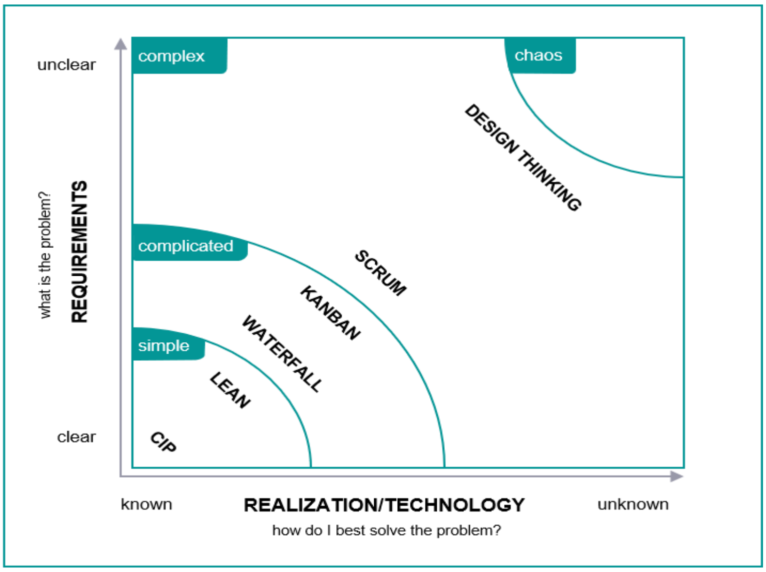

Figure 10: Stacy matrix[21]

Making the correct decision about which working methodologies are to be utilized in a project and involving the right experts will improve collaboration and project delivery.

Within Annex 1: Definitions and abbreviations you can find further descriptions and information about the work products.

[21] In accordance with the graphic of blu professionals GmbH

4 Siemens Project Management Plan (SPMP)

The procedures in the SPMP will determine the project Management prerequisites for each project.

Based on standards from the organization, complemented by project specifics, the SPMP supports the PM in efficiently collaborating with their organization and transparently communicating the expected contributions to their team and other stakeholders. It is a continuously evolving collection of project plans and a core element of every project. Each element of the SPMP is mentioned once (in the initial phase of its development). After that, the continuous updates to these elements appear only as activity under "update SPMP".

The SPMP should be initiated in the early sales phase and regularly updated with key data during the project lifecycle. Irrespective of the controlling of the project results, the content and progress of the project Management topics (described in the SPMP) need to be continuously updated. The right side of the SPMP (Figure 11) shows the deliverables which need to be included/referenced.

The format of the SPMP can vary, however it always needs to be up to date, clear and be produced to a high quality. The SPMP is mandatory for all projects in the categories A, B, C[22].

[22] Project categorization tool

	Key project deliverables
Project overview and bid deliverables	Project Overview
	Bid Strategy
	Bid Work Breakdown Structure
	Bid Time Schedule
	Bid Project Description
Contract Management	Change Log/Register
	Claim Log/Register
Requirement and Deliverable Management	Scope of work
	Project Work Breakdown Structure
	Roles and Responsibilities
	Requirement Collection (including breakdown and structure)
	Requirement Management
	(post-implementation service projects) Service Concept
Resource Management	Project Organizational Chart
	Resource Plan
	Project Target Agreement
Time Scheduling	Project Time Schedule
	Deployment Plan (incl. data migration if necessary)
Cost Management	Offer Calculation
	Order Entry Calculation
	Current Cost Calculation
Quality Management	Nonconformance Log
	Test Plan
	Design Freeze Agreement
	Quality Plan
	Milestone/Quality Gate Documentation
Risk and Opportunity Management	Risk and Opportunity Register
Procurement and Logistics Management	Supply Chain/Logistics Concept
	List of Suppliers
	Procurement Plan
Communication and Collaboration Management	Stakeholder Register
	Communication Matrix
Document Management/ Configuration Management	Document mgmt. including Plan and List (internal and external)
	Configuration Management Documentation (plan and tracking)
Project Security	Project Security Concept
Environment Health and Safety Management	Environment Protection, Health Management and Safety plan
	Site Procedures
Cybersecurity including PSS, Information Security	PSS Concept
	Protection Concept
	Concept for other Cybersecurity risks (e.g.IT/OT, supply chain)
Data Privacy	Data Privacy Plan (if required)
Lessons Learned	-

Figure 11: SPMP with procedures[23] and key project deliverables

[23] Procedures = organizational standards complemented by project specifics

5 Business and Project Types

Siemens has defined four Business Types: Product-, System-, Solution and Service Business.

Business type	Definition along management rationale	Comments
Product business	• Standardized products or components incl. standard software • Typically without customer specific adjustments/integration value add • Success factors: Scale, product cost and complexity, design to cost	• Synergies tend to be higher in **portfolio dimension**
System business	• Order, customer or industry specific combination of standard products or software components (may include internally sourced products) • Typically includes customer specific engineering/integration value add • Success factors: Platform design, modular architecture, IT competency	
Solution business	• Tailor made orders, customer or industry specific hardware or software deliveries (may include internally sourced products/systems) • Typically includes project mgmt., design, installation, functional warranty • Success factors: Project management and integration competency, experience, customer and process understanding, partner network	• Synergies tend to be higher in **customer dimension** • Key customer segments/industries to be defined as part of Business Mandate
Service business	• Activities to change or maintain the condition of a product or process • Encompasses product related services and value added services • Success factors: Integration competency, experience, customer and process understanding, partner network, local presence	

Project business/construction contracts

Project management

Other relevant terminology not affected by this definition

Figure 12: Siemens' business types

All of these business types include activities, which are organized as projects. However, the characteristics and challenges of these projects vary, depending on business type and complexity. Therefore, we need to have a closer look, at which project types are relevant for Customer Projects and PM@Siemens (having in mind there is not always a 1:1 correlation between the business and project types):

- **Solution Projects** (including System Projects)
  This type is characterized by the provision of a customer contract-based or industry-specific combination of products and systems (incl. hardware and/or software) that typically comprises – but is not limited to – the operations project Management, design, engineering, purchasing, constructing, commissioning, installation, and integration.
  Solution Projects are divided in two types:
  - Solution Projects (conventional): Main scope of project focus is on physical equipment, machinery, infrastructure and other "hardware" components
  - Software-driven Solution Projects: Main scope of project is Software.
- **Service Projects**
  This type is characterized by the provision of customer contract-based services to change or maintain the condition of a product, software or process (including product/software related support services and professional services). The ongoing services should be based on Service Level Agreements (SLA). Typical service project scenarios and service project variations can be considered. For more information on service business have a look at Service Business Excellence (including Service Guide)
- **Small Projects**
  These projects have the same attributes as the Solution or Service projects described above; however, they can be characterized by less complexity and therefore fewer risks and lower risk profiles. Accordingly, Small Projects are subject to a simplified PM process with a reduced set of Milestones to balance the risk with a higher value contribution.

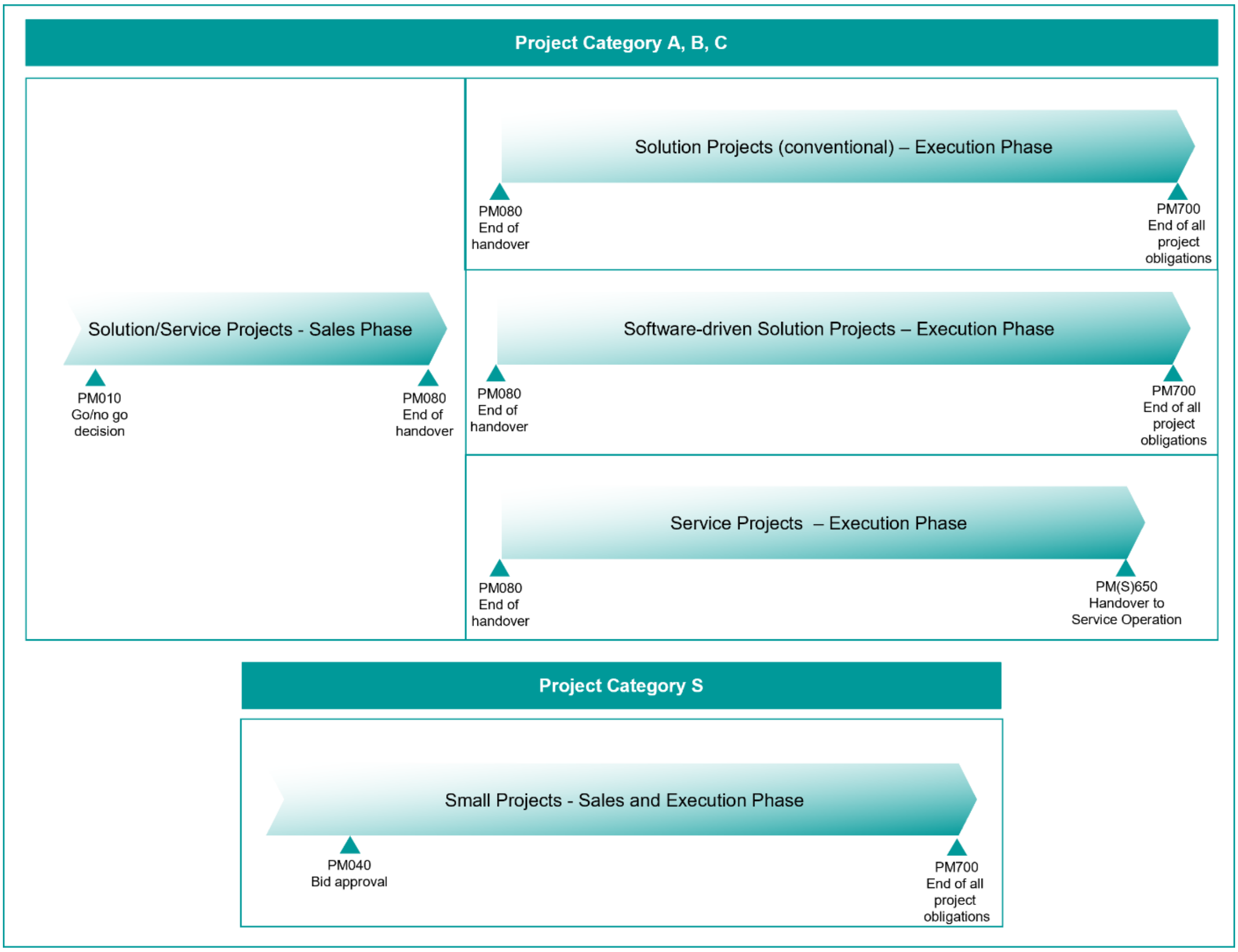

Figure 13: Specific PM@Siemens Milestone Models for defined business types

The figure above is an overview of the three project types and respective sub-types. It is the basis for the more detailed descriptions of the milestone-based activity (input/output) lists.

5.1 How to read/understand the Milestone processes

Milestones: a Milestone is a checkpoint to evaluate the achievement of defined outputs. It is the end of a specific phase, wherein all listed tasks must be completed. As soon as all the required outputs are to hand and accepted at the Milestone check the respective phase can be completed, the Milestone is reached, and the next phase begins.

Quality Gates (QG): some of the Milestones are defined as Quality Gates to verify the maturity of the outputs. The underlying objective of these gates is to proceed only once the requirements have been fulfilled or clear actions have been defined and assigned to close the gaps.

The following input/activity/output lists describe mandatory elements of the PM@Siemens Milestones/Quality Gates.

Mandatory input: mandatory inputs must be provided by the CEO and/or RBM. In addition to the inputs mentioned in the Milestones, the process framework and historical values must be provided over the entire project cycle.

The responsibility for the phase “Lead Management” lies with the line organization. There are no mandatory activities and outputs defined for the PM. The defined activities within the individual project phase (starting with the “Opportunity Development Phase” after PM010) guide the PM to achieve the mandatory outputs.

Mandatory output: the required outputs are minimum standards. Additional Milestones, activities and required outputs can be added to the respective Milestone descriptions according to business specific needs. However, nothing can be omitted.

Outputs are only described if they are not clear evident from the activity – unless the output is part of the SPMP (these are always included).

Mandatory supporting specifications/methodologies/tools: If no link is displayed, the * indicates that business specific regulations must be observed.

5.2 Solution-, Software-driven Solution- and Service Projects: Sales Phase

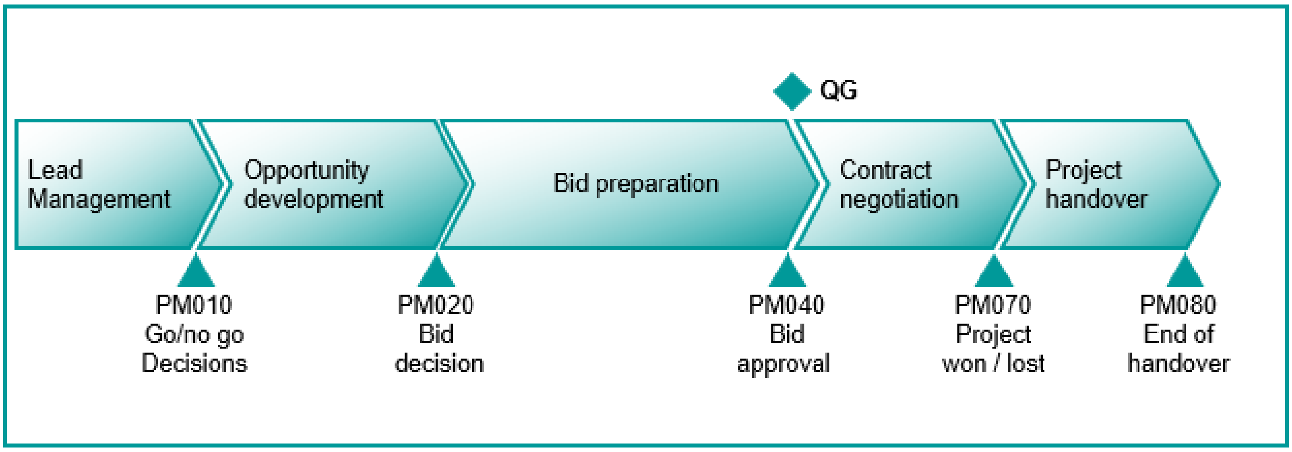

Figure 14: Sales Phase for Solution, Software-driven Solution and Service Projects – Categories A, B, and C

The responsibility for "Lead Management" (until PM010) lies with the line organization. There are no mandatory activities and outputs defined.

The defined activities start with the "Opportunity Development Phase" (after PM010).

5.2.1.1 PM020 – Opportunity development

Opportunity development phase		
Enable a bid decision by challenging a defined bid strategy, including benefits, risks, competencies		**Milestone PM020: Bid decision**
Mandatory input	Relevant activities	Mandatory output
• Organizational Strategy		
• Nomination of the Responsible Business Manager		
• Official appointment of Bid Manager	• Hand over from Sales representative to Bid Manager	
	• Document, evaluate, substantiate lead	• Requirement Collection (including break down and structure) (belongs to SPMP)
• Standard WBS	• Evaluate customer's business situation & competitive environment	• Project Overview (belongs to SPMP)
		• Project WBS (belongs to SPMP)
		• Bid WBS (belongs to SPMP)
• Customer Tender or Request for Quotation (RfQ)	• Evaluate ESG topics relevant for bid decision[24]	
	• Check feasibility, e.g., legal, commercial, technical, timely delivery	

[24] Only applicable for Siemens external customer projects

Opportunity development phase		
Enable a bid decision by challenging a defined bid strategy, including benefits, risks, competencies		**Milestone PM020: Bid decision**
Mandatory input	Relevant activities	Mandatory output
	• Check lead for cross-organizational business potential	• (If applicable) Cross-Organization Cooperation Agreement: project specific
	• Draft business model/business set up	
• Role Descriptions	• Define required human and other resources for bid project and confirm their availability with support of Responsible Business Manager (RBM) – including Project Procurement Manager (PPM) and Quality Manager in Projects (QMiP)	• Bid Time Schedule (belongs to SPMP)
	• Identify required human and other resources for execution project and check their availability	• Resource Plan (belongs to SPMP)
	• Evaluate availability of equipment for long lead items	
	• Consider commercial aspects incl. e.g.: invoicing, cash flow, financing, bank guarantees, INCOTERMS	
	• Provide information for possible contract setups and request tax advice (particularly for international projects in accordance with circular 202)	• Draft of Contract Structure (e.g. onshore-offshore-split)
	• Perform preliminary risk and opportunity analysis	• R&O register (belongs to SPMP)
	• Evaluate and derive the market price and define respective target costs	•
	• Analyze and document stakeholders (incl. conflicting interests)	• Stakeholder Register (belongs to SPMP)
		• Bid Strategy (belongs to SPMP)
	• Finalize and document Milestone "bid decision"	• Milestone Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Customer Relationship Management (CRM) tool*
- New Collaboration Model (NCM) (Link) (for international projects)
- Circular 222: Application for and approval of capital investments and divestment (Link)
- Project categorization tool (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- International Delegations and Global Delegation Centers (Link)
- Permanent Establishment (PE) (Link)
- ESG Radar (Link)
- Project Security (CE SEC) (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)) Data Privacy (Link)

5.2.1.2 PM040 – Bid preparation

Bid preparation phase		
Submission of a successful bid, with approved risk profile		**Quality Gate PM040: Bid approval**
Mandatory input	Relevant activities	Mandatory output
• Claim Strategy from the organization • Standard Cost Breakdown Structure (CBS)		
	• Set up and plan bid project	
	• Perform PACT workshop (mandatory for projects of category A, B)	
• Lessons Learned from previous projects	• Review previous Lessons Learned and implement if applicable	
	• Identify and classify requirements	
• Certified Project Manager from execution phase involved	• Establish commercial, legal, technical and delivery bid parts and assess feasibility (e.g. scope, design, software architecture incl. technical requirements)	• Bid Project Description (belongs to SPMP)
		• Concept Design for overall technical solution and Basic Design for critical components
	• Establish project execution organization chart (including nomination of key resources)	• Project Organizational Chart (belongs to SPMP)
	• Establish project execution time schedule	• Project Time Schedule (belongs to SPMP)
	• Identify Project Security Level (PSL) & check project security requirements (PS@S) • Check security requirements • Check EHS requirements	• Bid Security Concept (if applicable) • Travel security concept (if applicable) • EHS concept
	• Check requirements regarding Cybersecurity (incl. project-specific Product & Solution Security (PSS), Information Security and Data Privacy	• Cybersecurity Concept (incl. Project-specific Product & Solution Security (PSS) Concept and Information Security Data Privacy (belongs to SPMP))
• **T**ax advice and applicable contract setups (PATAC)	• Initiate and incorporate PATAC requirements (incl. taxes, customs), Permanent Establishment (PE) (if applicable), delegations, NCM, ECC, currency exchange risks and insurance into bid/offer	• Approved Contract Setup
		• (Cross business/cross border) Memorandum of Understanding
		• Bid progress/status reports
• **E**xisting (frame/standard) contracts with suppliers	• Identify further potential suppliers	• List of (potential) Suppliers (belongs to SPMP)
	• Send out supplier requests for binding offers	• binding offers from suppliers
	• Verify completion of ESG mitigation actions[25]	

[25] Only applicable for Siemens external customer projects

Bid preparation phase		
Submission of a successful bid, with approved risk profile		**Quality Gate PM040: Bid approval**
Mandatory input	Relevant activities	Mandatory output
• Standard contract terms and conditions and NDAs	• Prepare Supply Chain/Logistics Concept	• Supply Chain/Logistics Concept (belongs to SPMP)
	• Identify appropriate project Management methodology (e.g. Traditional, Agile, Hybrid) and integrate derived requirements in Bid/SPMP elements to deliver the solution to/with customer	
• LoA process and documentation requirements	• Perform LoA process[26]	• LoA Documentation and decision
	• Prepare Siemens Project Management Plan	
	• Produce bid (technical, commercial, contractual legal, delivery, responsibilities)	• Bid Documentation (technical, commercial, legal and delivery part)
	• Finalize Offer Calculation	• Offer Calculation (belongs to SPMP)
	• Finalize and document Milestone/ Quality Gate "bid approval"	• Milestone/Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Project Acceleration by Coaching and Teamwork (PACT) workshop (Link)
- Customer Relationship Management (CRM) tool*
- Project categorization tool (Link)
- Limits of Authority (LoA) Application (Link)
- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- New Collaboration Model (NCM) (Link) (for international projects)
- Scheduling software*
- Preferred supplier database "SCM STAR" (Link)
- Requirement Management tool*
- International Commercial Terms (INCOTERMS)*
- Circular 222: Application for and approval of capital investments and divestments (Link)
- Export Control (DAMEX) check (Link)
- Customs check "Global Internal Control Program Customs" (Link)
- Corporate Business Procedures (Zentrale Regeln Geschäftsverkehr) (ZRG) (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- International Delegations and Global Delegation Centre (Link)
- Project Security (CE SEC) (Link)
- Project Security@Siemens Tool (PS@S) (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)
- ESG Radar (Link)

[26] Only applicable for Siemens external customer projects

5.2.1.3 PM070 – Contract negotiation

Contract negotiation phase		
Conducting negotiations until a final decision, fully documenting and communicating the results		**Milestone PM070: Project won/lost**
Mandatory input	Relevant activities	Mandatory output
• Nomination of PM and CPM for the execution phase		
	• Finalize and hand over bid to customer, update as required	
	• Prepare and carry out contract negotiation	
	• Validate draft contract against LoA mandate[27]	
	• (If necessary) perform LoA and project categorization update	
	• Finalize and sign contract	• Customer Contract
	• Provide order confirmation to customer	
	• (If applicable) finalize and document agreement with Siemens parties (e.g. HQ - Regional units)	
	• Establish Communication Matrix	• Communication Matrix (belongs to SPMP)
	• Establish Contract Management Plan (Change log/register and Claim log/register	• Change Log/Register (belongs to SPMP) • Claim Log/Register (belongs to SPMP)
	• Transfer Offer Calculation into as-sold calculation	
	• If contract negotiation not successful/project lost: close bid activities (incl. all obligations and documentation)	
	• Perform bid Lessons Learned including win/loss analysis and provide them to the organization	• Lessons Learned (belongs to SPMP)
	• Archive bid documentation (including e.g. LoA-, ESG-, Offer tool)	
	• Finalize and document Milestone "project won/lost"	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Limits of Authority (LoA) Application (Link) • Project categorization tool (Link) • Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects) • ESG Radar (Link)		

[27] Only applicable for Siemens external customer projects

5.2.1.4 PM080 – Project handover

Project handover phase		
Finalizing the sales phase, ensuring a structured handover of all relevant information to the execution team		**Milestone PM080: End of handover**
Mandatory input	Relevant activities	Mandatory output
	• Consider Lessons Learned from sales phase	
	• Handover of bid documentation to project manager of execution phase	
	• Finalize LoA conformity check of the contract[28]	
• Appointment for an empowered PM and CPM (and WPC for Cross business projects)		• Empowerment Document
	• Validate the project baselines for requirements, schedule, calculation	• Project baselines
• All documents required for project execution according to contract	• Prepare a list of internal and external documents (from sales/for execution phase)	• Document List (belongs to SPMP)
	• Transfer as-sold calculation into Order Entry Calculation (OEC) (primarily volume and profit)	• Order Entry Calculation (belongs to SPMP)
	• Create transparency on contracts and offers between involved parties (e.g. NCM)	• finalized and signed contracts between parties involved (NCM)
	• (Cross-border staffing projects) complete Project Delegation Concept	
	• Screen and understand the risk assessment incl. all identified risks/opportunities and planned actions (including compliance, legal, technical, ESG ...)	• Risk and Opportunity Register (belongs to SPMP)
	• Finalize and document Milestone "end of handover"	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Limits of Authority (LoA) Application (Link) • New Collaboration Model (NCM) (Link) (for international projects) • Requirement Management tool* • International Delegations and Global Delegation Centre (Link)Compliance Handbook, chapter 5.3 Compliance in Project Business, (Link) • Compliance in Project Business process (Link)		

[28] Only applicable for Siemens external customer projects

5.3 Solution Projects (conventional): Execution Phase

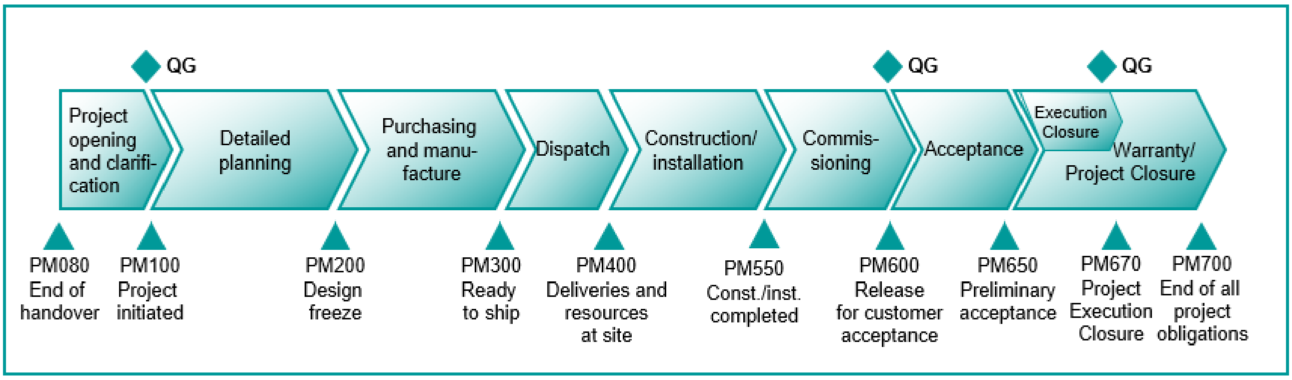

Figure 15: Execution phase of Solution Projects (conventional) in Categories A, B, and C

5.3.1.1 PM100 – Project opening and clarification

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		**Quality Gate PM100: Project initiated**
Mandatory input	Relevant activities	Mandatory output
• Role Descriptions • Project Organizational Chart • Resource Calendar • Standard WBS	• Set up project infrastructure and team • Book OEC into the ERP application, and set up the project WBS	
	• Agree the Project Target Agreement	• Project Target Agreement (belongs to SPMP)
• Standard CBS	• Prepare first Current Cost Calculation (CCC or MiKa)	• Current Cost Calculation (belongs to SPMP)
	• Establish the Procurement Plan	• Procurement Plan (belongs to SPMP)
	• Establish the Quality Plan	• Quality Plan (belongs to SPMP)
	• (If applicable) establish the Project Security Concept	• (if applicable) Project Security Concept (belongs to SPMP)
	• Establish the EHS Plan	• (if applicable) EHS Plan (belongs to SPMP)
	• Establish the Cybersecurity Concept	• Cybersecurity Concept (belongs to SPMP)
	• Perform project kick-off at least with core team	
	• Perform PACT workshop (mandatory for projects of category A, B)	
• Customer Contract	• Perform contract reading with project core team and identify critical topics	• Contractual risks (technically, commercially and legally) and mitigation measures updated
• Compliance in project execution process	• Update compliance risk analysis and define compliance mitigation measures	
• Siemens Project Management Plan procedures	• Complete and update Siemens Project Management Plan (SPMP) (Figure 11) as baseline for the project execution phase	

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		**Quality Gate PM100: Project initiated**
Mandatory input	Relevant activities	Mandatory output
	• Prepare billing schedule/ incorporate billing information into the Project Time Schedule	
	• Plan activities to secure and obtain necessary permits and licenses	
	• Break down targets for the team and initiate team performance evaluation	
• Non-conformance Management Procedure	• Start documenting non-conformances	• Non-conformance Log (belongs to SPMP)
	• Break down requirements & implement requirements tracing	• Requirements Collection with Requirements Traceability Matrix (belongs to SPMP)
• (If applicable) Cross-Organization Cooperation Agreement	• (Cross-business projects) identify synergies in project organization and site infrastructure and implement the benefit of these by sharing indicated functions for all involved parties	
	• (In case of a Permanent Establishment (PE)) prepare necessary input for Permanent Establishment	• Documentation for Permanent Establishment Registration
• Lessons Learned from previous projects	• Review previous/other projects for Lessons Learned and implement (if applicable)	
	• agree project execution Milestone process (conventional or software-driven)	
	• Finalize and document Milestone/ Quality Gate “project initiated”	• Milestone/Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Limits of Authority (LoA) Application I (Link)
- Project Acceleration by Coaching and Teamwork (PACT) workshop (Link)
- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- Compliance Handbook, chapter 5.3 Compliance in Project Business (Link)
- Compliance in Project Business process (Link)
- Project Security (CE SEC) (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)
- Cybersecurity (Link) incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.3.1.2 PM200 – Detailed planning

Detailed planning phase		
Detailed implementation plan for the final design/solution as agreed with the customer		**Milestone PM200: Design freeze**
Mandatory input	Relevant activities	Mandatory output
	• Specify requirements and technical solution	• Basic Design Documentation - customer approved (recommended)
	• Clarification of interfaces and business workflows	• Interface Documentation/Specification (logical /physical)
	• Inspect/analyze the installation environment	• Defined project infrastructure
	• Establish construction, installation, commissioning manuals	
• Reference Solutions	• Confirm/evaluate reference selection/reuse of existing standards/solutions	
	• Verify with customer on format and content of customer documentation	
	• Establish test concept	• Test Plan (belongs to SPMP)
	• Release engineering design	• Design Freeze Agreement (belongs to SPMP) - customer signature (recommended)
• Former training materials/documents/manuals and list of enabled trainers	• Define training needs for project team	
	• Define training needs for end user	• Training Plan
	• Assign human resources	
• Existing (frame/standard) contracts with suppliers		
• Standard contract terms & conditions and NDAs		
	• Prepare and release remaining supply requirements for purchasing negotiation	• Released supplier offers handed over to purchasing department
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “design freeze”	• Milestone Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Scheduling software*
- Requirement Management tool*
- Environment protection, Health Management and Safety (EHS) (Link)
- Siemens Non-conformance Cost (NCC) guideline (Link)
- International Delegations and Global Delegation Centre (Link)
- Project Security (CE SEC) (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.3.1.3 PM300 – Purchasing and manufacture

Purchasing and manufacture phase		
Preconditions for the first deliveries to the defined location are fulfilled		**Milestone PM300: Ready to ship**
Mandatory input	Relevant activities	Mandatory output
	• Place remaining purchase orders for deliverables and supporting services	• Supplier Contracts and updated Resource Calendar
	• Verify quality and completeness of deliveries	• Products released for dispatch (e.g. within Factory Acceptance Test (FAT))
	• Ensure required infrastructure on site for all contract deliverables	
	• Prepare dispatch and release shipment documentation (including ECC)	
	• Establish Site Procedures	• Site Procedures (belongs to SPMP)
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “ready to ship”	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• International Commercial Terms (INCOTERMS)* • Export Control – Digital Exports (Link), Export Control (DAMEX) check (Link) • Customs check “Global Internal Control Program Customs“ (Link) • Environment protection, Health Management and Safety (EHS) (Link)		

5.3.1.4 PM400 – Dispatch

Dispatch phase		
Site readiness declared; receipt of deliveries confirmed		**Milestone PM400: Deliveries and resources at site**
Mandatory input	Relevant activities	Mandatory output
• Resource Calendar	• Verify completeness of received deliverables including customer(s) and partner(s) • Implement Project Security Concept[29] • Implement EHS Concept[30] • Implement Cybersecurity Concept (including PSS, Information Security) and Data Privacy throughout the supply chain)[31] • Complete documentation for construction and installation • Update Siemens Project Management Plan (SPMP) (Figure 11) • Finalize and document Milestone "Deliveries and resources at site"	• Verified products/deliveries at site • Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Environment protection, Health Management and Safety (EHS) (Link) • Project Security (CE SEC) (Link) • Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)		

[29] to be implemented before first installation/commissioning activities

[30] to be implemented before first installation/commissioning activities

[31] to be implemented before first installation/commissioning activities

5.3.1.5 PM550 – Construction/installation

Construction/installation phase		
Construction and installation of the solution internally verified against the agreed scope		**Milestone PM550: Construction/Installation completed**
Mandatory input	Relevant activities	Mandatory output
	• Carry out construction and installation[32]	• Solution verified against scope
	• Commence the end user training	
	• Verify and amend drawings/documentation in line with "as built" installation	• Redline Documentation of construction and installation
	• Finalize preparation of commissioning phase	• Documentation for Commissioning (e.g. commissioning plan)
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “Construction/Installation completed”	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Environment protection, Health Management and Safety (EHS) (Link)		

[32] already starts with first deliveries at site (in parallel to PM400)

5.3.1.6 PM600 – Commissioning

Commissioning phase		
Solution validation concluded, ready for customer acceptance		**Quality Gate PM600: Release for customer acceptance**
Mandatory input	Relevant activities	Mandatory output
	• Finalize commissioning of solution	
	• Establish test plan and conduct internal tests	• Documentation of tests results
	• Compose "as built" Customer Documentation	
• Customer Contract	• Verify fulfillment of all contractual requirements	• List of Open Points (LOP)
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone/ Quality Gate “Release for customer acceptance”	• Milestone/Quality Gate Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Requirement Management tool* • Environment protection, Health Management and Safety (EHS) (Link) • Compliance Handbook, chapter 5.3 Compliance in Project Business (Link) • Compliance in Project Business process (Link)		

5.3.1.7 PM650 – Acceptance

Acceptance phase		
Customer handover achieved and risk transferred		**Milestone PM650: Preliminary acceptance**
Mandatory input	Relevant activities	Mandatory output
	• Conduct customer acceptance tests • Document Preliminary Customer Approval • Update Siemens Project Management Plan (SPMP) (Figure 11) • Finalize and document Milestone “customer acceptance”	• List of Open Points (LoP) • e.g. Provisional Acceptance Certificate (PAC) • Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Scheduling software* • Requirement Management tool* • Environment protection, Health Management and Safety (EHS) (Link) • Siemens Non-conformance Cost (NCC) guideline (Link) • International Delegations and Global Delegation Centre (Link)		

5.3.1.8 PM670 – Execution closure

Execution closure phase		
Concluding project closure of the execution project including experience transfer to the organization		**Quality Gate PM670: Project execution closure**
Mandatory input	Relevant activities	Mandatory output
	• Close out open points	
	• Ensure correct posting and accounting of surplus/scrap material	
	• Clarify and close out open claims with contract partners	
	• Finalize and hand over final Customer Documentation	
	• Demobilize site	• Documentation of site closure
	• Close Permanent Establishment (PE) or handover (if applicable)	
	• Issue and send the final (project) invoice	
	• Ensure that the payment of final invoice is received	
	• Request customer satisfaction feedback	
	• Check the completion of project targets and carry out Lessons Learned	• Lessons Learned reflected back to organization
	• Archive the documentation from project execution	
	• Establish Project closure report and prepare handover to warranty phase	• Project Closure Report
		• (If applicable) documentation of warranty handover
• (If applicable) official nomination of the responsible person for warranty • (If applicable) official release of the PM, CPM and rest of project team	• (If applicable) handover project to responsible for warranty	
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone/ Quality Gate "Project execution closure"	• Milestone/Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Document archiving system* • Permanent Establishment (Link), PE Manager Checklist (Link), PE closing request (Link)

5.3.1.9 PM700 – Warranty/project closure

Warranty/project closure phase		
Warranty concluded and final acceptance certificate (FAC) received with release of all project obligations and handing over to after sales		**Milestone PM700** **End of all project obligations**
Mandatory input	Relevant activities	Mandatory output
	• Perform warranty activities	
	• Document the end of warranty	• Final Acceptance Certificate (FAC)
	• Dissolve provisions	
	• Ensure that the bonds are returned	
	• Finalize the project calculation, including non-conformance costs	• Final Project Calculation
	• Archive documentation changed/established after PM670	
	• End all delegations	
	• Finalize and close project	• Project Closure Report
• Official release of the responsible person for warranty		
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone "end of all project obligations"	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Document archiving system*		

5.4 Software-driven Solution Projects: Execution Phase

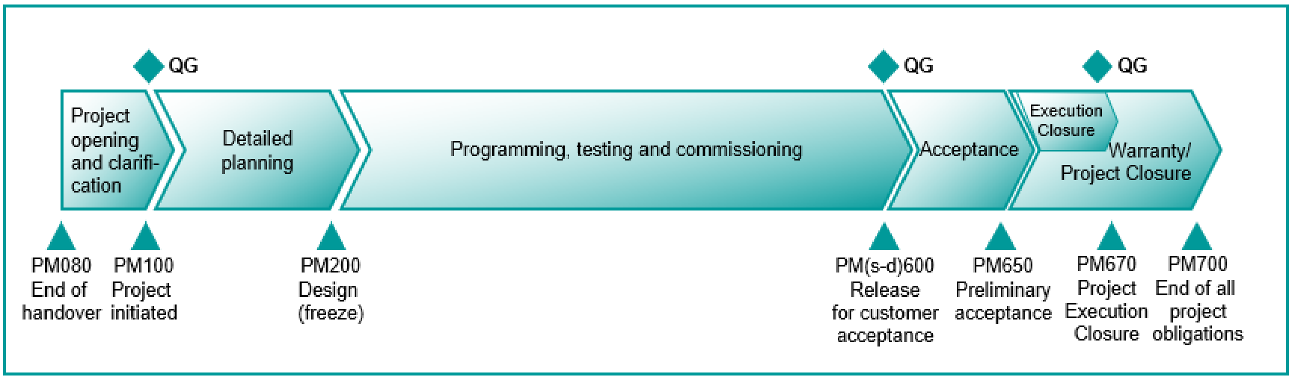

Figure 16: Execution phase of Software-driven Solution Projects in Categories A, B, and C

5.4.1.1 PM100 – Project opening and clarification

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		Quality Gate PM100: Project initiated
Mandatory input	**Relevant activities**	**Mandatory output**
• Role Descriptions • Project Organizational Chart • Resource Calendar • Standard WBS	• Set up project infrastructure and team	
	• Book OEC into the ERP application, and set up the project WBS	
	• Agree to on the Project Target Agreement	• Project Target Agreement (belongs to SPMP)
• Standard CBS	• Prepare first Current Cost Calculation (CCC or MiKa)	• Current Cost Calculation (belongs to SPMP)
	• Establish the Procurement Plan	• Procurement Plan (belongs to SPMP)
	• Establish the Quality Plan	• Quality Plan (belongs to SPMP)
	• (If applicable) establish the Project Security Concept	• (if applicable) Project Security Concept (belongs to SPMP)
	• Establish the EHS Plan	• (if applicable) EHS Plan (belongs to SPMP)
	• Establish the Cybersecurity Concept	• Cybersecurity Concept (belongs to SPMP)
	• Perform project kick-off at least with core team	
	• Perform PACT workshop (mandatory for projects of category A, B)	
• Customer Contract	• Perform contract reading with project core team and identify critical topics	• Contractual risks (technically, commercially and legally) and mitigation measures updated
• Compliance in project execution process	• Update compliance risk analysis and define compliance mitigation measures	

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		Quality Gate PM100: Project initiated
Mandatory input	**Relevant activities**	**Mandatory output**
• Siemens Project Management Plan procedures	• Complete and update Siemens Project Management Plan (SPMP) (Figure 11) as baseline for the project execution phase	
	• Prepare billing schedule/ incorporate billing information into the Project Time Schedule	
	• Plan activities to secure and obtain necessary permits and licenses	
	• Break down targets to the team and initiate team performance evaluation	
• Non-conformance Management Procedure	• Start documenting non-conformances	• Non-conformance Log (belongs to SPMP)
	• Break down requirements & implement requirements tracing	• Requirements Collection with Requirements Traceability Matrix (belongs to SPMP)
• (If applicable) Cross-Organization Cooperation Agreement	• (Cross-business projects) identify and initiate synergies in project organization and site infrastructure by sharing indicated functions for all involved parties	
	• (In case of a Permanent Establishment (PE)) prepare necessary input for Permanent Establishment	• Documentation for Permanent Establishment Registration
• Lessons Learned from previous projects	• Screen previous/other projects for Lessons Learned and implement if applicable	
	• Determine project execution Milestone process (conventional or software-driven)	
	• Finalize and document Milestone/Quality Gate “project initiated”	• Milestone/Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Limits of Authority (LoA) Application (Link)
- Project Acceleration by Coaching and Teamwork (PACT) workshop (Link)
- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- Compliance Handbook, chapter 5.3 Compliance in Project Business (Link)
- Compliance in Project Business process (Link)
- Project Security (CE SEC) (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.4.1.2 PM200 – Detailed planning

Detailed planning phase		
Detailed implementation plan for the final design/solution as agreed with the customer		**Milestone PM200: Design (freeze)**
Mandatory input	Relevant activities	Mandatory output
	• Specify requirements and technical solution	• Basic Design Documentation - customer approved (recommended)
	• Clarification of interfaces and business workflows	• Interface Documentation/Specification (logical /physical)
	• Inspect/analyze the installation environment	• Defined project infrastructure
	• Establish construction, installation, commissioning manuals	
• Reference Solutions	• Confirm/evaluate reference selection/reuse of existing standards/solutions	
	• Verify with customer on format and content of customer documentation	
	• Establish test concept	• Test Plan (belongs to SPMP)
	• Release engineering design	• Design (Freeze) Agreement (belongs to SPMP) - customer signature (recommended)
• Former training materials/documents/manuals and list of enabled trainers	• Define training measures for project team	
	• Define training measures for end user	• Training Plan
	• Assign human resources	
• Existing (frame/standard) contracts with suppliers		
• Standard contract terms & conditions and NDAs		
	• Prepare and release remaining supply chain for purchasing negotiation	• Released supplier offers handed over to purchasing department
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “design freeze”	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Environment protection, Health Management and Safety (EHS) (Link) • Siemens Non-conformance Cost (NCC) guideline (Link) • International Delegations and Global Delegation Centre (Link) • Project Security (CE SEC) (Link) • Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link) • Scheduling software* • Requirement Management tool*		

5.4.1.3 PM(s-d)600 – Programming, testing and commissioning

Programming, testing and commissioning phase		
Solution validation concluded, ready for customer acceptance		**Quality Gate: PM(s-d)600 Release for customer acceptance**
Mandatory input	Relevant activities	Mandatory output
	• Continuously update Siemens Project Management Plan (SPMP) (Figure 11) after every "Sprint"	
	• Place remaining purchase orders for deliverables and supporting services	
	• Verify the completeness of received deliverables including customer(s) and partner(s)	
	• (If applicable) implement EHS Concept	
	• Implement Cybersecurity Concept (including PSS, Information Security) and Data Privacy throughout the supply chain)	
	• (If applicable) prepare dispatch and release shipment documentation (including ECC)	
	• Perform Code Reviews	
	• Build solution per plan	
	• Identify out of the box (OOTB) software defects and issues	
	• Design and implement integration to other systems (interfaces)	
	• (If applicable) ensure required infrastructure for software integrations	
	• Update software design documents	
	• Establish test plan and conduct internal tests	• Documentation of tests results
	• Initiate the end user training	
	• Complete documentation for installation and commissioning	
	• Finalize commissioning of solution	
• Standard Software Documentation	• Compose "as built" customer documentation (including manuals and operating instructions)	
	• Finalize and document Milestone/Quality Gate "Release for customer acceptance"	• Milestone and Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools
• Compliance Handbook, chapter 5.3 Compliance in Project Business (Link) • Compliance in Project Business process (Link) • Environment protection, Health Management and Safety (EHS) (Link) • Export Control – Digital Exports (Link), Export Control (DAMEX) check (Link) • Project Security (CE SEC) (Link) • Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.4.1.4 PM650 – Acceptance

Acceptance phase		
Customer handover achieved and risk transferred		**Milestone PM650: Preliminary acceptance**
Mandatory input	Relevant activities	Mandatory output
	• Conduct customer acceptance tests	• List of Open Points (LoP)
	• Document Preliminary Customer Approval	• e.g. Provisional Acceptance Certificate (PAC)
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “Preliminary acceptance”	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Scheduling software* • Requirement Management tool* • Environment protection, Health Management and Safety (EHS) (Link) • Siemens Non-conformance Cost (NCC) guideline (Link) • International Delegations and Global Delegation Centre (Link)		

5.4.1.5 PM670 – Execution closure

Execution closure phase		
Concluding project closure of the execution project including experience transfer to the organization		**Quality Gate PM670: Project execution closure**
Mandatory input	Relevant activities	Mandatory output
	• Close out open points	
	• Ensure correct posting and accounting of surplus/scrap material	
	• Clarify and close out open claims with contract partners	
	• Finalize and hand over final Customer Documentation	
	• Demobilize site	• Documentation of site closure
	• Close Permanent Establishment (PE) or handover (if applicable)	
	• Issue and send the final (project) invoice	
	• Ensure that the payment of final invoice is received	
	• Request customer satisfaction feedback	
	• Evaluate the fulfillment of project targets and perform Lessons Learned	• Lessons Learned reflected back to organization
	• Archive the documentation/artifacts from project execution	
	• Establish Project closure report and prepare handover to warranty phase	• Project Closure Report
		• (If applicable) documentation of warranty handover
• (If applicable) official nomination of the responsible person for warranty	• (If applicable) handover project to the person responsible for warranty	
• (If applicable) official release of the PM, CPM and rest of project team		
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone/ Quality Gate “Project execution closure”	• Milestone/Quality Gate Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Document archiving system* • Permanent Establishment (Link), PE Manager Checklist (Link) and PE closing request (Link)		

5.4.1.6 PM700 – Warranty/project closure

Warranty/project closure phase		
Warranty concluded and final acceptance certificate (FAC) received with release of all project obligations and handing over to after sales		**Milestone PM700** **End of all project obligations**
Mandatory input	Relevant activities	Mandatory output
	• Perform warranty activities	
	• Document the end of warranty	• Final Acceptance Certificate (FAC)
	• Dissolve provisions	
	• Ensure that the bonds are returned	
	• Finalize the project calculation, including non-conformance costs	• Final Project Calculation
	• Archive documentation/artifacts changed/established after PM670	
	• End all delegations	
	• Finalize and close project	• Project Closure Report
• Official release of the responsible person for warranty		
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “end of all project obligations”	• Milestone Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Document archiving system*

5.5 Service Projects: Execution Phase

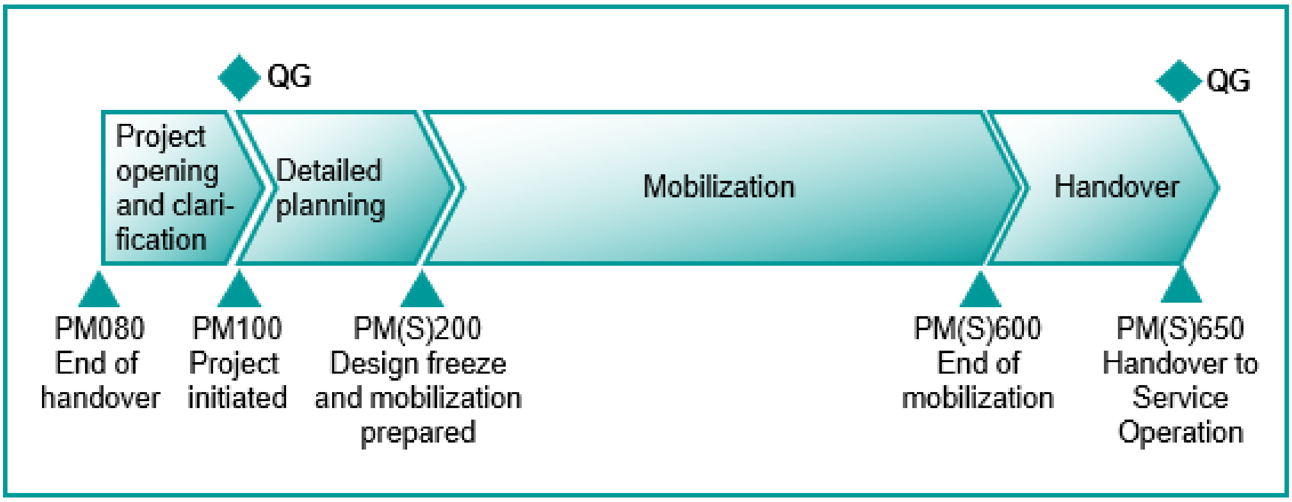

Figure 17: Execution phase of Service Projects in Categories A, B, and C

5.5.1.1 PM100 – Project opening and clarification

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		**Quality Gate PM100: Project initiated**
Mandatory input	Relevant activities	Mandatory output
• Role Descriptions • Project Organizational Chart • Resource Calendar • Standard WBS	• Set up project infrastructure and team	
	• Book OEC into the ERP application, and set up the project WBS	
	• Commit on the Project Target Agreement	• Project Target Agreement (belongs to SPMP)
• Standard CBS	• Prepare first Current Cost Calculation (CCC or MiKa)	• Current Cost Calculation (belongs to SPMP)
	• Establish the Procurement Plan	• Procurement Plan (belongs to SPMP)
	• Establish the Quality Plan	• Quality Plan (belongs to SPMP)
	• (If applicable) establish the Project Security Concept	• (if applicable) Project Security Concept (belongs to SPMP)
	• Establish the EHS Plan	• (if applicable) EHS Plan (belongs to SPMP)
	• Establish the Cybersecurity Concept	• Cybersecurity Concept (belongs to SPMP)
	• Perform project kick-off at least with core team	
	• Perform PACT workshop (mandatory for projects of category A, B and recommended for category C)	

Project opening and clarification phase		
Content clarified within the project boundaries and project execution initiated		**Quality Gate PM100: Project initiated**
Mandatory input	Relevant activities	Mandatory output
• Customer Contract	• Perform contract reading with project core team and identify critical topics	• Contractual risks (technically, commercially and legally) and mitigation measures updated
• Compliance in project execution process	• Update compliance risk analysis and define compliance mitigation measures	
• Siemens Project Management Plan procedures	• Complete and update Siemens Project Management Plan (SPMP) (Figure 11) as baseline for the project execution phase	
	• Prepare billing schedule/ incorporate billing information into the Project Time Schedule	
	• Plan activities to secure and obtain necessary permits and licenses	
	• Break down targets to the team and initiate team performance evaluation	
• Non-conformance Management Procedure	• Start documenting non-conformances	• Non-conformance Log (belongs to SPMP)
	• Break down requirements & implement requirements tracing	• Requirements Collection with Requirements Traceability Matrix (belongs to SPMP)
• (If applicable) Cross-Organization Cooperation Agreement	• (Cross-business projects) identify and initiate synergies in project organization and site infrastructure by sharing indicated functions for all involved parties	
	• (In case of a Permanent Establishment (PE)) prepare necessary input for Permanent Establishment	• Documentation for Permanent Establishment Registration
• Lessons Learned from previous projects	• Review previous/other projects for Lessons Learned and implement if applicable	
	• Determine project execution Milestone process (conventional or software-driven)	
	• Finalize and document Milestone/Quality Gate "project initiated"	• Milestone/Quality Gate Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Limits of Authority (LoA) Application (Link)
- Project Acceleration by Coaching and Teamwork (PACT) workshop (Link)
- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- Compliance Handbook, chapter 5.3 Compliance in Project Business (Link)
- Compliance in Project Business process (Link)
- Project Security (CE SEC) (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.5.1.2 PM(S)200 – Detailed planning

Detailed planning phase		
Detailed implementation plan for the design/service approved by the customer and mobilization plan finalized.		**Milestone PM(S)200: Design freeze and mobilization prepared**
Mandatory input	Relevant activities	Mandatory output
	• Define organizational setup for operations	
	• Perform site inspection	
• Lessons Learned from previous projects	• Screen previous Lessons Learned and implement if applicable	
	• Define training measures for project team	
	• Define training measures for end user	• Training Plan
• (List of) Core Competencies		
• Existing (frame/standard) contracts with suppliers		
• Standard contract terms and conditions and NDAs		
	• Define documentation to be handed over	
	• Determine key suppliers/service providers	
• Spare Part List from (Supply) project	• Place purchase orders for deliverables (including initial spares and consumables) and supporting services	• Supplier Contract
	• Establish and detail service concept	• Detailed Service Concept (belongs to SPMP)
	• Finalize Recruitment Plan with P&O	
	• Implement external/internal reporting system	
	• finalize Operations Concept with operations provider (internal or external)	
	• Establish Site Procedures	• Site Procedures (belongs to SPMP)
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “Design freeze and mobilization prepared”	• Milestone Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Requirement Management tool* • Environment protection, Health Management and Safety (EHS) (Link) • Siemens Non-conformance Cost (NCC) guideline (Link) • International Delegations and Global Delegation Centre (Link) • Project Security (CE SEC) (Link) • Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)		

5.5.1.3 PM(S)600 – Mobilization

Mobilization phase		
Service ready to perform maintenance and operations		**Milestone: PM(S)600 End of mobilization**
Mandatory input	Relevant activities	Mandatory output
	• Procurement of initial spares and consumables	
	• Agree with the stakeholders and finalize the Service Concept	
	• Pre-operational test of relevant maintenance systems and products	• Test Records
	• Ensure readiness of organizational setup	
	• Implement and test infrastructure (IT incl. CMMS, warehouse, offices, desks, etc.)	
• Customer Contract	• Implement service subcontracts with (local) service providers	
	• Execute project-specific staff training	
	• Implement Project Security Concept	
	• Implement EHS Concept	
	• Implement Cybersecurity (including PSS, Information Security) and Data Privacy throughout the supply chain)	
	• Train the customer for service interface (who, what, how, when, to whom)	
	• Specify and train customer regarding customer obligation to cooperate	
	• Mobilize people and ensure resources at site	
• Service Manuals from supplier	• Elaborate work instructions	
	• Update Siemens Project Management Plan (SPMP) (Figure 11)	
	• Finalize and document Milestone “End of Mobilization”	• Milestone Documentation (belongs to SPMP)

Mandatory supporting specifications/methodologies/tools

- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- Requirement Management tool*
- Environment protection, Health Management and Safety (EHS) (Link)
- Project Security (CE SEC) (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.5.1.4 PM(S)650 – Handover

Handover phase		
Carrying out handover to Service Operation Manager		**Quality Gate: PM(S)650 Handover to Service Operation**
Mandatory input	Relevant activities	Mandatory output
• Official nomination of the responsible person for service	• Complete/finalize on lessons learned from project execution phase • Assignment and empowerment of Service Operation Manager (e.g. by empowerment letter) • Handover of contract, SPMP and all change orders/agreements to responsible Service Operation Manager • Stand down the PM, CPM, and rest of project team • Start/launch service • Update Siemens Project Management Plan (SPMP) (Figure 11) • Finalize and document Milestone/Quality Gate “handover to service operation”	• Milestone and Quality Gate Documentation (belongs to SPMP)
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Compliance Handbook, chapter 5.3 Compliance in Project Business (Link) • Compliance in Project Business process (Link)		

The service project finishes after this Milestone is complete. Should the service end (and perhaps also demobilization be necessary), the following topics need to be considered as minimum requirements:

- the operational performance has to be validated and the performance verification report finalized
- the Lessons Learned regarding mobilization must be identified and documented
- the list of open points items has to be completely implemented
- the maintenance plan has to be reviewed
- the completion of operations has to be performed and the report finalized
- the demobilization has to be planned and performed
- all open points from and towards the customer have to be resolved
- contractually agreed return conditions have to be fulfilled
- the operation documentation has to be updated and handed over
- all commercial and P&O topics have to be closed
- the Permanent Establishment has to be closed, documented and handed over if applicable

After completing the above-mentioned topics, this will be the end of all obligations.

5.6 Cat Small (S) Projects

No PACT workshop is required for S projects. However, it is recommended to facilitate a PACT light workshop.

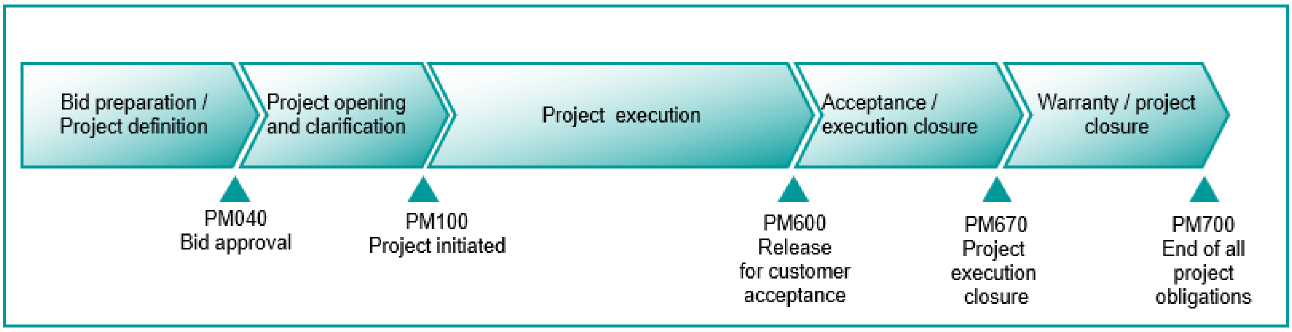

Figure 18: PM@Siemens Milestone Model for Category S projects

5.6.1.1 PM040 – Bid preparation/project definition

Bid preparation/project definition phase – S Projects		
Submission of a successful bid, with approved risk profile		**Milestone PM040: Bid approval**
Mandatory input	Relevant activities	Mandatory output
	• Document, evaluate and substantiate lead	• Requirement Collection (including break down and structure)
	• Evaluate customer's business situation & competitive environment	
	• Initiate bid	• Project Overview
	• Establish Project WBS	• Project WBS
• Lessons Learned from previous projects	• Document deviations from standard contracts and procedures	• List of deviations from standard contracts and procedures
	• Check security requirements	
• Tax advice and applicable contract setups (PATAC)	• Drive Permanent Establishment Decision (if applicable)	• Approved contract setup
	• Finalize and document Milestone "bid approval"	• Milestone Documentation

Mandatory supporting specifications/methodologies/tools

- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- International Commercial Terms (INCOTERMS)*
- Export Control (DAMEX) check (Link)
- Customs check "Global Internal Control Program Customs" (Link)
- Corporate Business Procedures (Zentrale Regeln Geschäftsverkehr) (ZRG) (Link)
- Process Adjustment Tax Advice and Compliance (PATAC) process (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- International Delegations and Global Delegation Centre (Link)
- Project Security (CE SEC) (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)
- Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)

5.6.1.2 PM100 – Project opening and clarification

Project opening and clarification phase – S Projects		
Content clarified within the project boundaries and project execution initiated		**Milestone PM100: Project initiated**
Mandatory input	Relevant activities	Mandatory output
• Official appointment of PM, CPM, and rest of project team for execution phase		
	• Set up project infrastructure and team	
	• Book order entry into ERP application, and set up the WBS/CBS	• Order Entry Calculation
	• Agree on the Project Target Agreement	• Project Target Agreement
	• Update R&O register	
• Compliance in project execution process (relevant for designated projects)	• Perform joint compliance risk analysis and define compliance mitigation measures	• (For designated projects) compliance risks and mitigations, included in R&O register
	• Prepare first Current Cost Calculation	• Current Cost Calculation
	• (In case of a PE) initiate and perform • Permanent Establishment setup	• Documentation for Permanent Establishment registration
	• Finalize and document Milestone "project initiated"	• Milestone Documentation

Mandatory supporting specifications/methodologies/tools

- Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)
- New Collaboration Model (NCM) (Link) (for international projects)
- Environment protection, Health Management and Safety (EHS) (Link)
- Compliance Handbook, chapter 5.3 Compliance in Project Business (Link)
- Compliance in Project Business process (Link)
- Permanent Establishment (Link), PE Manager Checklist (Link)

5.6.1.3 PM600 – Project execution

Project execution phase – S Projects		
Solution validation concluded, ready for customer acceptance		**Milestone PM600: Release for customer acceptance**
Mandatory input	Relevant activities	Mandatory output
	• Detailed planning	
	• Purchasing & manufacture	
	• Update and implement security requirements	
	• Dispatch	
	• Construction/installation	
	• Commissioning	
	• Conduct internal tests for the plant/ solution	• Test records • “As built” Customer Documentation
• Compliance in project execution process (for relevant projects)	• Perform joint compliance analysis focused on customer acceptance and final check that all identified compliance mitigation measures are implemented	• (for relevant projects) concluded compliance risks and mitigations, documented in R&O register
• Customer Contract	• Verify fulfillment of all contractual requirements	• List of Open Points (LOP)
	• Finalize and document Milestone “release for customer acceptance”	• Milestone Documentation
Mandatory supporting specifications/methodologies/tools		
• Environment protection, Health Management and Safety (EHS) (Link) • Compliance Handbook, chapter 5.3 Compliance in Project Business (Link) • Project Security (CE SEC) (Link) • Cybersecurity (Link) (incl. Product and Solution Security (Link), Information Security (Link)), Data Privacy (Link)		

5.6.1.4 PM670 – Acceptance and execution closure

Acceptance and execution closure phase – S Projects		
Customer handover achieved and project closure of the execution project concluded		**Milestone PM670: Project execution closure**
Mandatory input	Relevant activities	Mandatory output
	• Establish final customer documentation	
	• Perform acceptance	• Acceptance Certificate
• (If applicable) official nomination of the responsible person for warranty/service		
	• Close LOP	
	• Perform Lessons Learned	• Lessons Learned performed
• (If applicable) official release of the PM, CPM, and rest of project team		
	• Perform Permanent Establishment closing or handover (if applicable)	• Documentation for closing PE
	• Finalize and document Milestone "project execution closure"	• Milestone Documentation
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link) • Permanent Establishment (Link), PE Manager Checklist (Link), PE closing request (Link)		

5.6.1.5 PM700 – Warranty/project closure

Warranty/Project closure phase – S projects		
Warranty concluded and final acceptance certificate received with release of all project obligations and handing over to after sales		**Milestone PM700:** **End of all project obligations**
Mandatory input	Relevant activities	Mandatory output
	• Perform and close warranty activities	• Final Acceptance Certificate (FAC)
	• Finalize the project calculation, including non-conformance costs	• Final Project Calculation
	• Archive project data	
• Official release of the responsible person for warranty/service		
	• Finalize and document Milestone “end of all project obligations”	• Milestone Documentation
Mandatory supporting specifications/methodologies/tools		
• Siemens Financial Reporting Guidelines (FRG) and Implementation Guidelines (IG) (Link)		

6 Annex 1: Definitions and abbreviations

Term	Description	Link
Acceptance certificate	Confirms in writing the preliminary and/or final acceptance of the complete contracted solution or service by the customer prior to the end of warranty. The warranty period ends with the FAC, in most cases. If there are any open issues lasting beyond project closure, for example, guarantees, assigned personnel, or other obligations, these need to be closed before the end of warranty.	
Baseline	The approved version of a work product that can be changed using formal change control procedures and that is used as a basis for comparison to actual results. (Source: PMI Lexicon Version 3)	
Basic Engineering	Defined phase of Engineering: Basic Engineering results in a technically verifiable design (Basic Design).	
Bid	A formal and in most cases, binding set of documents sent to a customer with the goal of being awarded an order for a project. The set of documents usually includes information such as the scope, requirements, time schedule, and price of the project.	
Bid progress/ **status reports**	Bid progress/status reports Collection of information showing the progress of the bid project, at least in terms of time, cost, open points, risks.	
Bid project description	Comprises descriptive information and boundary conditions for creating the bid. It is used to define the responsibilities, budget, and resources for bid preparation	
Bid security concept	Describes the security risks and expectedly required security measures. It also provides cost indicators to support the bid calculation.	Project Security
Bid strategy	An agreed approach for how the bid will be executed and offered to the customer in order to beat the competition.	
Bid time schedule	Structured representation of activities, Milestones, and deliverables for the bid project, with the corresponding start dates, end dates, and assigned resources.	
Bid Work Breakdown Structure	Structured representation of the work that needs to be performed as part of bid preparation.	
Bi-directional requirement tracing	Approach for the consistent Management of requirements, including linkage between source requirements and their lower-level requirements and vice versa (vertical), and between requirements and tests (horizontal). This helps determine whether all requirements have been completely addressed.	
Business case	Describes the expected business and benefits for the project stakeholders.	
Business Risk Class (BRC)	The business specific Business Risk Classification subdivides projects into Business Risk Classes 1 to 3 (1 =high risk, 3 = low risk). It is a further factor in deciding the Escalation Level, including at least the technical LoA aspects and the commercial / contractual LoA aspects that are relevant from the specific viewpoint of the business	
Change Control Board (CCB) method	A recommended method for managing changes within the lifecycle of the project extending beyond one discipline and changing the project baseline. It involves a group of defined project team members empowered to decide whether the proposed changes should be implemented. Approved changes are then implemented and baselined by the project team.	

Term	Description	Link
Change log/register	Complete record of change requests and approved changes related to the project baseline, in terms of time, cost, scope, and quality. It must include at least: source of the change, impact, measures, responsible person, due date, date of identification and notification, description of change, affected scope, status, and individual change order strategy.	
Change order	Formal addition to the contract, approved by the customer and/or supplier, stating a change to the original signed contract (including impact on time, cost, scope, and quality).	
Change request	An official demand to make an adjustment to the project baseline.	
Claim log/register	Collection of the potential and actual project claim information, both incoming and outgoing. It must include at least: source of the claim, impact, measures, responsible person, due date, date of identification and notification, description of claim and evidence, affected scope, classification, counterpart of claim, claimed value, probability, expected value, related contract clauses, status, and individual claim strategy	
Commercial Project Manager (CPM)	Core member in a project, responsible for all commercial and financial topics. The CPM will be nominated by the business responsible in writing.	
Communication and collaboration Management plan	Describes the “How” of managing the interaction between (internal and external) stakeholders. The issues to be addressed must include at least: communication matrix, stakeholder register, results from stakeholder analyses, reporting requirements (including frequencies, deliverables, and those involved and responsible), and escalation approach including paths and timeframes.	
Communication matrix	Represents the content and format of the information exchange in the form of an “n:n” relationship. It must mention at least: the stakeholders providing the information, those receiving the information, the type of information (for example, status report), type of communication (for example, team meeting, telephone conference, e-mail, electronic document) and the frequency.	
Compliance Handbook	Mandatory process used to identify and mitigate compliance risks in the project. Specific measures are mandatory for projects for which the PM@Siemens LoA process is applicable and that have been defined as relevant for the execution phase. Identified compliance risks and mitigation measures shall be included in the Risk and Opportunity Register	Compliance Handbook, chapter 5.3 Compliance in Project Business
Compliance Officer (CO) (in LoA Process)	As the expert on compliance, the respective Compliance Officer supports the Project Manager. The Compliance Officer contributes his or her knowledge of the regional and industry-specific Compliance risks (e.g. corruption risks, money-laundering risks, human rights risks). The Authorized Compliance Officer (ACO) is determined for each project automatically by the LoA Application based on the ARE (Siemens entity).	
Computerized Maintenance Management System (CMMS)	Software that centralizes maintenance information and simplifies the processes of maintenance operations.	
Concept Engineering	Defined phase of Engineering: Concept Engineering results in a technically feasible design (Concept Design).	
Condition survey	Analysis of the condition of the infrastructure and site that is to be taken over from the customer.	
Configuration List	Representation of the solution configuration, including all the configuration items (hardware and software), versions, part numbers, and other relevant information in the immediate environment of the solution.	
Configuration Management	Configuration Management is a Management activity that applies technical and administrative direction over the lifecycle of a product and service, its configuration identification and status, and related product and service configuration information.	

Term	Description	Link

Configuration of Software	Using the included capabilities of the off-the-shelf software to adjust the software without altering the software code. Configuration allows customers to select available options to adjust the software to align the off-the-shelf product behavior with the identified requirements. - changing parameters, do not change the software
	
	Examples (non-exhaustive list):
	• Creation of workflows
	• Checking boxes, entering values in dialog field
	• Using BMIDE (Business Modeler IDE interface) including:
	  • Adding or adjusting properties and classes to align with customer business needs.
	  • Adding or adjusting options, conditions, and constants to align with customer processes.
	  • Adding or adjusting standard dialogs to align with customer processes.
	  • Setting preferences to affect behavior of the software to align with customer processes.
	  • Utilizing access rules and organizational structures to align with security and data access requirements.
	  • Editing or creating Style Sheets to align with customer processes, in a supported language.
	  • Adaptation of the user interface to simplify or present specific information for a certain class of users.
	  • Adjusting revision rules and closure rules to filter information into the form best utilized by customer users.	

Construction and installation documentation	Consists of detailed solution-specific information necessary to execute the construction and installation, for example, detailed installation drawings, interface descriptions, and method statements. Supplements the construction documentation plan.	
Continuous improvement Management plan	Describes how the project will utilize the latest assets and knowledge of the organization, and how it will contribute towards the organization further learning and improving.	
Continuous Improvement Process (CIP)	A way of thinking that aims to strengthen the competitiveness of companies with constant improvements in small steps. CIP relates to product, process and service quality.	
Contract Management plan	Describes the “How” of contract Management, including change orders and claims. The issues to be addressed must include at least: contract negotiation strategies, claim strategies, change order strategies, and the roles responsible for the various contracts, claims, and change orders.	
Contract reading	A meeting (following a preparation phase) during which contractual criticalities (for example, deviations from the standard technical solution, deviations from the standard terms and conditions) in each subject area and their consequences for the project execution are transparently analyzed and discussed.	
Controlling and updating Management plan	Describes the “How” of the Management of the project’s progress. The issues to be addressed must include at least: how the changes to the project will be handled, how the baselines in terms of time, cost, quality, risk, and opportunity requirements and other factors will be captured and used to control the progress of the project.	
Core competencies	Key knowledge, abilities, and skills that the project organization possesses that contribute to fulfilling the organization’s strategy and delivering additional value to the customer.	

Term	Description	Link
Core team	Consists of at least the PM and CPM. Depending on the project's complexity, the following additional project Management roles should be allocated to dedicated persons: Scheduler, Technical Project Manager, Site Manager, Risk and Opportunity Manager, Project Procurement Manager (PPM), Contract and Claim Manager, EHS Manager in Projects, Quality Manager in Projects (QMiP), Logistics Project Manager, Solution Architect for software-driven projects.	PM/CPM QMiP PPM LPM
Cost breakdown structure (CBS)	Table of all the associated project costs in line with the WBS to ensure clear cost identification and control.	
Cost Management plan	Describes the 'How' project costs and key financial parameters are to be handled. The issues to be addressed must include at least: project account, accounting type, CBS, tools to be used, frequency for updating the current cost calculation, financial reporting requirements.	
Cross-Organization cooperation agreement	Determines the cooperation between organizations prior to bid approval.	
Current Cost Calculation (CCC)	Detailed description of the planned as well as current costs and revenues for the project, constantly updated during project execution. It shall be based on the order entry calculation. It must include at least: cash flow planning, breakdown of project costs and revenue planned until the completion of the project.	
Custom Software	Purely custom software that is not a modification of our pre-existing software, not developed using our APIs or tools, and is not intended to be incorporated into our software now or in the future.	
Customer deliverables	Contractually defined scope of the project, which needs to be provided to the customer to fulfill the contract.	
Customer documentation	Contractually defined documentation which needs to be provided to the customer to fulfill the contract.	
Customer Relationship Management (CRM)	The consistent orientation of a company towards its customers and the systematic design of customer relationship processes	
Customization of Software	Adding capabilities, features or functionalities that are not included in the off-the-shelf software and are not available after the configuration of the off-the-shelf software by writing additional software code leveraging the APIs exposed by software (for the purpose of customization) in order to meet the customer's unique requirements. - keep IP within Sie, some coding added to off-the shelf software but not to the core - optional The customer is responsible for any maintenance of the customizations after the final deliverable. Customizations are largely reusable after upgrades to future releases by slightly modifying the code, depending on changes to APIs. Examples (non-exhaustive list): • Building customer specific applications and extension using APIs • Programming languages (C, C++, Java etc.) in a services' project, if so agreed to with the customer.	
Customization Source Code or Source Code for Customizations	New source code that is developed to create customizations. In order to maintain the customization applications and extensions, it needs to be in an un-compiled and readable format. This is also software run through an API. This is Siemens intellectual property, and the customer is licensed to use it. Note: In order to avoid a misunderstanding with the term "Source Code" the usage of this full term in cases involving customization is recommended. -Siemens keeps the IP but the customer can use it-	
Cybersecurity	means the protection against harm caused by (mainly, but not limited to) digital attacks against the confidentiality, integrity, availability, authenticity, reliability of information (physical and digital) and assets (IT, OT, Product, Solutions & Services). It includes but is not limited to the organization, collection of resources, processes, and structures to assure an end-to-end security across the full supply chain and partner network. It comprises and combines Information Security based on ISO/IEC 27001,	Siemens Information Security Policy / Circular 236

Term	Description	Link
	IT/OT/Product/Solution/Service Security, Industrial Security based on IEC 62443 and external Cybersecurity Service Offering	SfeRA-Security Framework and Regulation
Data Machine Export control (DAMEX)	The system **DAMEX** (**DA**ta **M**achine **EX**port control) is a modular application for checking ultimate destination, end-use, end-user, as well as items-specific export controls. DAMEX implements applicable legal and Siemens-internal export control requirements and enables automatic export control to be incorporated into Siemens' processes and procedures.	
Data Privacy / Data Protection (DP)	Data privacy or data protection concerns the proper handling of personal data in compliance with data privacy and data protection laws – the legal basis for the collection, privacy by design and default requirements, notice requirements, technical and organizational measures to protect the personal data and further regulatory obligations.	
Data Privacy Manager	Supports all employees to comply with Data Privacy laws and regulations. Thereby, the Data Privacy Manager proactively contributes to the reduction of risk in the Siemens group.	Data Privacy Manager
Deployment of Software	It is all of the activities that make a software system available for use.	
Design freeze	Formal agreement between the contractual parties to accept the current design as the basis (or baseline). This takes place once a contractually defined level of maturity of the design has been reached.	
Detailed Engineering	Defined phase of Engineering: Detailed Engineering results in a design ready for implementation (Detailed Design).	
Division of Responsibilities	Describes the work split between the involved parties, based on the WBS.	
Document list	Collection of documents which are necessary to be created during the project. It must include at least: the name of the document, purpose of the document, its reference number, status, creator and reviewer, review method and deadline, document type (e.g. internal, external, contract relevant, drawing), deadline for its creation.	
Documentation for Commissioning	Consists of detailed solution specific information necessary to carry out the commissioning, e.g. customer specific procedures or interfaces	
Documentation for commissioning	Describes the "How" of commissioning. The issues to be addressed must include at least: a manual for commissioning, EHS aspects, prerequisites for beginning with commissioning, tests that need to be performed, and criteria for verifying the completion of the commissioning phase.	
Documentation for construction	Describes the "How" of executing the construction. The issues to be addressed must include at least: a manual for construction, EHS aspects, prerequisites for beginning the construction, tests that need to be performed (if applicable), and criteria for verifying the completion of the construction phase.	
Documentation for installation	Describes the 'How' installation is to take place. The issues to be addressed must include at least: manual for installation, EHS aspects, pre-requisites for beginning with installation, tests which need to be performed (if applicable) as well as criteria for verifying the completion of the installation phase.	

Term	Description	Link
Documentation Management plan	Describes the 'How' project documentation is to be managed. The issues to be addressed must include at least: naming convention, archiving (including legal requirements), document formats, protection, document Management tools, review of documents, customer documentation handling including transmission, contractual document deliverables.	
Engineering documentation	Contains all Engineering deliverables to establish the technical solution according to the requirements specified in the customer contract. This typically comprises concepts, technical calculations, drawings, designs and specifications for internal and external manufacturing, software. Usually involved functions are system, mechanical, electrical and automation/software engineering.	Siemens Engineering Community
Enterprise Risk Management (ERM)	Holistic risk Management approach that combines traditional risk Management with a risk-oriented capital analysis and takes into account the dependencies between individual risks. ERM is to be understood as company-wide, holistic or strategic risk Management. In contrast to traditional risk Management, risk measurement in the ERM is not carried out in isolation for individual business areas and risk types, but at the overall company level; risk control is centralized and both risks and opportunities are considered.	
Environment protection, Health Management and Safety (EHS) Management plan	The EHS Management Plan documents how EHS requirements, plans and activities will be developed, documented, and managed throughout the project. Components of this plan should include: • How EHS requirements will be identified, analyzed, planned, tracked, and reported • How EHS activities will be planned, tracked, and reported • How changes to EHS requirements, plans and activities will be managed	
Environment protection, Health Management and Safety (EHS) plan	The EHS plan defines all elements affecting the Environment, Health and Safety during the execution of the project. The EHS plan is the basis for the internal/external partners to execute, verify and trace EHS-related activities.	EHS Standard for project business
Export Control and Customs (ECC)	For a number of reasons, e.g. political or economic issues, the import or export of certain goods, software, technology, services or capital underlies controls and regulations. In the following they are referred to as items. All these regulations apply to certain countries, individuals, entities and organizations, as well as to some categories of goods. As a consequence, **exports and imports may be prohibited, subject to authorization or restricted.** Furthermore, **imports may require customs payments.** These **Export Control and Customs regulations differ in each country or economic area**. They are subject to a very large number of legal changes every year. Therefore, domestic activities as well as cross-border activities, such as imports and exports, must continuously be monitored and legal requirements must be complied with in order to achieve project success.	Customs Export Control
Factory Acceptance Test (FAT)	It validates the operation of the equipment and makes sure the customers' purchase order specifications and all other requirements have been met.	
Final Acceptance Certificate (FAC)	A certification system work process that is issued by an owner to a contractor after successfully completed the final acceptance criteria such as a performance test including all the contractual obligations and requirements.	
Financial Reporting Guidelines (FRG)	The basis for preparing Siemens consolidated financial statements in a uniform and compliant manner. It includes classification and measurement rules for external reporting based on the International Financial Reporting Standards (IFRS) as well as internal reporting requirements. The FRG are one part of the House of Financial Reporting Guidelines.	House of Financial Reporting Guidelines
Global Learning Portal (GLP) 2.0	Global Learning Portal 2.0 is the central Siemens application for planning, documenting and reporting on learning activities.	GLP 2.0
Global Learning World (GLW)	Global Learning World is the global network of learning organizations in Siemens.	
Headquarter (HQ)	Headquarter means the global headquarter of each Business, e.g. ADV, DI, SI, SMO.	
Historical values	Typical values (e.g. for durations, efforts of activities, costs for products, services) which have been collected in the past and consolidated accordingly. These are provided by the organization to support the estimations in projects.	

Term	Description	Link
Implementation Guidelines (IG)	Practical examples of the implementation of the end account entries. The IGs are one part of the House of Financial Reporting Guidelines.	House of Financial Reporting Guidelines
Implementation of Software	Implementation of software consists of leveraging commercial off-the-shelf software to improve business processes. The implementation of software encompasses activities to support integrating the software into a customer's processes and IT landscape. Software implementation typically includes, but is not limited to, performing an analysis of requirements, software installation and configuration, data loading, testing and user training. An important part of software implementation is the analysis to determine if the out of the box software can be configured to meet the customers' requirements. If the software is not able to meet unique customer requirements, then Customization of the off-the-shelf software may be required. It is the process of putting a decision or plan into effect; execution.	
Information Security Incident	An 'Information Security Incident' is defined as the direct or indirect compromise of confidentiality, integrity and or availability of information according to its classification.	
Installation of Software	Installing core products/solutions into an environment (somewhere) - is not doing anything (In general, software installation is a simple task that does not include configuration or customization of the software)	
Integration of Software	process of providing communication between two or more parts (to provide functionality) -> include interfaces	
Interface	A common boundary where information is exchanged. Interfaces can be provided to the customer as a standard interface (not altered) or as a customized interface (see customization). Types of Interfaces (non-exhaustive list): • Hardware interface – consists of cables, connectors, and ports that link devices • Software interface – consists of commands, codes, and messages (application program Interface) that enable different programs to communicate with each other and the operating system. • User interface – consists of command line, menus, prompts, dialog boxes, icons, wizards, etc., that enable a user and a computer to communicate with each other	
International Commercial Terms (INCOTERMS)	It is the official policy of the International Chamber of Commerce (ICC) which contains international rules for the interpretation of trade clauses to facilitate international trade. They are usually referred to in (international) contracts.	Link to ICC
International Project Management Association (IPMA)	The International Project Management Association (IPMA) is a global project Management association. IPMA A – D are the certification levels for project managers that are aligned to the various PM@Siemens certification levels.	IPMA
Investment application ("I-Antrag")	Describes the investment proposal required to be approved prior to the conclusion of the negotiations.	Investment & Divestment application (SR_222)
Kick-off	Meeting of the project team at the project start including internal and optionally external stakeholders, to present the project basis, scope, specifics, targets, communication / collaboration approach, rules of engagement, to reach a common understanding. The individuals which must take part include at least the core team	
Learnings in Project Management Plan	Describes the 'How' Lessons Learned will be collected and communicated during the project, e.g. during which events, how often, by whom.	
Lessons Learned	Learnings gained from previous projects. Lessons Learned may be identified at any point in time. They can be derived from positive as well as negative experiences which mean that they describe both, risks, and optimization possibilities. The information exchange shall consider both pull (requesting information) and push (under the assumption that the information might be helpful) approaches.	
Letter of Empowerment (LoE)	Formal nomination of the responsibilities and authority of the PM and CPM. It must include at least: project designation, period of validity, obligations, and rights.	
Limits of Authority (LoA)	Process defining an internal approval procedure. It is used to define and delegate the authority to approve the submission of bids to specific Management levels. Hereby a set	LoA Application

Term	Description	Link
	of factors and risks must be analyzed to identify the necessary Management level that will make the final decisions.	
Limits of Authority (LoA) document requirements	Documents which must be prepared for each project to obtain a LoA-approval. The degree of details for these documents and of the information to be documented depends on the project's complexity and risks. The mandatory LoA documents include: • project description (a summary) with project key data • project Structure (contract setup and project Organization) • key financials (including calculations summary and cash flow) • statements on legal / financial / technical and compliance risks (if applicable)	
Limits of Authority (LoA) risk report	Standardized summary of the results of the technical, commercial, legal and compliance risk analyses. It is an integral part of the LoA documentation.	
Line manager (functional departments)	Line Managers in functional departments who make resources available to projects (such as Engineering, Procurement, EHS) are responsible to provide the required expertise for individual projects. Line Managers use the growth talks for developing individuals in alignment with the respective Project Managers, promote individual career development, give room for continuous learning and support personal growth.	
List of open points (LOP)	Record of all relevant points needed to achieve acceptance, which need to be addressed and solved to obtain preliminary and later final acceptance. The internal version of this should include both internally as well as externally (e.g. through the customer) identified points.	
List of suppliers	Documented record of selected suppliers, including information about each supplier and the deliverables which each of them is to deliver.	
Maintenance manuals	Describe how the equipment being serviced is to be maintained, including e.g. frequencies for checks or replacements, special issues to be considered, how repairs are to be managed.	
Maturity in Project Management (MPM)	The progressive development of an enterprise-wide project Management approach, methodology, strategy, and decision-making process.	
Milestone	Event of special importance along the project lifecycle. Often it is located at the transition of one phase to another. Milestones focus on major progress or interface points supporting the achievement of project goals.	
Milestone fulfillment and decision documentation	List of criteria for fulfilling a certain Milestone (organization specific) as well as the protocol of the decision made. This can be supported by checklists, tools, LoA documents, etc.	
Milestone Trend Analysis (MTA)	Method for systematic Milestone monitoring in a project with the goal of easily and graphically recognizing deviations from the schedule baseline.	
New Collaboration Model (NCM)	Outlines ways of cooperation between a Business Unit (BU) and a Regional Unit (RU), aligns risk and profit allocation in international intercompany business and further standardizes internal cooperation. The NCM was introduced for legal, tax and transparency reasons. Its goal is to avoid permanent establishments, fulfill transfer pricing requirements and being able to handle inter-national business within the Siemens organization better.	NCM
Non-Conformance Cost (NCC)	Non-Conformance costs (NCC) are costs which are found to deviate from the calculated order costs of the OIC over the whole period of the project, including the warranty period, i.e. NCCs are additional costs (including provisions for the future), less cost reductions and any cost reimbursements arising from claims.	
Non-conformance log	Record of non-conformances identified during the project (regardless of whether they cause a financial impact or not). It must include at least: identification numbers, descriptions, categories, financial impacts (non- conformance costs), impacted WBS elements, sources of the non-conformance, the planned responses with deadlines and those responsible, status.	

Term	Description	Link
Non-conformance Management procedure from organization	Describes the 'How' to deal with non-conformances encountered during the projects. The issues to be addressed must include at least: how the non- conformances can be identified, how they shall be evaluated, documented, approved, assigned, addressed, communicated, and resolved. Non-conformances with both recorded financial impacts (NCCs) and without (often called errors) need to be addressed.	
Offer calculation	Financial calculation for the offer, also the basis for the negotiations with the customer. It must include at least: budgeted costs (for the offered scope and customer requirements), contingencies, price, EBIT. This information needs to be for the entire project being offered to the customer and broken down to an adequate level of detail.	
Off-the-shelf-software	Commercial off-the-shelf (COTS) is a term used to describe the purchase of products that are standard manufactured products rather than custom, or bespoke, products. A synonym is Out of the Box (OOTB).	Open-Source Software (OSS) Open-Source Software
Online Competency Analysis (OCA)	Online Competency Analysis is a process to determine if a project manager has the right level of experience before carrying out a PM@Siemens certification (A,B,C level).	OCA
Operated infrastructure	Operated infrastructure includes Siemens operated assets (e.g. networks, clouds, systems for development and manufacturing).	
Operation Manuals	Describe how to perform tasks and activities during operations, e.g. how to start up a device, how to shut down, things which should be avoided.	
Operations concept	Document describing the modes of operations. This is needed to prepare the service concept.	
Opportunity	Is the potential for gain caused by an event (or series of events) that can positively affect the achievement of the project objectives.	
Order entry calculation	Financial calculation baseline (following contract signature) including budgeted costs, contingencies, price, payment information for the entire project broken down to an adequate level of detail. It shall be within the approved negotiation mandate and shall be based on the offer calculation.	
Organizational Strategy	A high-level plan stating how the organizational goals will be achieved.	
Out-of-the-box (OOTB)	OOTB is a feature or functionality of a product that works immediately after or even without any special installation without any configuration or modification.	
P&L Unit	Please see definition according to GM TO.	
PACT (workshop)	Project Acceleration by Coaching and Teamwork / PACT workshops are workshops normally held at the start or at other key points of a project and are facilitated by professionally trained PACT Facilitators.	PACT Link
PATAC process	PATAC is a standard for tax works in the international project business. The process defines the mandatory steps for the involvement of tax functions in project business. This ensures that a project has an optimized project structure to meet tax compliance standards. It is applicable to all Siemens entities that carry out international project business.	PATAC
Patching of Software	It is like the Update of Software, but it is bug fixing for an individual case.	
PE Manager	The PE Manager is the representative of the responsible unit for the respective PE. It is always the responsibility of the accountable unit to ensure that a valid PE Manager is appointed. The PE Manager must be named in the PE approval request. In case of single-project PEs, the function of the PE-Manager is usually taken over by the CPM. The PE Manager will be nominated by the business responsible with the form provided by CF T.	PE Manager PE Manager (checklist)
People & Organization (P&O)	Refers to the area of business Management that deals with the production factor work and with personnel. People & Organization is an existing function in all organizations,	

Term	Description	Link
	the core tasks of which are the provision and target-oriented deployment of personnel. Formerly known as Human resources.	
Percentage of completion (PoC) method	Under the PoC method, revenue, costs, and income are recognized throughout the contract duration according to the extent of progress made towards completion	PoC method in FRG
Performance verification report	Documents the results proving the fulfillment of the agreed performance targets, regarding e.g. availability, performance.	
Permanent Establishment (PE)	A Permanent Establishment (PE) is a legally dependent business form of the registered presence of a non-resident company (parent entity) in a country of activity different to the country of residence. Depending on local legal and / or tax view it can be perceived as an independent entity and is thus subject to the local law of the country of activity.	Permanent Establishment PE Homepage CF T
PM@Siemens Coordinator	Drives and supports the implementation of PM@Siemens in own area of responsibility on behalf of respective CEO	
Process Framework	Processes and tools intended to serve as guidance for the organization as well as the projects.	
Procurement Management plan	Describes the 'How' procurement is to be managed. It should at least include information about the involved external parties, how they are embedded in the workflow of the organization.	
Procurement plan	Describes the project specific procurement issues, and must include at least: make or buy decisions, project specific procurement strategy, preferred supplier list (or link), supplier selection criteria, tools to be used, supplier evaluation and Management measures (e.g. expediting), supplier reporting requirements, list of critical items to purchase in the project, change order and claim strategies for the suppliers, how logistics will be carried out.	
Product and Solution Security (PSS)	Focuses on security risks that are inherent to Siemens' products, solutions, and services which might arise from their delivery and deployment. The PSS Initiative aims at improving the Siemens' portfolio from a security point of view.	Relation between PSS and IT/OT security Product and Solution Security Initiative
Project baselines	References (officially declared at a certain point in time) against which project progress or deviations are measured. At least the following baselines shall be established: time schedule baseline, calculation baseline, WBS baseline, requirement baseline.	
Project closure report	Information about how and why the project was lost or how it was implemented or completed. The issues to be addressed must include at least: win or loss analysis, the main learnings gathered (e.g. risks, materialized opportunities, stakeholders, claims, change orders, non-conformance costs), the fulfillment of financial and other targets, site demobilization (e.g. compliance, scrap material, spare parts), minutes of site closure, contact people for the warranty phase.	
Project delegation concept	Applicable for cross-border staffing projects, describes the delegations deemed to be necessary for the project. The issues to be addressed must include at least: requirements on delegation, delegation types, definition of necessary contractual basis for a delegation, tax liabilities, consideration of the obligation to contribute to Social Security / Insurance, immigration requirements and preconditions (work / residence permit).	(Link)
DRIVE+	hosted by SMO (former Project Management Basic Learning Program PM1) - target level: Project Director/Commercial Project Director.	DRIVE+
Project Manager (PM)	Overall responsible for the project/delivering the solution.	PM
PM2	Project Management Core Learning Program - target level: Senior Project Manager.	PM2

Term	Description	Link
PM Professional	Project Management Core Learning Program – target level: Project Manager and Project Manager for S Projects. (Replacing PMF and PM3)	
PM3	Project Management Core Learning Program - target level: Project Manager.	PM3
PMF	Project Management Fundamental training - target level: Project Manager for S Projects.	PMF
Project Management Institute (PMI)	is a U.S.-based professional organization for project Management.	PMI
Project Management Professional (PMP)	is a certification title granted by PMI.	PMP
Project Management Office (PMO)	is a function of Business Excellence with the objective of supporting the successful delivery of projects.	
Project organizational chart	Representation showing the defined roles, names, and hierarchy for a project.	
Project overview	Initial description of the execution project delivered prior to the opportunity development phase. Some of the issues, which this initial version must include are: expected scope and deliverables, customer expectations, business case, risks and opportunities. Starting with the opportunity development phase, it is extended by the bid manager (and later updated by the Project Manager) to include at least the following issues: background information, reasoning behind the intention to bid, project purpose, project objectives (with success criteria), project boundaries, project assumptions, escalation routes(s) of the Project Manager.	
Project Portfolio Management	Coordinated approach for strategically managing selected and aligned components to achieve organizational objectives in a more effective way. The main goal is to optimize the resource mix (e.g. human, technological, financial), while at the same time considering constraints coming from markets, customers, strategic objectives, other environmental factors.	
Project Procurement Manager (PPM)	Integrates Procurement in the project Management process and project teams at an early involvement during sales phase. Handles, controls and monitors all Procurement and if required Purchasing activities in a project. Establishes and maintains preferred supplier lists (e.g. FPL) and introduces Procurement/Sourcing strategies targeting maximum contribution to the operating result under the consideration of product/service availability & quality and guaranteeing the observance of the Procurement processes described and optimizing deployment of resources.	
Project proposal (P-Vorlage)	Describes the proposed project investment to receive an approval for negotiation prior to the start.	Investment & Divestment application (SR_222)
Project security concept	Describes the required security measures for the project. It addresses issues like e.g. awareness by the employees, accommodation, transportation, site security, emergency/crisis preparedness	Project Security
Project Security@Siemens Tool	ProjectSecurity@Siemens (PS@S) is a joint project security management platform used across divisions, regions and functions. The platform supports Bid Managers and Project Managers to request required input from SEC and to manage security aspects of their projects.	PS@S
Project Target Agreement (PTA)	Formal confirmation of an execution project documenting agreed targets regarding e.g. deadlines, scope, costs, and margin.	
Project time schedule	Structured representation of activities, Milestones and deliverables with the corresponding start and end dates. Its goal is to show transparency about the sequence which needs to be followed to fulfill the project goals, and to be used as a basis for tracking the progress during the entire project duration. It must include at least: WBS, Milestones (contractual, payment, PM@Siemens, etc.), customer obligations, supplier inputs / outputs, constraints, links, representation of the critical path. Recommended good practices include activity lists, activity durations, network diagrams, assumptions.	

Term	Description	Link
	The controlling of the project progress can be supported using a Milestone Trend Analysis (MTA).	
Project-specific Product & Solution Security (PSS) Concept	Describe and define the tailored set of PSS security activities and all derived high-level protection goals (e.g. specify which of the customer solution's data and functionality must be protected from loss of confidentiality, integrity, and availability.) Note: In SMO the term "PSS plan" is used instead. It is equivalent to the term PSS concept."	Security Activities Overview for Solution Business

Term	Description	Link
Quality Gate	Located at selected Milestones (in the sales and execution phases of projects) that could have serious implications for the project success, e.g. handover of responsibilities, points at which preventive measures could be taken most effectively. A Quality Gate requires the involvement of a QMiP as well as the approval of the Responsible Business Manager.	
Quality Gate fulfillment and decision documentation	List of criteria for fulfilling a certain Quality Gate (organization specific) as well as the protocol of the decision made. This can be supported by checklists, tools, LoA documents, etc.	
Quality Management plan	Describes the 'How' quality is to be managed It should at least include information about the involved parties, how they are embedded in the workflow of the organization.	-
Quality Manager in Projects (QMiP)	Establishes the defined quality Management system tailored/aligned to the project-specific requirements to ensure the quality in process and product quality in service/solution project to achieve customer satisfaction and business success. Acts as a member of the project core team with special focus on failure prevention. Advocates to a higher quality aspiration.	QMiP
Quality plan	Describes the project specific quality issues so that in the end the customer deliverables have the necessary quality to achieve acceptance and reach customer satisfaction. Both internal and external (to Siemens) aspects are addressed. The issues to be addressed must include at least: quality goals and requirements for the project, relevant norms and standards addressed, schedule and plan for the necessary Milestones and Quality Gates, a plan for reviews (also for audits if necessary) of the project or its deliverables, quality relevant issues regarding suppliers (e.g. expediting concept, FATs, audits, quality reporting), quality relevant issues regarding the customer (e.g. validation of requirements, witnessing of tests, acceptance procedures).	
Ramp up plan	Description of the allocation of resources and steps necessary from the first customer acceptance to full system operational.	
Realization concept	Description of how the service is to be carried out, including e.g. procurement as well as dispatch of the necessary goods and services, expectations on site, infrastructure, necessary materials, staff requirements, schedule, downtimes.	
Redline	Correction or adjustment to a document which is temporary, prior to its' incorporation into the final version. This term comes from using a red pen to mark changes or corrections which need to be implemented into the final 'clean' version of a file or document.	
Reference scope solutions	Pre-defined solutions for standard project scope, which should be used for optimizing the sales and execution phase of a project. An organization should consciously select or design such solutions. The issues to be addressed as part of a reference solution must include at least: WBS, list of interfaces, standard products to be used with drawings and bills of material, cost breakdown, typical schedule including all necessary effort and duration breakdown (including e.g. creation, installation, programming), performance information, all relevant documentation, standard suppliers, information about impacts of changes to the scope or interfaces or performance.	
Request for Quotation (RfQ)	The non-binding gathering of information, in particular for the purpose of submitting an offer.	
Requirement collection (including break down and structure)	Structured compilation and classification of customer needs and expectations based on the contract and broken down to a manageable level. The issues to be addressed must include at least: requirement baselines, clear labeling of critical requirements, requirement traceability matrix.	
Requirement traceability matrix (potentially in selected tool)	Structured representation of the relationships between the requirements which need to be fulfilled and the tests which prove that these requirements are fulfilled. It must include at least: mapping of requirements to each other and test cases, dependencies, basis for tracing the requirements and their fulfillment bi-directionally.	

Term	Description	Link
Requirements and Deliverables Management Plan	The Requirement and Deliverable Management Plan documents how requirements, deliverables and work scope will be analyzed documented and managed throughout the project. Components of this plan can include, but are not limited to: • How requirements activities will be planned (including prioritization and tracing), tracked and reported • How activities regarding deliverables (including acceptance criteria) will be planned, tracked, approved, and reported • How activities regarding work scope (including WBS and work split) will be planned, tracked, approved, and reported • How changes to requirements, deliverables and work scope will be initiated, how impacts will be analyzed, how they will be traced, tracked, and reported, as well as the authorization levels required to approve these changes.	
Resource calendar	Representation of the current availability of the various resources, which are utilized by projects.	
Resource Management plan (human and others)	Describes the 'How' resources are to be managed. The issues to be addressed must include at least: necessary resources and their availability, restrictions or constraints for their utilization, workload planning and documentation, documentation of actual resource availability, individual and team appraisal, project specific training needs.	
Resource Plan	Representation of the planned and agreed availability of the specific project resources.	
Responsible Business Manager (RBM)	Mandatory role, established by an organizational unit for each project, throughout the whole project lifecycle. Role ensuring that the project organization can carry out projects (project business aspects) and that the projects themselves receive the necessary support. They empowers the PMs and makes decisions and solves conflicts beyond the Project Manager's authority. In some organizations, the RBM acts as Project Sponsor.	
Retrospective	Retrospectives provide quick and easy lessons learned from what is happening in the project (team) and identify useful improvement measures.	SDS Wiki Agile@PB
Review method	Approach used to review a deliverable. Some of the typical review methods are e.g. buddy, 4 eyes, walkthrough.	
Risk	Uncertain event (with considerable probability and impact) or condition whose occurrence will have a negative impact on the fulfillment of one or more project objectives. Risk categories are e.g. compliance, technical, commercial.	
Risk and Opportunity (R&O)	Risk is the potential for loss caused by an event (or series of events) that can adversely affect the achievement of the project´s business objectives. Opportunity is the potential for gain caused by an event (or series of events) that can positively affect the achievement of the project´s business objectives.	
Risk and Opportunity register	Record of identified risks and opportunities. It must include at least: identification numbers, descriptions, categories, results of quantitative and qualitative analyses (financial impact x probability before and after measures), the planned responses with deadlines and those responsible, status.	
Risk contingencies	Financial provisions for potential risks, which may occur during the project.	
Role description	Documents what responsibilities a certain role (a function, not a person) has and what it is authorized to do and prerequisites to be able to fulfill this role. An individual can carry out several roles (sometimes used is the term: wear several hats), depending on a project or situation.	
Service concept	Describes how the service will be carried out. It considers the operations concept and includes the ramp-up plan (if applicable), as well as e.g. information about which components need to be stored or ordered, lead times for repairs.	
Service project scenarios	• The customer buys a system from Siemens; concurrently, the customer signs a (long-term) service agreement.	

Term	Description	Link
	• Installed base where Siemens is taking over the maintenance from the customer or other previous service providers • Siemens is taking over the maintenance of third-party products • Extension or enhancements of existing Siemens service contract (time or scope).	
Service project variations	Long Term Service Projects: • Long term Service-Related Solutions are offers to customers that optimize and implement new applications, services, and business models for an extensive period. • Long term service projects have both a Mobilization and Operations phase in their life cycle. While the Mobilization phase occurs only once in a long-term project (the project character ends with the PM(S)650), the operations phase covers repetitive activities over an extensive length of time. Short Term Service Projects: • Solutions are offered to customers for planning and executing of event(s) that occur over a short period of time (few months). These projects although might be lengthy in the planning stage are performed rather quickly in the timeframe of the project.	
Service work instructions	Detailed information for the staff on site describing how the various service tasks and activities are to be performed.	
Shipment documentation	Includes a selection of the following: air waybill, bill of lading or truck bill of lading, commercial invoice, certificate of origin, insurance certificate, packing list, or other documents required to clear customs and take delivery of the goods.	
Siemens Project Management Plan (SPMP)	Consists of: 1. Project overview as well as key project deliverables relevant for the bid 2. Siemens Project Management Plan Procedures as well as key project deliverables relevant for the project execution. The procedures need to be adjusted for the project and describe how it is to be executed, monitored, controlled, and closed. It can be either summary level or detailed, however, it must address all the prescribed issues, which are specifically detailed to the extent required by a project.	
Siemens Project Management Plan (SPMP) Procedures	The procedures describe how a project in an organization is generally to be executed, monitored, controlled, and closed. These need to be adjusted to include project specifics. The following procedures belong to the Siemens Project Management Plan: • Contract Management Plan • Requirement and Deliverable Management Plan • Resource Management Plan • Schedule Management Plan • Cost Management Plan • Quality Management Plan • Risk and Opportunity Management Plan • Procurement and Logistics Management Plan • Communication and Collaboration Management Plan • Documentation Management Plan • EHS) Management plan • Learnings in Project Management Plan • Controlling and Updating Management Plan	
Siemens Source Code	This term refers exclusively to the source code on which our core, standard products are based, e.g. NX, Teamcenter. This is Siemens intellectual property. Own IP - will not be handed over to customer	

Term	Description	Link
Site procedure	Description of the most important issues which need to be considered or followed when present or working on the site. Some of the issues which need to be addressed include site induction, work processes, EHS responsible people, emergency procedures, security procedures.	
Software development	The process of conceiving, specifying, designing, programming, documenting, testing, and bug fixing involved in creating and maintaining applications, frameworks, or other software components. It is a process of writing and maintaining the source code, but in a broader sense, it includes all that is involved between the conception of the desired software through to the final manifestation of the software, sometimes in a planned and structured process. Therefore, software development may include research, new development, prototyping, modification, reuse, re-engineering, maintenance, or any other activities that result in software products.	
Software Utility Program/Tool	A utility program or tool is a specific type of tool that performs general purpose work, e.g. administration or data migration activity. Some utilities are included in a COTS package, others are not. Typically, they are standalone programs that are run to perform a specific function. Such a tool can be pre-existing, is created as part of the implementation activity or independent of any customer engagement. Examples (non-exhaustive list): A Siemens utility program that supports the data cleansing and migration process for one of our product installations.	
Stakeholder	Person, group, or organization that has interests in, or can affect, be affected by, or perceive itself to be affected by, any aspect of the project (Source: ISO21500).	
Stakeholder register	Collection of information about stakeholders outside the project organization, which have an interest in or are affected by the project and its outcome. It must include at least: stakeholder names, their positions, results of an analysis of their interest and influence on the project, actions (with deadlines and responsible people) for addressing the stakeholders.	
Statement of Work (SOW)	Agreements and scope of work to be delivered as part of customer contract	
Strategic Workforce Planning (SWP)	Repetitive approach for proactively planning and realizing the mid and long-term workforce needs of an organization, adopted from the strategy. SWP aims at identifying competency gaps (quantitative and qualitative) based on the current workforce and the future workforce demand. It enables the organization to initiate adequate actions (e.g. employee development, recruiting, training) in a proactive manner to close the identified competency gaps.	
Style Sheet Language	A computer language that expresses the presentation of structured documents, e.g. HTML, XML.	
Supply chain / logistics concept	Described how the internal and external procurements will be transferred from their origin to their destination. The issues to be addressed must include at least: transport and hand over of the various goods to the next interface (including e.g. road surveys, selected freight forwarders), good storage and bundling, customs, quality and completeness verification, site readiness, receipt and storage on the final site.	
Sustainable Project Management	Sustainable Project Management is the planning, monitoring and controlling of project delivery and support processes, with consideration of the social, environmental and economic aspects throughout the entire project life cycle.	
System architecture	The structural design of systems (can include software and hardware).	Architecture (PLM&I)
Target costs	These are the costs which, based on the market price, need to be reached to make it profitable to bid for a specific project.	
Test plan	Description of which test phases as well as tests need to be performed. The issues to be addressed must include at least: test scopes, time schedules with the necessary resources for the tests, test locations, customer and supplier obligations, assurance of EHS requirements.	

Term	Description	Link
Schedule Management plan	Describes the 'How' the project schedule is to be managed. The issues to be addressed must include at least: the process used for updating the schedule including responsibilities and frequencies, the tools to be applied, the WBS, descriptions of the analyses required for the reporting.	
Total Cost of Ownership (ToCo)	Approach for comparing supplier offers based not only on the financial values but also considering other factors such as experience, financial health, risks, necessary expediting.	
Training plan	Includes the trainings which need to be performed by the internal as well as external employees who will operate and service the equipment in the future once customer's solution has been implemented. It should include e.g., time frames, who will perform the trainings, who shall attend the trainings, information about the training content.	
Update of Software	Maintenance - Customer systems are adapted due to small software - all the functionalities are running smoothly and are up to date perhaps little improved (incl. Bug-fixing)	
Upgrade of Software	Customer will pay Changing customer system due to new product release - adding functionalities for customer Bug fixing with new features - for all customers who are paying	
Vulnerability Monitoring	Vulnerability monitoring and update management is part of the security activities to be performed in the customer project execution process. Monitoring is an ongoing activity, and in case of a vulnerability, it needs to be managed, typically by deploying a security patch or suitable countermeasure.	Vulnerability Monitoring in Siemens
Work Breakdown Structure (WBS)	Structured representation of the project scope. It must include at least: the structure of the project down to the work package level, a dictionary explaining the various work packages, bill of quantity, assumptions, and boundaries.	
World Project Controller (WPC)	The World Project Controller (WPC) is a role at Siemens that is required to be fulfilled when a project meets a certain criterion, e.g. the involvement of at least two legal entities or two P&L Units or particular technical and/ or financial risks. Usually, these projects are executed in a complex contractual environment. The WPC is not an additional function in the project organization, it is a role that is staffed by one of the parties involved. Generally, the WPC is the Commercial Project Manager (CPM) of one of the units involved, i.e., Business Unit headquarters (BU HQ) or Regional Unit (RU) in case of the involvement of two legal entities. The WPC will be nominated by the business responsible in writing.	WPC tasks according FRG

7 Annex 2: Career Model and Development for Project Manager and Commercial Project Manager

General Disclaimer: All gender-specific terms are intended to refer to any gender, except when specifically referring to an individual.

7.1 Introduction

Siemens AG maintains a company-wide internal qualification and certification program for project management (PM@Siemens).

This Annex, titled "Career Model and Development for Project Manager and Commercial Project Manager", specifies company-wide regulations to ensure a sustainable development of Project Managers (PMs) and Commercial Project Managers (CPMs), and a reliable proof of competence through a dedicated certification process. The Annex outlines mandatory minimum certification requirements, in line with the project management policy as it is defined in the PM@Siemens Guide.

Regional amendments to this guideline, necessitated by local legal requirements, can only be implemented by the respective PM@Siemens Coordinator in alignment with the PM@Siemens Corporate Governance owner. Such amendments shall be limited to provisions adjusted to meet local legal requirements.

Businesses and regional companies are responsible for the implementation of this Annex.

The current version supersedes all previous versions of the Annex.

7.2 Target Group

This Annex addresses all relevant stakeholders in the project management environment, supporting the certification and qualification of project managers and commercial project managers executing external customer projects, as defined in 1.3.1 of the PM@Siemens Guide.

The target group of this Annex includes:

- Project managers, commercial project managers, and employees designated to assume a project manager or commercial project manager role
- Responsible business managers
- Project management excellence representatives and P&O employees responsible for the implementation of this framework
- Key support roles contributing to the implementation of PM@Siemens

7.3 Project Manager and Commercial Project Manager Career Model

Project managers act as entrepreneurs taking full responsibility for project results. This guiding principle is the core of the PM@Siemens standard. Commercial project managers bear full responsibility for commercial and financial matters of the project.

To support this entrepreneurial culture and based on predefined job profiles, Siemens has established an attractive career path for project managers and commercial project managers. This includes a comprehensive qualification landscape and an internal certification program.

A typical journey to becoming a Project Director as the highest certification level typically spans approximately 10 years and follows a structured progression through various roles and responsibilities. The path begins at an operational level, where candidates spend their first one to two years in sales, engineering, or site management positions, followed by about 2 years of project core team membership which allows them to gain essential knowledge on project methodology and collaborative skills. The next step is a transition into a (Sub-) Project Manager position in S projects (PM-S/CPM-S) for about one year, marking their first step into management responsibilities.

As the next step in the project management path professionals advance to the Project Manager role on C level or Multi S level (PM/CPM/PM/CPM Multi S) for about three years, allowing them to handle more complex projects and larger teams before they can target Senior Project Manager assignment (SPM/SCPM), a role which is typically executed for about 5 years and prepares the candidates for the final and highest level in project leadership, which is Project Director level (PDir/CPDir).

Throughout this progression, candidates evolve from purely operational roles to project management positions with increasing responsibility for budgets, teams, and strategic decisions. It further ensures that Project Directors possess comprehensive understanding and hands-on experience across all aspects of project management at each level.

The above-mentioned periods to be spent in every development step are an approximation only as the duration varies significantly between different types of project business.

1) An approximation only, as the duration varies significantly between different types of project business

Figure 19: Career path for project managers and commercial project managers

7.4 PM@Siemens Learning and Development Landscape

7.4.1 My Learning World

Continuous learning is one of the key factors in maintaining a high level of project execution quality. It is an integral part of project managers' everyday work-life and career development.

To provide the best learning experience globally, anytime and anywhere, PM@Siemens has established the PM@Siemens - Project Management channel on "My Learning World".

This channel provides colleagues interested in project management with all the relevant information needed to enhance their project management expertise. It consists of the following sections:

- **Project Management Topics**: This section contains recommended learning courses on general project management topics and related soft skills.
- **Qualifications and Personal Development**: This section contains PM@Siemens core learning programs (for details refer to 7.4.2), the finance curriculum (for details refer to 7.4.3), and the PM@Siemens roles.
- **Useful links**: This section contains links relevant for PM@Siemens, e.g., PM@Siemens Guide, certification, LoA.

7.4.2 PM@Siemens Core Learning Programs

PM@Siemens core learning programs are comprehensive training courses covering all areas of personal development and project management skills, as outlined in the respective job profiles and required for projects of specific complexity, e.g., project management methodologies, business administration and leadership competencies. (Figure 20).

The programs are delivered depending on local requirements with an onsite, online or blended learning approach. They ensure that project management standards are put into practice throughout Siemens, lessons learned from projects are shared globally, and personal skills and competencies are developed continuously.

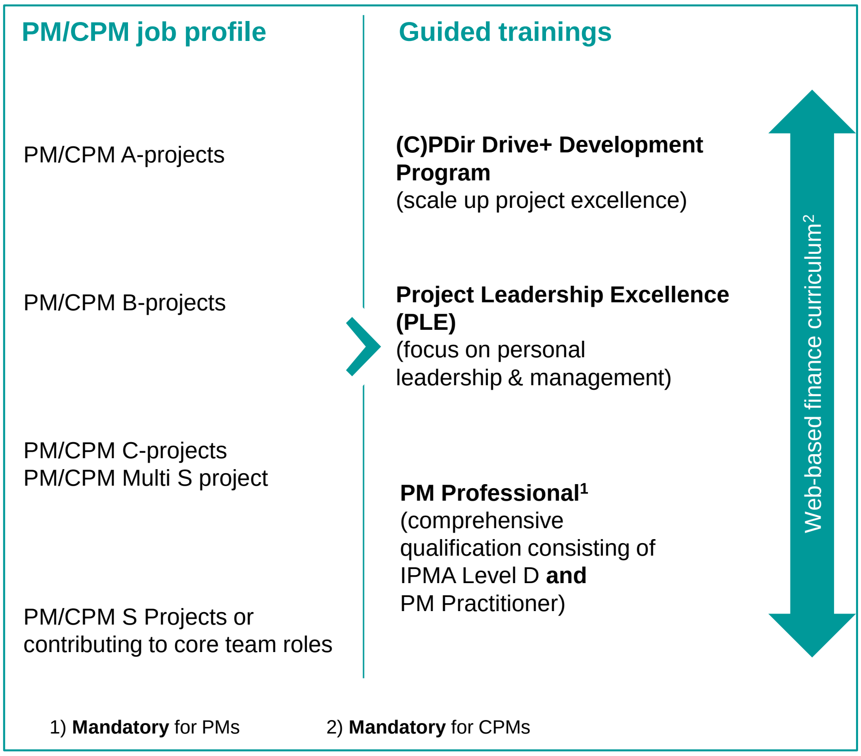

Figure 20: Core learning programs in relation to PMs/CPMs development level

The PM Professional course offers a transformative learning journey tailored for qualified project managers. This comprehensive program integrates IPMA Level D certification in combination with the PM Practitioner, in which Siemens specific PM knowledge is imparted. With an emphasis on competencies and delivered by expert trainers proficient in multiple languages, PM Professional equips participants with the tools and knowledge needed to excel in today's dynamic project environments. For further details on the PM Professional training course refer to PM Professional FAQ

Web-based finance curricula are mandatory training courses for CPM certification only.

The Core Learning Programs PLE and DRIVE+(former PM1) are recommended elements for development towards Senior Project Manager and Project Director levels.

Participation in any of the above-mentioned training courses does not replace PM@Siemens certification.

7.4.3 Web-Based Finance Curriculum

Management of project financials is the core responsibility of commercial project managers.Project managers, who bear overall responsibility for project results, require a deep understanding of project financials.

In cooperation with CF R, PM@Siemens maintains a web-based learning platform called fiT Desk, which offers training courses on all relevant finance topics, e.g., risk and opportunity management, cost calculation, and NCM.

Fit training courses are offered as a fixed set of courses, e.g., CPM S, CPM Multi S, CPM and SCPM curricula, to cater for different project complexities. These courses are a mandatory part of the certification path for commercial project managers at respective certification levels (for details refer to 7.5.2.1).

The fiT platform also offers recommended curricula for project managers and PM/CPM line managers (for details refer to 7.5.2.1).

7.5 PM@Siemens Certification Program

Given the risk specifics of project business, Siemens requires that every customer project (in the category A, B and C) including its subprojects in the execution phase must be staffed with a certified Project Manager and a certified Commercial Project Manager in accordance with the project category they are responsible for. For this purpose, PM@Siemens maintains the following certification program.

7.5.1 Target Group

The target group for certification includes:

- Active project managers with full responsibility for project results, and active commercial project managers with full responsibility for commercial and financial matters, or
- Employees designated to take over execution responsibility within the next 12 months for a project category equivalent to the intended certification level.

7.5.2 PM@Siemens Certification Process

The PM@Siemens certification process consists of the following steps:

Process step	Description
Nomination and Approval	• Identification of potential candidates based on defined job profiles • Candidate applies for certification through their line manager, who approves the application and thereby nominates the candidate for certification • PM@Siemens Coordinator verifies and approves the nomination as the single point of contact for certification issues • PM@Siemens Coordinator documents the nomination in GLP 2.0
Verification of Certification Prerequisites Online competency analysis, mandatory trainings and knowledge tests	In the frame of certification, candidates need to prove: • Fulfillment of project history requirements, experiences, and competences confirmed by the PM@Siemens Online Competency Analysis (OCA) in the PMCAT tool (for details refer to 7.5.2.2) • Completion of mandatory training courses, i.e., PM Professional course for PMs and WBT Curricula for CPMs (for details refer to 7.5.2.1) • PM Knowledge Test or equivalent external certificate (for details refer to 7.5.2.3) • Finance Expert Knowledge Test (FEKT) for SCPM and CPDir certifications (for details refer to 7.5.2.3)
Certification Board	• Certification is confirmed by the PM@Siemens certification board (not required for PM S or CPM S Level certification, for details refer to 7.5.2.4)

7.5.2.1 General Prerequisites for Certification

For the admission to the certification process, candidates need to

- Have full (commercial) responsibility for project results in the project category equivalent to the intended certification level or higher, or be designated to take over this responsibility within the next 12 months, and

- Have comprehensive knowledge of project management standards as well as experience in project sales and execution (for details refer to 7.5.2.2).

To achieve the target certification level, PM candidates need to fulfill the following requirements:

Level	PM Certification Requirements				
	Nomination	**Online Competency Analysis**	**PM Professional**	**PM Knowledge Test**	**Certification Board**
PM S	x	n/a	x	(x)	n/a
PM Multi S	x	x	x	(x)	x
PM	x	x	x	(x)	x
SPM	x	x	x	(x)	x
PDir	x	x	x	(x)	x

For details on specific requirements when progressing from one certification level to the next one refer to the Overview of certification requirements PM on the PM@Siemens SharePoint General notes:

- Online competency analysis is performed in the PMCAT tool, for details refer to 7.5.2.2
- For PM S certification, the line manager confirms the candidate's project experience in managing 3 S projects or adequate subprojects over at least 12 months via workflow in the PMDOC tool
- PM Professional is required only once at a lower certification level, e.g., if attended on PM S level, it does not need to be repeated on PM level.
- PM Knowledge Test is only required if IPMA level D certificate, e.g. from the PM Professional course, is not available. PM Knowledge Test can be substituted by a valid PMI certificate. The PM Knowledge Test is required only once at a lower certification level, e.g., if passed at PM S level, it does not need to be repeated at PM level. For details refer to 7.5.2.3.

To achieve the target certification level, CPM candidates need to fulfill the following requirements:

Level	CPM Certification Requirements					
	Nomination	**Online Competency Analysis**	**PM Knowledge Test**	**WBT Curricula**	**FEKT**	**Certification Board**
CPM S	x	n/a	x	x	n/a	n/a
CPM Multi S	x	X	x	x	n/a	x
CPM	x	X	x	x	n/a	x
SCPM	x	X	x	x	x	x
CPDir	x	X	x	x	x	x

For details on specific requirements when progressing from one certification level to the next one refer to the Overview of certification requirements CPM on the PM@Siemens Sharepoint

General notes:

- Online competency analysis is performed in the PMCAT tool, for details refer to 7.5.2.2
- For CPM S certification, the line manager confirms the candidate's commercial experience in managing projects or bids over at least 12 months via workflow in the PMDOC tool
- The PM Knowledge Test is required only once at a lower certification level, e.g., if passed at CPM S level, it does not need to be repeated at CPM level. For details refer to 7.5.2.3.

- WBT Curricula requirements:
  - For CPM S certification, WBT Curriculum for S projects is required
  - For CPM Multi S and CPM certification, both CPM Curriculum for S projects and CPM Curriculum are required
  - For SCPM and CPDir certifications CPM Curriculum for S projects, CPM Curriculum and Curriculum SCPM are required.
  - WBT Curricula are to be completed only once at a lower certification level, e.g., if CPM Curriculum for S projects is completed at CPM S level, it does not need to be repeated at CPM level. For details refer to 7.5.2.1.
- FEKT is required only once, e.g., if passed at SCPM level, it does not need to be repeated at CPDir level.

7.5.2.2 Online Competency Analysis

Except for PM/CPM S level, the PM@Siemens online competency analysis is a mandatory part of the PM and CPM certification process. It supports the analysis of competencies defined in the PM and CPM job profiles essential for a successful execution of projects in the respective project categories.

The PM@Siemens online competency analysis is performed in the Project Management Competency Analysis Tool (PMCAT) and consists of three major elements:

- Verification of minimum project history requirements
- Candidate’s self-evaluation of competencies and experiences based on the respective job profile
- Line manager’s evaluation of the candidate’s competencies and experiences based on the respective job profile

For C-level and Multi S level certification (PMs/CPMs and PM/CPM Multi S), the Lead Country (LC)/ PM@Siemens Coordinator:

- Reviews the OCA report
- Decides whether the intended certification level can be confirmed, and the candidate can proceed on the certification path
- Provides feedback to the candidate and line manager on the final report decision.

For A- and B-level certifications (Senior and Director level), the Lead Country PM@Siemens Coordinator confirms the OCA result together with the headquarters PM@Siemens Coordinator of the business.

In all cases, PM@Siemens Coordinators are responsible for the appropriate application and evaluation of the OCA reports in accordance with the certification requirements and for the feedback to candidates and their line managers on OCA results.

A positive OCA result is a prerequisite for candidates to proceed on the intended certification path.

For further details, please refer to the PMCAT (OCA) Homepage.

Minimum Project History Requirements for PM certification:

Criteria	Experience as PM with full responsibility	Experience as core team member	Team size managed	International experience[33]	International experience gained in
Project Manager Multi S (PM Multi S) (S-Projects)	• Managing ≥ 2 S-Projects/bids with a volume ≥ EUR 250,000 each or equivalent responsibility for a subproject within a C-Project • Additionally, > EUR 1'500.000 as the overall volume of multi projects/bids within 3 years	n/a	1 year ≥ 4 internal/external parties/stakeholders within the project ecosystem (e.g., construction managers, sub-suppliers)	n/a	n/a
Project Manager (PM) (C-Projects)	1 year managing ≥ 2 S-Projects/bids with a volume ≥ EUR 500,000 each or equivalent responsibility for a subproject within a C-Project	Additional 2 years • Core team member in A-, B- or C-Projects/bids **or** • Equivalent responsibility for individually categorized subprojects ≥ EUR 500,000 **or** • PM in S- Projects ≥ EUR 500,000 each	1 year ≥ 4 members	n/a	n/a
Senior Project Manager (SPM) (C-, B-Projects)	• 3 years managing ≥ 2 C-Projects or equivalent responsibility for subprojects within a B- Project	Additional 2 years • Core team member in A-, B- or C-Projects/bids **or** • Equivalent responsibility for individually categorized subprojects ≥ EUR 500,000 **or** • PM in S- Projects ≥ EUR 500,000 each	2 years ≥ 6 members	≥ 1 year	Recommended: ≥ 1 projects • Delegation as responsible PM in another country, **or** • Managing a project with full responsibility across at least 2 business units/countries **or** • Leading an intercultural team and managing suppliers from different countries
Project Director (PDir) (C-,B-, A-Projects)	6 years managing ≥ 2 ≥ B-Project/bids or equivalent responsibility for subprojects within an A-Project	n/a	6 years ≥ 8 members	≥ 2 years	Mandatory: 2 projects • Delegation as responsible PM in another country, **or** • Managing a project with full responsibility across at least 2 business units/countries, **or** • Leading an intercultural team and management of suppliers from different countries

[33] Experience in international or cross-organizational projects, and/or with intercultural teams and/or international suppliers

Minimum Project History Requirements for CPM certification

Criteria	Experience as PM with full responsibility	Experience as a core team member	International experience[34]	International experience gained in
Commercial Project Manager Multi S (CPM Multi S) (S-Projects)	• Managing ≥ 2 S-Projects/bids with a volume ≥ EUR 250,000 each or equivalent responsibility for a subproject within a C-Project • Additionally, > EUR 1'500.000 as the overall volume of multi projects/bids within 3 years	n/a	n/a	n/a
Commercial Project Manager (CPM) (C-Projects)	• 1 year commercially managing external or intercompany projects/bids (≥1 projects or ≥2 bids) with a volume ≥ EUR 500,000 each or • Equivalent responsibility for a ≥1 subproject in execution	Additional 2 years • Commercial team member in ≥1 ≥ C-Project in execution or • Responsible CPM in ≥ 1 S-Project in execution ≥ EUR 500,000 each or • Commercially managing ≥2 ≥S-bids ≥ EUR 500,000 each	n/a	n/a
Senior Commercial Project Manager (SCPM) (C-, B-Projects)	• 3 years commercially managing ≥ C-Projects (≥1 project or ≥2 bids) or • Equivalent responsibility for ≥1subproject in execution	Additional 2 years • Commercial team member in ≥1 ≥ C-Project in execution or • Responsible CPM in ≥ 1 S-Project ≥ EUR 500,000 in execution or • Commercially managing ≥1 ≥C-bid	≥ 1 year	Recommended ≥ 2 projects • Delegation as responsible CPM in another country, or • Managing a project with full commercial responsibility across at least 2 business units/countries or • Leading an intercultural (commercial) team and managing suppliers from different countries
Commercial Project Director (CPDir) (C-, B-, A-Projects)	6 years commercially managing ≥ B-Project/bids (≥ 1 projects or ≥ 2 bids) or equivalent responsibility for ≥ 1 subproject in execution	n/a	≥ 2 years	Mandatory: ≥ 2 projects • Delegation as responsible CPM in another country, or • Managing a project with full commercial responsibility across at least 2 business units/countries, or • Leading an intercultural team and management of suppliers from different countries

General notes:

- In case of PM certification, bid projects shall not exceed 50% of the overall project management history
- Except for Multi S level, project/bids must have a minimum volume of EUR 500,000 each to be considered in the project history
- A maximum of 10 years of experience are considered in the project history

[34] Experience in international or cross-organizational projects, and/or with intercultural teams and/or international suppliers

- Calendar months spent in the role of PM/CPM or core team member (see PM@Siemens Guide Annex 1 “Definitions and Abbreviations” for role definition) are counted for these criteria and not the aggregated duration of all projects managed. Overlapping months are counted only once.

7.5.2.3 Knowledge Tests

PM Knowledge Test

The Project Management Knowledge Test (PM KT) proves the candidate’s proficiency in PM methodology and is required only once at the first certification level. The test is mandatory for PMs and CPMs and is arranged by the responsible PM@Siemens Coordinator and conducted by an authorized PM KT Facilitator, either in a classroom environment or in a supervised online setting.

The PM KT Facilitator role is granted to PM@Siemens Coordinators or proxies by the PM@Siemens corporate governance owner. For further details, refer to PM@Siemens Knowledge Test (PM KT).

The PM Knowledge Test can be substituted with an IPMA Level D certificate or higher or PMI PMP certificate or higher. For details on IPMA Level D certificate in connection with the PM Professional training course, refer to 7.4.2.

Finance Expert Knowledge Test

The Finance Expert Knowledge Test (FEKT) proves for SCPM or CPDir candidates their proficiency in financial and controlling matters. iThe test is arranged by the responsible PM@Siemens Coordinator and conducted by a FEKT Facilitator, either in a classroom or in a supervised online setting.

The FEKT Facilitator role is granted to PM@Siemens Coordinators or proxies by the PM@Siemens corporate governance owner. For further details, refer to PM@Siemens FEKT.

7.5.2.4 Certification Boards (CB)

Duration, key elements and participants of the certification boards differ depending on the certification level:

Project Manager (PM)/ Project Manager Multi S Commercial Project Manager (CPM) /Commercial Project Manager Multi S	**Senior Project Manager (SPM) Senior Commercial Project Manager (SCPM)**	**Project Director (PDir)/ Commercial Project Director (CPDir)**
• Duration: **2.5 hours** per candidate • Short **project presentation** and **interview**/ questions & answers • **2 PM or CPM Assessors** 1 additional P&O Assessor is optional but recommended • The **Lead** role is taken by one of the 2 PM/CPM Assessors	• Duration: **4.5 hours** per candidate • Short **project presentation** and **interview**/ questions & answers • **3 SPM or SCPM Assessors**, including: 1 entitled Lead Assessor[1], 1 Business Assessor and 1 P&O Assessor/ Psychologist	• Duration: **8 hours** per candidate • **Project presentation**, **case study** and **interview**/ questions & answers • **4 PDir or CPDir Assessors**, including: 1 Psychologist/ Lead Assessor, 3 Business Assessors

Silent observers are allowed for training purposes only, e.g., Assessor training

[1]Designated Lead Assessor holds a Lead Assessor Entitlement

Figure 22: Structure of a Certification Board

PDir/CPDir Certification Boards are organized by P&O TO LG ECO. Applications are filed by the respective headquarters PM@Siemens Coordinator of the Business.**SPM/SCPM Certification Boards** are arranged by the PM@Siemens Coordinator.

Regional certification boards without a PM@Siemens Coordinator must be aligned with the headquarters PM@Siemens Coordinator of the respective business.

PM/CPM or PM/CPM Multi S Certification Boards are arranged by business or regional companies if entitled PM/CPM Assessors[35] are available.

Online Certification Boards

Certification boards can be conducted remotely[36], if the following prerequisites are met:

- Availability of an appropriate conferencing medium on site with high-quality video and audio signals
- A 30-minute time extension is considered to cover additional topics, e.g., transfer of photos and handling potential disruptions
- Assessors provide digital signatures on the feedback/result sheet

General Rules for the Certification Boards

- **PDir/CPDir** and **SPM/SCPM** certification boards are conducted in **English**
- Decisions are unanimous, all Assessors have equal voting rights
- Assessors are authorized by entitlement letters up to the level for which they have been trained and approved. Authorized Assessors and lead Assessors are published on the PM@Siemens SharePoint.
- PM/SPM/PDir Assessors can participate as co-Assessors in CPM/SCPM/CPDir certification boards and vice versa. Assessors authorized for PM/CPM certification boards are authorized for PM/CPM Multi S Certification Boards
- To maintain objectivity, Assessors must remain independent and unbiased by having no personal or professional relationship with the candidate and refraining from reviewing detailed candidate information (such as OCA results or project history) before the certification board meeting
- Lead Assessors should be external to the candidate's Business Unit (headquarters candidates), and regional company (regional candidates)
- Once started, a certification board can only be terminated by mutual agreement with the candidate for serious reasons like medical conditions, force majeure or when it becomes obvious that the prerequisites for taking the certification board are not fulfilled. The latest moment for termination is at the end of the question & answer session that follows the project presentation. Termination does not equal failure.
- The PM@Siemens Coordinator or the P&O partner of the respective company unit is the first point of contact for any queries of the candidate. The PM@Siemens corporate governance owner acts as the final escalation authority.
- The certification board is conducted strictly for the level the candidate is nominated for. If a certification board (e.g., SPM) is not successful, a lower level (e.g., PM) cannot be granted even though the scores achieved might be sufficient for the lower level.
- If failed, the certification board can be repeated twice per level, earliest 12 months post last attempt, requiring re-planning in GLP 2.0

[35] Entitlements also include authorization to conduct certification boards on lower level

[36] In alignment with country specific labor law and regulations only

Criteria to Pass a Certification Board

During the certification board, candidates need to demonstrate their expertise and competence in the following competence areas, in accordance with the master job profiles:

For Project Managers:

- Driving for results and entrepreneurship
- Strategic orientation and decision making, incl. analytical skills
- Value orientation
- Customer focus
- Requirement and quality management
- Project finance
- Project planning, monitoring, and closure
- Bid-, contract- and claim management
- Risk, opportunity, and issue management
- Collaboration and influencing, negotiation and conflict management[37]
- Leadership and team development

For Commercial Project Managers:

- Driving for results, entrepreneurship & strategic orientation, decision-making incl. analytical skills
- Value orientation
- Customer focus
- Project finance
- Project planning, monitoring and closure
- Bid-, contract- and claim management
- Risk, opportunity, and issue Management
- Collaboration and influencing, incl. negotiation and conflict management
- Leadership and team development

To facilitate a certification board, PM@Siemens corporate governance owner provides observation sheets with recommended questions and expected behavioral anchors for assessing candidates.

Assessors evaluate each competence area individually in accordance with the following scheme:

[37] Including building networks and partnerships on PDir level only.

Level 1: Serious gaps	Level 2: Gaps identified	Level 3: Target level	Level 4: Exceeding target level
Missing essential elements	Proving a good foundation but not fulfilling all expectations	Clearly meeting the requirements	Consistently exceeding expectations

Figure 23: Assessment levels for PM@Siemens certification

To successfully pass a certification board, project managers need to achieve the following minimal results:

PM/PM Multi S	SPM	PDir
• **7 of 11** competence areas evaluated at **level 3 or above** •	• **8 of 11** competence areas evaluated **at level 3 or above** • The following **competence areas** must be evaluated at level 3 or above: • Driving for results and entrepreneurship; strategic orientation and decision making incl, analytical skills: leadership and team development •	• **9 of 11** competence areas evaluated at **level 3 or above** The following **competence areas** must be evaluated at level 3 or above: • Driving for results and entrepreneurship • Collaboration and influencing, incl. negotiation and conflict management • Leadership and team development • Strategic orientation and decision making, incl. analytical skills
No competence areas evaluated as **serious gaps**	**No** competence areas evaluated as **serious gaps**	**No** competence areas evaluated as **serious gaps**

To successfully pass a certification board, commercial project managers need to achieve the following minimal results:

CPM/CPM Multi S	SCPM	CPDir
6 of 9 criteria evaluated at **level 3** or **above**	• **7 of 9** criteria evaluated at **level 3** or **above**. The following **competence area** must be evaluated at level 3 or above: • Driving for results and entrepreneurship & strategic orientation and decision-making incl. analytical skills	• **8 of 9** criteria evaluated at **level 3** or **above** The following 2 **competence areas** must be evaluated at level 3 or above: • Driving for results, entrepreneurship & strategic orientation, decision-making incl. analytical skills • Collaboration and influencing, incl. negotiation and conflict management
No competence areas evaluated as **serious gaps**	**No** competence areas evaluated as **serious gaps**	**No** competence areas evaluated as **serious gaps**

Documentation of the Certification Board

The result of the certification board is documented in the decision and feedback sheet, which is handed over to the candidate.

In addition and at the end of the certification board, the Lead Assessor provides verbal feedback to the candidate regarding identified strengths, improvement areas, and suggested development actions.

A copy of the decision sheet is sent to the responsible PM@Siemens Coordinator. Other documents created before or during the certification board, e.g., project presentation, Assessor notes, are destroyed immediately after the certification board.

It is at the candidate’s discretion to inform their line manager about the feedback received.

7.5.3 Certificates (issuance, validity, recognition of existing certificates and titles)

Issuance

After successful completion of the certification process, the responsible PM@Siemens Coordinator initiates the issuance of the certification certificate using the PMDOC tool. A copy of the certificate is stored in GLP 2.0.

Validity

PM@Siemens certificates have lifetime validity. Since internal and external rules and regulations, as well as project management requirements evolve over time, it is recommended to regularly refresh project management knowledge. This can be achieved by attending new and revised training courses (e.g. financial WBTs, master classes) to follow the principle of lifelong learning.

Recognition of existing PM@Siemens certificates

Commercial project managers who hold a PM certificate issued by PM@Siemens, do not need to perform the OCA and commercial certification board to obtain a CPM certificate at the respective level. However, they are required to complete the mandatory Web Based Trainings and, if applicable, pass the Finance Expert Knowledge Test.

Certified PM Multi S or CPM Multi S project managers who want to obtain the PM or CPM certification must proof that they have managed at least two projects with a volume ≥ EUR 500,000 each for a minimum duration of one year. A new OCA is needed to confirm these experiences, however, a certification board for PM/CPM level is not required.

Recognition of external project management certifications

For fulfilling the PCMB requirements on adequate staffing (for details refer to SC No. 200, PCMB Control Requirement 2.3.1.4), certain external project management certifications are considered equivalent in accordance with Figure 24. .

PM@Siemens Certificate	External PM-Certificate
PDir	IPMA Level A
SPM	IPMA Level B
PM	IPMA Level C
PM Multi S	IPMA Level C
PM S	IPMA Level D

Figure 24 External Certifiation Acknowledgement Rules

External PM certificates do not automatically translate into PM@Siemens certificates. PM/ CPM holding external certifications can obtain corresponding PM@Siemens certificates if they meet the certification requirements for the respective levels (refer to 7.5.2.1).

Titles

A successful completion of the certification entitles Project Managers/ Commercial Project Managers to bear the following titles:

- **PM S/CPM S certification**: Project Manager/Commercial Project Manager S.
- **PM Multi S/CPM Multi S certification**: Project Manager Multi S/Commercial Project Manager Multi S
- **PM/CPM certification**: Project Manager/Commercial Project Manager
- **SPM/SCPM certification**: Senior Project Manager/Senior Commercial Project Manager
- **PDir/CPDir certification**: Project Director/Commercial Project Director

7.5.4 PM@Siemens Assessors

PM@Siemens certification boards are carried out by authorized Assessors in accordance with their entitlement:

- PM/PM Multi S and CPM/CPM Multi S Assessors
- SPM and SCPM Assessors
- SPM Lead and SCPM Lead Assessors
- P&O Assessors[38]
- PDir and CPDir Assessors
- PDir/CPDir P&O Lead Assessors[39]

Prerequisites for the Assessor role are described in the respective PM@Siemens Assessor profiles.

7.5.4.1 How to Become a PM@Siemens Assessor

To become a PM@Siemens Assessor, the candidate applies via the PM@Siemens Coordinator to the PM@Siemens corporate governance owner.

The Assessor"s application is verified based on the defined Assessor profile. If the application is approved by the corporate governance owner, the candidate must attend an Assessor training conducted by the authorized Assessor trainer as described in 7.5.4.2.

Post training, the candidate is invited to participate one time as a silent observer and two times as co-Assessor in three certification boards.

After successful observing and co-assessing, the candidate is granted official entitlement for a period of three years. To reinstate the Assessor entitlement after three years, refresher training conducted by an authorized trainer must be attended.

Entitlement and re-entitlement for PM/CPM and SPM/SCPM Assessors is granted by the PM@Siemens Coordinator.

Entitlement and re-entitlement for PDir/CPDir Assessors is granted by PM@Siemens corporate governance owner.

A database of all authorized and active Assessors is published by the PM@Siemens corporate governance owner.

7.5.4.2 Assessor Training

Initial and refresher trainings for PM/CPM as well as SPM/SCPM Assessors are provided by an authorized trainer appointed by the PM@Siemens corporate governance owner

Initial Assessor trainings for PDir/CPDir Assessors are solely provided by P&O TO LG ECO. Refresher trainings can also be conducted by employees of the PM@Siemens corporate governance owner who hold an entitlement as Assessor trainer and PDir/CPDir Assessor.

7.5.4.3 PM@Siemens Lead Assessors

In PM/CPM/PM Multi S/CPM Multi S certification boards, the Lead Assessor role is assumed by the more experienced Assessor.

In SPM/SCPM as well as PDir/CPDir certification boards, Lead Assessors are required to have an official entitlement to act in that capacity.

Lead Assessor Role:

Lead Assessors act as moderators in certification boards by:

[38] Can be active in PM/PM Multi S, CPM/CPM Multi S, SPM **and** SCPM Certification Boards

[39] Can be active in PDir **and** CPDir Certification Boards

- Clarifying and aligning roles between co-Assessors and silent observers
- Guiding the candidate through the certification board
- Ensuring the time-schedule is adhered to
- Aligning content and responsibilities for the interviews within the Assessor panel
- Evaluating the candidate's performance based on behavioral patterns
- Seeking consensus within the Assessor panel
- Ensuring that all Assessors have equal opportunities to ask questions and vote
- Preparing the feedback and decision sheet with all Assessors
- Communicating decision and feedback to the candidate
- Ensuring compliance with the certification process and confidentiality rules

Prerequisites to become a Lead Assessor:

The prerequisites to become a Lead Assessor are specified in the respective PM@Siemens Lead Assessor profiles.

Assessors seeking entitlement as Lead Assessors for SPM/SCPM Certification Boards must be nominated by their respective business and approved by the PM@Siemens corporate governance Owner, in line with the defined Lead Assessor Profile.

If the application is approved by the corporate governance owner, the Lead Assessor candidate must attend a training conducted by the Corporate Governance Owner.

Post- training, the candidate is invited to conduct one certification board in the Lead Assessor role supervised by the Corporate Governance Owner. In case of positive feedback provided by the Corporate Governance Owner, the candidate is granted official entitlement as Lead Assessor for the same period as the Assessor Entitlement.

Lead Assessors in PDir/CPDir certification boards are mandated and trained by P&O TO LG ECO only.

7.6 Description of Roles Involved in Certification

Roles in Certification	Description
People and Organization (P&O) representative	Consults the Management of operational businesses
Knowledge Test Facilitator	Employees entitled to conduct the Knowledge Test
FEKT Test Facilitator	Employees entitled to conduct the Finance Expert Knowledge Test
Line Manager or Responsible Business Manager	Has the disciplinary responsibility for the candidate
PM@Siemens corporate governance owner	CF A PEX
PM/CPM Assessor	Is entitled to act as co-Assessor in a certification board of the respective certification level
PM/CPM Lead Assessor	Is entitled to act as Lead Assessor in a certification board of the respective level
PM@Siemens Coordinator	Drives and supports the implementation of PM@Siemens in the business or a regional company on behalf of the respective CEO/CFO

7.7 Applicable Regulations

CP-Circular No. 34/01 - Betriebspsychologische Eignungsuntersuchungen (only Germany)

Master job profiles for Project Managers and Commercial Project Managers

Role description for PM@Siemens Coordinator

Role descriptions for PM@Siemens Assessors

Referenced documents (restricted – only accessible for authorized persons)

Application sheets for PM@Siemens Assessors Entitlements for PM@Siemens Assessors

Schedule for PM/CPM, PM/CPM Multi S, SPM/SCPM and PDir/CPDir Certification Board

PM@Siemens Online Competency Analysis Tool

PM@Siemens Knowledge Test

Observation sheets for PM/CPM, PM/CPM Multi S, SPM/SCPM and PDir/CPDir

Additional business specific requirements

Advanta (ADV)

t.b.d.

Digital Industries (DI)

Chapter 1.3.1 and 3.1 – 3.14: DI Circular No. 2_C_22, Project Management

Chapter1.3.1 and Opportunity Management and Risk Management: DI Circular No. 2_C_24, Risk and Opportunity Management in Project Business

Global Business Services (GBS)

t.b.d.

Smart Infrastructure (SI)

PM @ SI - Home (sharepoint.com)

Specific Requirements for CPM Small and CPM certification: Candidates need to attend SI specific classroom trainings

Siemens Mobility (SMO)

Mobility-specific supplement to LoA- and PM@Siemens Guide

Webbook: Siemens Mobility PM-Guideline

Link page

Application for and approval of capital investments and divestment (SC 222)

Architecture (PLM&I) – Definition

Asset Classification & Protection (ACP)

Compliance – Business Partners

Compliance – Third Party Risk Management

Compliance Requirements for Third-Party Intermediaries

Compliance – Collective Action

Compliance – Export Control

Compliance – Handbook

Compliance in Project Business

Contract, Change and Non-Conformance Management for Projects (DI C23)

Corporate Business Procedures (Zentrale Regeln Geschäftsverkehr) (ZRG)

Customs

Cybersecurity – Contact of Cybersecurity Officers

Cybersecurity - Policies, Rules and Regulations

Cybersecurity risks in the Supply Chain

DAMEX

Data Privacy – Contact of DP Managers

Data Privacy – Intranet

Data Privacy by Design – Toolkit

Delegation – Overview and Policies

DRIVE+

Enterprise Risk Management (ERM)

Environment, Social, Governance (ESG) – Intranet

Environmental, Social, Governance (ESG) – Radar

Environment Health and Safety (EHS) – Intranet

Finance Training (fiT)

Financial Reporting Guidelines and Implementation Guidelines

GLP 2.0

Information Security - SFeRA (Security Framework and Regulations Application)

Internal Control Program Export Control – Digital Exports

International Chamber of Commerce

International Project Management Association (IPMA)

Investment & Divestment application (SR 222)

Limits of Authority (LoA) – Intranet

Limits of Authority (LoA) – Tool

My Learning World

New Collaboration Model (NCM)

Non-conformance Cost (NCC) - Guideline

Open-Source Software (OSS) – Definition

Online Competency Analysis (OCA)

PM@Siemens Coordinator – Role

Permanent Establishment (PE)

Permanent Establishment (PE) – Closing Request

Permanent Establishment (PE) – Manager Checklist

PM@Siemens Certification board – Schedule

PM@Siemens Guide – Intranet

PM@Siemens – People Excellence

PM@Siemens Sharepoint

PM@Siemens – Certification

PM – Role (MySkills)

Process Adjustment Tax Advice and Compliance (PATAC)

Product and Solution Security (PSS)

Project Acceleration by coaching and Teamwork (PACT)

Project Categorization tool

Project Categorization Intranet

Project Management Core Learning Program PM2

Project Management Core Learning Program PMPRO

Project Management Institute (PMI)

Project Management Professional (PMI PMP)

Project Management in DI (DI Circular C22)

Project Security (CE SEC) – Intranet

PSS Knowledge Base (PSSKB)

QMiP – Role

Relation between PSS and IT/OT security

Risk and Internal Control (RIC)

Risk and Opportunity Management in DI Projects (DI Circular C24)

Security Activities Overview for Solution Business

Security Framework and Regulation – SfeRA

Service Business Excellence (including Service Guide)

Siemens Business Conduct Guidelines

Siemens Engineering Community

Siemens Engineering Council

Siemens Information Security Policy (SC 236)

Siemens Organization Handbook

Supply Chain Management Procurement Compendium – SCM STAR

SC213 „Procurement principles at Siemens"

World Project Controller (WPC) - Tasks according to FRG

Version Control

Version	Release Date	Changes to previous version
6.0.0	18.03.2016	New PM@Siemens Guide
6.0.1	27.06.2016	Update with minor textual corrections and updating of some links
6.0.2	04.10.2016	Update with minor textual corrections and updating of some links on corporate level; Change in wording from EHS implementation concept to EHS plan (according to EHS standard) and corresponding rephrasing of the definition in chapter 6.1 Integration of a requirements and deliverables Management plan in chapter 6.1 Integration of PACT (workshop) in chapter 6.1 and additional information in chapter 3.12 Shift of the output „Bid team reward recognition" to input of the organizational unit within phase of PM080
6.0.3	11.01.2017	Minor textual corrections and updating of some links Mandatory establishment of a Project Sponsor for every project throughout the whole lifecycle of a project, starting with PM020 Adjusted allocation of Milestone requirements for the Milestones PM670 (now called: Project execution closure) and PM700 (now called: End of all project obligations)
6.0.4	24.02.2017	Minor textual corrections with consistency validation of meaning in both languages (de/en) Update of links due to SIENET transfer
6.0.5	31.03.2017	Update of links and minor textual corrections Implementation of adjusted Compliance requirements within the project business
6.0.6	30.06.2017	Update of links and minor textual corrections
6.0.7	29.09.2017	Update of links (e.g. EHS) and minor textual corrections
6.0.8	12.01.2018	Update of Glossary and minor textual corrections Specification of Engineering outputs within PM@Siemens Milestone model
6.0.9	14.08.2018	Update of links and minor textual corrections Integration of activities regarding Permanent Establishment in Milestone tables Additional information regarding configuration Management in SPMP
6.0.10	07.02.2019	Update of links and minor textual corrections Elimination of mandatory LoA for S projects
6.0.11	07.06.2019	Update of organizational structure Update of links and minor textual correction
6.0.12	10.12.2019	Update of organizational structure Update of links and correction of text
6.0.13	02.11.2020	Update of organizational structure Update of links and correction of text Elimination of the electronic version of the PM@Siemens Guide

Version	Release Date	Changes to previous version
6.1	02.11.2021	Alternative execution process for Software-driven Solution Projects Streamlining of execution process for service projects Integration of Cybersecurity (incl. Data Privacy and “buy decision”) and ESG in milestones Adaption of the SPMP Harmonization of Annex 1 and 3 Integration of Annex 2 (elimination of respective version control) Adjustment of links and textual corrections Transfer from pdf to Theum All the changes of the PM@Siemens Guide 6.1 compared to 6.0 are applicable/binding latest Apr 1, 2022 for each new project. Textual corrections and adjustments of links will from now on be done continuously without adding the date here in the version control
6.1.1	01.10.2022	Minor content adaption will only be implemented every ½ year (but adaption of links) All what was marked in grey before will now be not highlighted – only the new adaptions Adaption of the PM Guide is marked by the third digit: 6.1.1
6.1.2	01.06.2023	Adaption of Headlines in chapter 3.4 (now Time scheduling) and 3.10 (now Document Management) as well as in all figures and the SPMP Adaption of chapter 3.7 to create awareness towards Opportunity Management next to Risk Management Information Security integrated as part of Cybersecurity throughout the document (w/o being marked in grey) Minor editorial changes (in spelling and point settings)
6.2	06.08.2024	Multi S Certification: project managers of - in parallel - managed 'S' projects can be certified Disclaimer: PM Professional to replace PM3 OCA included One-pager for Sustainability included Update SPMP Structure Compliance Links updated
6.3	25.03.2025	One-pager for Cybersecurity included Update of links and minor textual corrections Clarifications in Annex 2 regarding Multi S certification Wording PBE Coordinator changed to PM@Siemens Coordinator
6.4	15.10.2025	Annex 2 restructured and reworded. No changes to content Minor wording adjustments across the Guide Reference to Generative AI Risk Management process included in Risk Management chapter